Regular expressions and finite automata, Context-free grammars and push-down automata, Regular and context-free languages, Pumping lemma, Turing machines and undecidability.

**Mark Distribution in Previous GATE**

| Year | 2026 - 1 | 2026 - 2 | 2025 - 1 | 2025 - 2 | 2024 - 1 | 2024 - 2 | 2023 | 2022 | 2021 - 1 | 2021 - 2 | Minimum |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 Mark Count | 2 | 1 | 2 | 3 | 1 | 1 | 3 | 2 | 2 | 3 | 1 |
| 2 Marks Count | 2 | 2 | 4 | 2 | 2 | 3 | 3 | 3 | 3 | 4 | 2 |
| Total Marks | 6 | 5 | 10 | 7 | 5 | 7 | 9 | 8 | 8 | 11 | **5** |

Welcome to the "Theory of Computation" chapter, a cornerstone of Computer Science that delves into the fundamental capabilities and limitations of computational models. This subject is crucial for the GATE CS exam as it builds a strong theoretical foundation for understanding algorithms, programming languages, and system design. It typically carries a weightage of 8-12 marks, with questions ranging from conceptual understanding of language classes and machine models to problem-solving involving the design of automata, grammar conversions, and decidability proofs. Expect a mix of Multiple Choice Questions (MCQs), Multiple Select Questions (MSQs), and Numerical Answer Type (NAT) questions, often requiring a deep grasp of definitions, properties, and theorems.

## Topic-wise Key Concepts

## Finite Automata (FA)

Finite Automata are the simplest computational models, used to recognize regular languages. They have a finite number of states and transitions based on input symbols, without any auxiliary memory. They are fundamental for lexical analysis in compilers and designing simple control systems.

- **Definition:** A Deterministic Finite Automaton (DFA) is formally defined as a 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$, where:
  1. $Q$ is a finite set of states.
  2. $\Sigma$ is a finite set of input symbols (alphabet).
  3. $\delta : Q \times \Sigma \to Q$ is the transition function.
  4. $q_0 \in Q$ is the initial state.
  5. $F \subseteq Q$ is the set of final (accepting) states.
- **Key Properties:**
  - Every NFA has an equivalent DFA.
  - DFA and NFA recognize the same class of languages (Regular Languages).
  - DFA can be minimized to a unique (up to isomorphism) minimal state DFA.
- **Common Pitfalls:** Incorrectly handling epsilon transitions in NFA to DFA conversion; confusing acceptance criteria for NFA/DFA.
- **Problem-Solving Techniques:**
  - **Subset Construction:** For converting NFA to DFA.
  - **Table Filling Algorithm** / **Myhill-Nerode Theorem:** For DFA minimization.
  - **State Elimination Method** / **Arden's Theorem:** For converting FA to Regular Expression.

## Finite State Machines (FSM)

Finite State Machines are mathematical models of computation used to design systems that can be in one of a finite number of states. While often used interchangeably with FA, FSMs typically include an output function, making them suitable for modeling sequential circuits and control logic.

- **Definition:**
  - **Mealy Machine:** $M = (Q, \Sigma, \Delta, \delta, \lambda, q_0)$, where $\Delta$ is the output alphabet, and $\lambda : Q \times \Sigma \to \Delta$ is the output function. Output depends on current state and input.
  - **Moore Machine:** $M = (Q, \Sigma, \Delta, \delta, \lambda, q_0)$, where $\lambda : Q \to \Delta$ is the output function. Output depends only on the current state.
- **Key Properties:**
  - Any Mealy machine can be converted to an equivalent Moore machine, and vice-versa.
  - Moore machine output is delayed by one clock cycle compared to Mealy.
- **Common Pitfalls:** Incorrectly converting between Mealy and Moore machines, especially handling initial state outputs.
- **Problem-Solving Techniques:** State diagram construction, state table analysis, conversion algorithms between

Mealy and Moore.

## Regular Expression (RE)

Regular Expressions are a powerful notation for describing regular languages. They provide a concise way to specify patterns of strings, widely used in text processing, search engines, and lexical analysis.

- **Definition:** A Regular Expression is built using basic symbols and three operations:
  - **Union:** $R_1 + R_2$ (or $R_1|R_2$) represents strings in $L(R_1)$ or $L(R_2)$.
  - **Concatenation:** $R_1R_2$ represents strings formed by concatenating a string from $L(R_1)$ with one from $L(R_2)$.
  - **Kleene Star:** $R^*$ represents zero or more concatenations of strings from $L(R)$, including the empty string $\epsilon$.
- **Key Identities:**
  - $R + \emptyset = R$
  - $R\epsilon = \epsilon R = R$
  - $(R^*)^* = R^*$
  - $\emptyset^* = \epsilon$
  - $R(S + T) = RS + RT$
  - $(RS)^*R = R(SR)^*$
  - $(R + S)^* = (R^*S^*)^* = (R^* + S^*)^*$
- **Common Pitfalls:** Operator precedence (Kleene star > concatenation > union); incorrect application of identities.
- **Problem-Solving Techniques:**
  - **Arden's Theorem:** For solving equations of the form $R = Q + RP$ to get $R = QP^*$.
  - **State Elimination Method:** For converting FA to RE.

## Regular Grammar (RG)

Regular Grammars are a type of formal grammar that generates exactly the class of regular languages. They are restricted in their production rules, making them less powerful than Context-Free Grammars but simpler to parse.

- **Definition:** A grammar $G = (V, \Sigma, P, S)$ is regular if all its production rules are of one of the following forms:
  - **Right-linear:** $A \to aB$ or $A \to a$ or $A \to \epsilon$, where $A, B \in V$ (non-terminals), $a \in \Sigma$ (terminal).
  - **Left-linear:** $A \to Ba$ or $A \to a$ or $A \to \epsilon$, where $A, B \in V$, $a \in \Sigma$.
- **Key Properties:**
  - A grammar cannot be both right-linear and left-linear simultaneously (unless it generates a finite language).
  - Regular Grammars, Regular Expressions, and Finite Automata are all equivalent in terms of language recognition power.
- **Common Pitfalls:** Mixing right-linear and left-linear rules in the same grammar; incorrectly converting between FA and RG.
- **Problem-Solving Techniques:** Constructing an FA from a RG, and vice-versa, by mapping non-terminals to states and productions to transitions.

## Regular Language (RL)

Regular Languages are the simplest class of languages in the Chomsky Hierarchy, recognized by Finite Automata, described by Regular Expressions, or generated by Regular Grammars. They represent patterns that can be matched without requiring memory beyond the current state.

- **Definition:** A language is regular if and only if it can be recognized by a Finite Automaton.
- **Key Properties (Closure Properties):** Regular languages are closed under:
  - Union ($L_1 \cup L_2$)
  - Intersection ($L_1 \cap L_2$)
  - Complement ($\Sigma^* \setminus L$)
  - Concatenation ($L_1L_2$)
  - Kleene Star ($L^*$)
  - Reversal ($L^R$)
  - Homomorphism, Inverse Homomorphism
- **Common Pitfalls:** Misidentifying non-regular languages as regular; incorrectly applying closure properties.
- **Problem-Solving Techniques:**
  - **Pumping Lemma for Regular Languages:** The primary tool to prove a language is NOT regular.
  - **Myhill-Nerode Theorem:** Can be used to prove regularity or non-regularity, and for DFA minimization.

## Non Determinism

Non-determinism in automata allows for multiple possible transitions from a given state on a given input symbol, or transitions on an empty string (epsilon transitions). It often simplifies the design of automata, though it doesn't increase their computational power for regular languages.

- **Definition:** A Non-deterministic Finite Automaton (NFA) is a 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$, where $\delta : Q \times (\Sigma \cup \{\epsilon\}) \to \mathcal{P}(Q)$ is the transition function, mapping to a set of states.
- **Key Properties:**
  - Every NFA has an equivalent DFA (recognizes the same language).
  - An NFA can be exponentially more concise than its equivalent DFA (e.g., an NFA with $n$ states might require up to $2^n$ states in the equivalent DFA).
  - Non-determinism is crucial for Pushdown Automata (NPDA is more powerful than DPDA).
- **Common Pitfalls:** Incorrectly handling epsilon closures; missing possible paths in non-deterministic transitions.
- **Problem-Solving Techniques:**
  - **Subset Construction Algorithm:** The standard method to convert an NFA (with or without epsilon transitions) to an equivalent DFA.
  - Understanding how to trace paths in an NFA to determine language acceptance.

## Number of States

The "number of states" refers to the cardinality of the set of states $Q$ in an automaton. This concept is critical for understanding the complexity and minimality of machine models, particularly for finite automata.

- **Key Properties:**
  - For an NFA with $n$ states, the equivalent DFA can have up to $2^n$ states.
  - A minimal DFA for a given regular language is unique (up to isomorphism) and has the fewest possible states.
  - The number of states in a minimal DFA is equal to the number of equivalence classes of the Myhill-Nerode relation.
- **Common Pitfalls:** Failing to identify redundant or unreachable states, which leads to non-minimal DFAs.
- **Problem-Solving Techniques:**
  - **DFA Minimization Algorithms:** Partitioning states into distinguishable/indistinguishable sets.
  - **Myhill-Nerode Theorem:** Using equivalence classes of strings to determine the minimum number of states.

## Minimal State Automata

A Minimal State Automaton (specifically, a Minimal DFA) is a Deterministic Finite Automaton that recognizes a given regular language using the smallest possible number of states. It is unique for any given regular language and is crucial for efficient implementation.

- **Definition:** A DFA $M'$ is minimal if there is no other DFA $M''$ such that $L(M'') = L(M')$ and $|Q''| < |Q'|$.
- **Key Properties:**
  - Every regular language has a unique minimal DFA (up to isomorphism).
  - All states in a minimal DFA must be reachable from the start state and must be distinguishable from each other.
- **Common Pitfalls:** Incorrectly merging distinguishable states; not removing unreachable states before minimization.
- **Problem-Solving Techniques:**
  - **Table Filling Algorithm (or Partitioning Algorithm):**
    1. Initialize a table of all pairs of states $(p, q)$. Mark all pairs $(p, q)$ where $p \in F$ and $q \notin F$ (or vice versa) as distinguishable.
    2. Iteratively mark $(p, q)$ as distinguishable if for some input symbol $a$, $(\delta(p, a), \delta(q, a))$ is already marked distinguishable.
    3. Repeat until no new pairs can be marked. All unmarked pairs are indistinguishable and can be merged.
  - **Myhill-Nerode Theorem:** States that a language $L$ is regular if and only if the number of equivalence classes of the Myhill-Nerode relation $\equiv_L$ is finite. The number of states in the minimal DFA is equal to the number of these equivalence classes.

## Pumping Lemma (for Regular Languages)

The Pumping Lemma for Regular Languages is a fundamental theorem used to prove that a language is NOT regular. It states that all sufficiently long strings in a regular language can be "pumped" (i.e., a middle section of the string can be repeated any number of times) and the resulting string will still be in the language.

- **Theorem:** If $L$ is a regular language, then there exists some integer $p \geq 1$ (the pumping length) such that for any string $s \in L$ with $|s| \geq p$, $s$ can be divided into three parts $s = xyz$ satisfying the conditions:
  1. $|y| > 0$ (the middle part cannot be empty).
  2. $|xy| \leq p$ (the pumpable part must occur within the first $p$ symbols).
  3. For all $i \geq 0$, the string $xy^iz \in L$.
- **Key Properties:**
  - It is a necessary condition for regularity, not a sufficient one. It can only be used to prove non-regularity.
  - The proof strategy is usually by contradiction (an "adversary game").
- **Common Pitfalls:**
  - Trying to prove regularity using the Pumping Lemma (it's impossible).
  - Incorrectly choosing the string $s$ or the partition $xyz$ to break the lemma.
  - Forgetting the condition $|y| > 0$ or $|xy| \leq p$.
- **Problem-Solving Techniques:**
  1. Assume $L$ is regular and let $p$ be the pumping length.
  2. Choose a "hard" string $s \in L$ such that $|s| \geq p$, typically one that grows with $p$ (e.g., $a^pb^p$).
  3. Consider all possible ways to divide $s$ into $xyz$ satisfying $|y| > 0$ and $|xy| \leq p$.
  4. For each division, find an $i \geq 0$ such that $xy^iz \notin L$. This contradicts the lemma, proving $L$ is not regular.

## Closure Property

Closure properties describe whether a class of languages remains within that class after applying certain operations. Understanding these properties is vital for classifying languages and proving relationships between language classes.

- **Definition:** A class of languages $\mathcal{C}$ is closed under an operation $\mathcal{O}$ if, for any language $L_1, L_2 \in \mathcal{C}$, the result of $\mathcal{O}(L_1, L_2)$ (or $\mathcal{O}(L_1)$) is also in $\mathcal{C}$.
- **Key Properties (to memorize):**

| Operation | Regular Languages | Context-Free Languages | Recursive Languages | Recursively Enumerable Languages |
|---|---|---|---|---|
| Union | Yes | Yes | Yes | Yes |
| Intersection | Yes | No | Yes | Yes |
| Complement | Yes | No | Yes | No |
| Concatenation | Yes | Yes | Yes | Yes |
| Kleene Star | Yes | Yes | Yes | Yes |
| Reversal | Yes | Yes | Yes | Yes |
| Homomorphism | Yes | Yes | Yes | Yes |
| Inverse Homomorphism | Yes | Yes | Yes | Yes |

- **Common Pitfalls:** Confusing closure properties between different language classes, especially for intersection and complement.
- **Problem-Solving Techniques:** Use closure properties to deduce the class of a new language based on known languages and operations. For non-closure, use counterexamples.

## Context Free Grammar (CFG)

Context-Free Grammars are more powerful than regular grammars and are used to describe Context-Free Languages. They are fundamental to defining the syntax of programming languages and parsing. Productions allow a non-terminal to be replaced by a string of terminals and non-terminals, regardless of its context.

- **Definition:** A CFG is a 4-tuple $G = (V, \Sigma, P, S)$, where:
  1. $V$ is a finite set of non-terminal symbols.
  2. $\Sigma$ is a finite set of terminal symbols.
  3. $P$ is a finite set of production rules of the form $A \to \alpha$, where $A \in V$ and $\alpha \in (V \cup \Sigma)^*$.
  4. $S \in V$ is the start symbol.
- **Key Forms:**
  - **Chomsky Normal Form (CNF):** All productions are of the form $A \to BC$ or $A \to a$, where $A, B, C \in V$ and $a \in \Sigma$. $\epsilon$-productions are allowed only for the start symbol if it doesn't appear on the RHS.
  - **Greibach Normal Form (GNF):** All productions are of the form $A \to a\alpha$, where $A \in V$, $a \in \Sigma$, and $\alpha \in V^*$.
- **Common Pitfalls:** Detecting ambiguity (a string having more than one leftmost/rightmost derivation or parse tree); incorrect conversion to CNF/GNF.

- **Problem-Solving Techniques:**
    - **Simplification of CFG:** Removing useless symbols, $\epsilon$-productions, and unit productions.
    - **Conversion to CNF/GNF:** Systematic procedures involving variable substitution and rule modification.
    - Derivation trees, leftmost/rightmost derivations.

## Context Free Language (CFL)

Context-Free Languages are recognized by Pushdown Automata and generated by Context-Free Grammars. They can describe languages with nested structures, such as balanced parentheses or properly nested HTML tags, which regular languages cannot handle.

- **Definition:** A language is Context-Free if and only if it can be generated by a Context-Free Grammar.
- **Key Properties:**
    - **Closure Properties:** Closed under union, concatenation, Kleene star, reversal, homomorphism, inverse homomorphism.
    - **Non-Closure Properties:** NOT closed under intersection, complement.
    - Deterministic CFLs (DCFLs) are a proper subset of CFLs and are closed under complement.
- **Common Pitfalls:** Misidentifying non-CFLs as CFLs; incorrectly applying closure properties.
- **Problem-Solving Techniques:**
    - **Pumping Lemma for CFLs:** The primary tool to prove a language is NOT context-free.
    - Designing CFGs or PDAs for given languages.

## Pushdown Automata (PDA)

Pushdown Automata are an extension of Finite Automata with an additional stack memory. This stack allows PDAs to recognize Context-Free Languages, which require memory to handle nested structures, unlike regular languages.

- **Definition:** A Pushdown Automaton is formally defined as a 7-tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$, where:
    1. $Q$ is a finite set of states.
    2. $\Sigma$ is the input alphabet.
    3. $\Gamma$ is the stack alphabet.
    4. $\delta : Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma \to \mathcal{P}(Q \times \Gamma^*)$ is the transition function.
    5. $q_0 \in Q$ is the initial state.
    6. $Z_0 \in \Gamma$ is the initial stack symbol.
    7. $F \subseteq Q$ is the set of final states.
- **Key Properties:**
    - Non-deterministic PDAs (NPDAs) are more powerful than Deterministic PDAs (DPDAs). NPDAs recognize all CFLs, while DPDAs recognize only DCFLs (a proper subset of CFLs).
    - PDAs can accept languages either by final state or by empty stack. These two modes are equivalent for NPDAs.
- **Common Pitfalls:** Incorrectly handling stack operations (push/pop); confusing acceptance criteria; designing a DPDA when an NPDA is required.
- **Problem-Solving Techniques:**
    - Designing PDAs for given CFLs, focusing on how the stack is used to remember context.
    - Converting CFG to PDA and vice-versa.

## Dpda (Deterministic Pushdown Automata)

A Deterministic Pushdown Automaton (DPDA) is a PDA where for any given state, input symbol (or epsilon), and stack top, there is at most one possible transition. DPDAs recognize Deterministic Context-Free Languages (DCFLs), which are a proper subset of CFLs.

- **Definition:** A PDA is deterministic if its transition function $\delta$ satisfies:
    1. For any $(q, a, X) \in Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma$, $|\delta(q, a, X)| \leq 1$.
    2. If $\delta(q, \epsilon, X) \neq \emptyset$, then $\delta(q, a, X) = \emptyset$ for all $a \in \Sigma$. (No conflict between epsilon and input transitions).
- **Key Properties:**
    - DPDAs recognize DCFLs. All regular languages are DCFLs.
    - DCFLs are closed under complement (unlike general CFLs).
    - Not all CFLs are DCFLs (e.g., $\{ww^R \mid w \in \{a, b\}^*\}$ is a CFL but not a DCFL).
- **Common Pitfalls:** Failing to ensure determinism in all transition cases; confusing DCFLs with general CFLs.
- **Problem-Solving Techniques:** Designing DPDAs requires careful consideration to avoid non-deterministic choices. Often involves lookahead or specific stack symbol usage.

## Pumping Lemma (for CFLs)

The Pumping Lemma for Context-Free Languages is a more generalized version of the Pumping Lemma for Regular Languages. It's used to prove that certain languages are NOT context-free, by showing that sufficiently long strings in a CFL can be "pumped" in a specific way.

- **Theorem:** If $L$ is a Context-Free Language, then there exists some integer $p \geq 1$ (the pumping length) such that for any string $s \in L$ with $|s| \geq p$, $s$ can be divided into five parts $s = uvwxy$ satisfying the conditions:
  1. $|vx| > 0$ (at least one of $v$ or $x$ must be non-empty).
  2. $|vwx| \leq p$ (the pumpable parts must occur relatively close to each other).
  3. For all $i \geq 0$, the string $uv^iwx^iy \in L$.
- **Key Properties:**
  - It is a necessary condition for context-freeness, not a sufficient one. It can only be used to prove non-context-freeness.
  - The proof strategy is by contradiction (an "adversary game"), similar to the regular pumping lemma.
- **Common Pitfalls:**
  - Trying to prove context-freeness using the Pumping Lemma.
  - Incorrectly choosing the string $s$ or the partition $uvwxy$.
  - Forgetting the conditions $|vx| > 0$ or $|vwx| \leq p$.
- **Problem-Solving Techniques:**
  1. Assume $L$ is CFL and let $p$ be the pumping length.
  2. Choose a "hard" string $s \in L$ such that $|s| \geq p$, typically one that involves three dependent counts (e.g., $a^pb^pc^p$).
  3. Consider all possible ways to divide $s$ into $uvwxy$ satisfying $|vx| > 0$ and $|vwx| \leq p$.
  4. For each division, find an $i \geq 0$ such that $uv^iwx^iy \notin L$. This contradicts the lemma, proving $L$ is not CFL.

## Turing Machine (TM)

The Turing Machine is the most powerful and widely accepted model of computation, capable of simulating any algorithm. It consists of a finite control unit, a read/write head, and an infinite tape. It forms the basis of the Church-Turing Thesis, which states that any effectively computable function can be computed by a Turing Machine.

- **Definition:** A Turing Machine is formally defined as a 7-tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$, where:
  1. $Q$ is a finite set of states.
  2. $\Sigma$ is the input alphabet (not containing $B$).
  3. $\Gamma$ is the tape alphabet ($\Sigma \subseteq \Gamma$, and $B \in \Gamma$).
  4. $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$ is the transition function (L for left, R for right).
  5. $q_0 \in Q$ is the initial state.
  6. $B \in \Gamma$ is the blank symbol.
  7. $F \subseteq Q$ is the set of final (halting) states.
- **Key Properties:**
  - Turing Machines can recognize Recursively Enumerable Languages.
  - Variations of TMs (multi-tape, multi-head, non-deterministic) have equivalent computational power to the basic TM.
  - The Halting Problem for Turing Machines is undecidable.
- **Common Pitfalls:** Designing complex TMs; understanding the implications of the halting problem; confusing decidability with recognizability.
- **Problem-Solving Techniques:**
  - Designing TMs for various languages (e.g., $\{a^nb^nc^n\}$, palindrome recognition).
  - Understanding the Church-Turing Thesis and its implications for computability.

## Recursive and Recursively Enumerable Languages

These two classes of languages represent the limits of what Turing Machines can compute. They are central to the study of computability and decidability.

- **Definition:**
  - **Recursively Enumerable Language (RE / Recognizable):** A language $L$ is RE if there exists a Turing Machine $M$ that halts and accepts every string $w \in L$, but may either halt and reject or loop indefinitely for strings $w \notin L$.

- **Recursive Language (Decidable):** A language $L$ is Recursive if there exists a Turing Machine $M$ that halts on all inputs. For $w \in L$, $M$ accepts; for $w \notin L$, $M$ rejects.
- **Key Properties:**
  - Every Recursive Language is also Recursively Enumerable.
  - A language $L$ is Recursive if and only if both $L$ and its complement $\bar{L}$ are Recursively Enumerable.
  - RE languages are closed under union, intersection, concatenation, Kleene star, homomorphism, inverse homomorphism. They are NOT closed under complement.
  - Recursive languages are closed under union, intersection, complement, concatenation, Kleene star, homomorphism, inverse homomorphism.
- **Common Pitfalls:** Confusing the definitions of "halts on all inputs" (decidable) vs. "halts and accepts" (recognizable); misinterpreting closure properties.
- **Problem-Solving Techniques:** Using closure properties to classify languages; applying reduction to prove undecidability/non-recognizability.

### Decidability

Decidability refers to whether an algorithm (a Turing Machine that always halts) exists to solve a given problem. A problem is decidable if there's an algorithm that can always give a "yes" or "no" answer in finite time for any instance of the problem.

- **Definition:** A problem is decidable if the language representing all "yes" instances of the problem is a Recursive Language.
- **Key Properties (Examples of Decidable/Undecidable Problems):**
  - **Decidable:**
    - Membership problem for Regular Languages (Is $w \in L(FA)$?).
    - Emptiness problem for Regular Languages (Is $L(FA) = \emptyset$?).
    - Equivalence problem for Regular Languages (Is $L(FA_1) = L(FA_2)$?).
    - Membership problem for CFLs (Is $w \in L(CFG)$?).
    - Emptiness problem for CFLs (Is $L(CFG) = \emptyset$?).
  - **Undecidable:**
    - Halting Problem for TMs (Does TM $M$ halt on input $w$?).
    - Equivalence problem for CFLs (Is $L(CFG_1) = L(CFG_2)$?).
    - Ambiguity problem for CFGs (Is CFG $G$ ambiguous?).
    - Membership problem for RE languages (Is $w \in L(TM)$? - this is RE, but not Recursive).
    - Emptiness problem for TMs (Is $L(TM) = \emptyset$?).
    - Post Correspondence Problem (PCP).
- **Common Pitfalls:** Assuming all problems are decidable; misremembering the decidability status of common problems.
- **Problem-Solving Techniques:**
  - **Reduction:** The primary method to prove a problem is undecidable.
  - Understanding Rice's Theorem (all non-trivial properties of RE languages are undecidable).

### Reduction

Reduction is a powerful technique used to prove the undecidability of problems. If problem A can be "reduced" to problem B, it means that an algorithm for solving B could be used to solve A. Therefore, if A is known to be undecidable, B must also be undecidable.

- **Definition:** A problem $A$ is reducible to problem $B$ (denoted $A \leq_m B$ for many-one reduction) if there exists a computable function $f$ that transforms every instance $x$ of problem $A$ into an instance $f(x)$ of problem $B$ such that $x$ is a "yes" instance of $A$ if and only if $f(x)$ is a "yes" instance of $B$.
- **Key Properties:**
  - If $A \leq_m B$ and $A$ is undecidable, then $B$ is undecidable.
  - If $A \leq_m B$ and $B$ is decidable, then $A$ is decidable.
  - If $A \leq_m B$ and $A$ is non-RE, then $B$ is non-RE.
  - If $A \leq_m B$ and $B$ is RE, then $A$ is RE.
- **Common Pitfalls:** Reducing in the wrong direction (e.g., trying to reduce an undecidable problem to a decidable one); constructing an incorrect transformation function $f$.
- **Problem-Solving Techniques:**
  - Identify a known undecidable problem (e.g., Halting Problem, Emptiness Problem for TMs) to reduce FROM.

- Construct a transformation function $f$ that takes an instance of the known undecidable problem and converts it into an instance of the target problem.
- Prove that this transformation preserves the "yes/no" answer.

### Countable Uncountable Set

These concepts from set theory are crucial for understanding the limits of computation, particularly why there exist languages that are not recursively enumerable. It relates to the cardinality of sets.

- **Definition:**
  - **Countable Set:** A set is countable if it is finite or if its elements can be put into a one-to-one correspondence with the set of natural numbers $\mathbb{N}$ (i.e., it can be enumerated). Examples: integers $\mathbb{Z}$, rational numbers $\mathbb{Q}$, set of all finite strings $\Sigma^*$.
  - **Uncountable Set:** A set is uncountable if it is infinite but cannot be put into a one-to-one correspondence with $\mathbb{N}$. Examples: real numbers $\mathbb{R}$, power set of $\mathbb{N}$, set of all languages over $\Sigma$.
- **Key Properties:**
  - The set of all Turing Machines is countable.
  - The set of all possible languages over any alphabet $\Sigma$ is uncountable.
  - Since there are more languages than Turing Machines, there must exist languages that cannot be recognized by any Turing Machine (i.e., non-RE languages).
- **Common Pitfalls:** Confusing finite sets with infinite countable sets; misapplying diagonalization.
- **Problem-Solving Techniques:**
  - **Diagonalization Argument:** A standard technique to prove a set is uncountable (e.g., Cantor's diagonalization for real numbers).
  - Understanding the implications for computability theory.

### Identify Class Language

This topic refers to the ability to correctly classify a given language within the Chomsky Hierarchy, which organizes formal languages by their generative power and the type of automaton that recognizes them.

- **Chomsky Hierarchy (from least to most powerful):**
  1. **Type 3: Regular Languages (RL):** Recognized by Finite Automata (FA). Generated by Regular Grammars (RG).
  2. **Type 2: Context-Free Languages (CFL):** Recognized by Pushdown Automata (PDA). Generated by Context-Free Grammars (CFG). (Deterministic CFLs (DCFLs) are a proper subset of CFLs, recognized by DPDAs).
  3. **Type 1: Context-Sensitive Languages (CSL):** Recognized by Linear Bounded Automata (LBA). Generated by Context-Sensitive Grammars (CSG). (Less common in GATE).
  4. **Type 0: Recursively Enumerable Languages (RE):** Recognized by Turing Machines (TM). Generated by Unrestricted Grammars. (Recursive Languages are a proper subset of RE languages, recognized by TMs that always halt).
- **Relationships:** $\text{RL} \subset \text{DCFL} \subset \text{CFL} \subset \text{CSL} \subset \text{Recursive} \subset \text{RE}$.
- **Common Pitfalls:** Incorrectly applying pumping lemmas; misjudging the memory requirements of a language; confusing closure properties.
- **Problem-Solving Techniques:**
  - **Pumping Lemma:** First attempt to prove non-regularity or non-context-freeness.
  - **Closure Properties:** Use them to combine or transform languages and deduce their class.
  - **Machine Models:** Try to design the simplest machine (FA, PDA, TM) that can recognize the language.
  - **Grammar Rules:** Analyze the structure of production rules if a grammar is given.

### Medium (Computational Models)

In the context of Theory of Computation, "Medium" refers to the abstract computational models or machines used to define and study computation. These models, such as Finite Automata, Pushdown Automata, and Turing Machines, represent different levels of computational power and memory capabilities.

- **Definition:** Computational models are abstract machines that define the capabilities and limitations of algorithms. Each model specifies a set of operations and memory structures it can use.
- **Key Models and Their Power:**
  - **Finite Automata (FA):** No auxiliary memory. Recognizes Regular Languages. Simplest model.
  - **Pushdown Automata (PDA):** Finite control + a stack (LIFO memory). Recognizes Context-Free Languages. More powerful than FA.

- **Turing Machine (TM):** Finite control + an infinite tape (random access memory). Recognizes Recursively Enumerable Languages. Most powerful model, equivalent to any real-world computer.
- **Key Properties:**
  - The hierarchy of computational power (FA < PDA < TM) directly corresponds to the Chomsky Hierarchy of languages.
  - Each model defines the class of languages it can recognize (its "computational power").
- **Common Pitfalls:** Confusing the memory capabilities of different models; overestimating or underestimating the power of a specific model.
- **Problem-Solving Techniques:**
  - Analyze the memory requirements of a language to determine the minimum computational model needed to recognize it.
  - Understand how each model's memory structure (or lack thereof) limits its ability to recognize certain patterns (e.g., FA cannot count arbitrary numbers of symbols).

## Quick Formula Reference

- **DFA:** $M = (Q, \Sigma, \delta, q_0, F)$
- **NFA:** $M = (Q, \Sigma, \delta, q_0, F)$, where $\delta : Q \times (\Sigma \cup \{\epsilon\}) \to \mathcal{P}(Q)$
- **Regular Expression Operations:** Union $(R_1 + R_2)$, Concatenation $(R_1 R_2)$, Kleene Star $(R^*)$
- **Arden's Theorem:** If $R = Q + RP$, then $R = QP^*$.
- **Regular Grammar (Right-linear):** $A \to aB$ or $A \to a$ or $A \to \epsilon$
- **Pumping Lemma for Regular Languages:** For $s = xyz \in L$ with $|s| \geq p$: $|y| > 0$, $|xy| \leq p$, $\forall i \geq 0, xy^i z \in L$.
- **Context-Free Grammar (CFG):** $G = (V, \Sigma, P, S)$, with productions $A \to \alpha$.
- **Chomsky Normal Form (CNF):** $A \to BC$ or $A \to a$.
- **Greibach Normal Form (GNF):** $A \to a\alpha$.
- **Pushdown Automata (PDA):** $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$, with transitions $\delta(q, a, X) = \{(q', Y_1 Y_2 \dots Y_k)\}$.
- **Pumping Lemma for Context-Free Languages:** For $s = uvwxy \in L$ with $|s| \geq p$: $|vx| > 0$, $|vwx| \leq p$, $\forall i \geq 0, uv^i wx^i y \in L$.
- **Turing Machine (TM):** $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$, with transitions $\delta(q, X) = (q', Y, D)$.
- **Chomsky Hierarchy:** $\text{Regular} \subset \text{DCFL} \subset \text{CFL} \subset \text{CSL} \subset \text{Recursive} \subset \text{RE}$.
- **Closure Properties (Summary):**

| Operation | RL | CFL | Recursive | RE |
|---|---|---|---|---|
| Union | Y | Y | Y | Y |
| Intersection | Y | N | Y | Y |
| Complement | Y | N | Y | N |
| Concatenation | Y | Y | Y | Y |
| Kleene Star | Y | Y | Y | Y |
| Reversal | Y | Y | Y | Y |

## Important Tips for GATE

1. **Master the Chomsky Hierarchy:** Understand the relationships between language classes (Regular, CFL, Recursive, RE) and their corresponding machine models (FA, PDA, TM). This is fundamental for classifying languages and solving many conceptual questions.
2. **Memorize Closure Properties:** Create a table and commit to memory which language classes are closed under which operations (union, intersection, complement, concatenation, Kleene star, reversal). This is a frequent source of direct questions.
3. **Practice Pumping Lemma Proofs:** The Pumping Lemma for both Regular and Context-Free Languages is a crucial tool. Practice choosing the "hard" string and demonstrating the contradiction for various non-regular/non-CFLs. Remember, it only proves non-membership.
4. **Understand NFA to DFA Conversion and Minimization:** Be proficient in subset construction for NFA to DFA conversion and the table-filling algorithm for DFA minimization. These are common problem-solving tasks.
5. **Distinguish Decidability vs. Recognizability:** Clearly understand the difference between Recursive (decidable) and Recursively Enumerable (recognizable) languages, especially regarding the halting behavior of Turing Machines. Know the decidability status of common problems (Halting Problem, Emptiness for TMs, Equivalence for CFLs, etc.).
6. **Focus on Design Principles:** While complex designs are rare, understand the core principles of designing simple FAs, PDAs, and TMs for basic languages. Pay attention to how memory (or lack thereof) is utilized.
7. **Pay Attention to Formal Definitions:** GATE questions often test your understanding of the precise formal

definitions of automata, grammars, and languages. Small details like epsilon transitions, initial stack symbols, or blank tape symbols can be critical.

8. **Time Management for Tricky Questions:** Some ToC problems, especially those involving complex reductions or intricate Pumping Lemma applications, can be time-consuming. Practice identifying these quickly and decide whether to attempt them or mark for review.

## 6.1 Closure Property (10)

✍ **Practice Test:** Test 1 (12Q)

### 6.1.1 Closure Property: GATE CSE 1989 | Question: 3-ii

Context-free languages and regular languages are both closed under the operation (s) of :

A. Union B. Intersection C. Concatenation D. Complementation

gate1989 easy theory-of-computation closure-property multiple-selects

Answer key☟

### 6.1.2 Closure Property: GATE CSE 1992 | Question: 16

Which of the following three statements are true? Prove your answer.

i. The union of two recursive languages is recursive.
ii. The language $\{O^n \mid n \text{ is a prime}\}$ is not regular.
iii. Regular languages are closed under infinite union.

gate1992 theory-of-computation normal closure-property proof descriptive

Answer key☟

### 6.1.3 Closure Property: GATE CSE 2002 | Question: 2.14

Which of the following is true?

A. The complement of a recursive language is recursive
B. The complement of a recursively enumerable language is recursively enumerable
C. The complement of a recursive language is either recursive or recursively enumerable
D. The complement of a context-free language is context-free

gatecse-2002 theory-of-computation easy closure-property

Answer key☟

### 6.1.4 Closure Property: GATE CSE 2013 | Question: 17

Which of the following statements is/are **FALSE**?

1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine.
2. Turing recognizable languages are closed under union and complementation.
3. Turing decidable languages are closed under intersection and complementation.
4. Turing recognizable languages are closed under union and intersection.

A. $1$ and $4$ only
B. $1$ and $3$ only
C. $2$ only
D. $3$ only

gatecse-2013 theory-of-computation normal closure-property

Answer key☟

### 6.1.5 Closure Property: GATE CSE 2016 | Set 2 | Question: 18

Consider the following types of languages: $L_1$: Regular, $L_2$: Context-free, $L_3$: Recursive, $L_4$: Recursively enumerable. Which of the following is/are **TRUE ?**

I. $\overline{L_3} \cup L_4$ is recursively enumerable.

II. $\overline{L_2} \cup L_3$ is recursive.
III. $L_1^* \cap L_2$ is context-free.
IV. $L_1 \cup \overline{L_2}$ is context-free.

A. I only.
B. I and III only.
C. I and IV only.
D. I, II and III only.

gatecse-2016-set2 theory-of-computation regular-language context-free-language closure-property normal

Answer key

### 6.1.6 Closure Property: GATE CSE 2017 | Set 2 | Question: 04

Let $L_1, L_2$ be any two context-free languages and $R$ be any regular language. Then which of the following is/are CORRECT?

I. $L_1 \cup L_2$ is context-free
II. $\overline{L_1}$ is context-free
III. $L_1 - R$ is context-free
IV. $L_1 \cap L_2$ is context-free

A. I, II and IV only
B. I and III only
C. II and IV only
D. I only

gatecse-2017-set2 theory-of-computation closure-property

Answer key

### 6.1.7 Closure Property: GATE CSE 2018 | Question: 7

The set of all recursively enumerable languages is:

A. closed under complementation
B. closed under intersection
C. a subset of the set of all recursive languages
D. an uncountable set

gatecse-2018 theory-of-computation closure-property easy one-mark

Answer key

### 6.1.8 Closure Property: GATE CSE 2025 | Set 2 | Question: 20

Consider the two lists List I and List II given below:

| **List I** | **List II** |
|---|---|
| Context-free languages | Closed under union |
| Recursive languages | Not closed under complementation |
| Regular languages | Closed under intersection |

For matching of items in List I with those in List II, which of the following option(s) is/are CORRECT?

A. (i) - (a), (ii) - (b), and (iii) - (c)
B. (i) - (b), (ii) - (a), and (iii) - (c)
C. (i) - (b), (ii) - (c), and (iii) - (a)
D. (i) - (a), (ii) - (c), and (iii) - (b)

gatecse2025-set2 theory-of-computation closure-property match-the-following multiple-selects easy one-mark

Answer key

### 6.1.9 Closure Property: GATE CSE 2026 | Set 1 | Question: 41

Let $L_1$ and $L_2$ be two languages over a finite alphabet, such that $L_1 \cap L_2$ and $L_2$ are regular languages.

Which of the following statements is/are always true?

A. $L_1$ is regular
B. $L_1 \cup L_2$ is regular
C. $\overline{L_2}$ is context-free
D. $L_1$ is context-free

gatecse-2026-set1 theory-of-computation closure-property two-marks multiple-selects

Answer key

### 6.1.10 Closure Property: GATE IT 2006 | Question: 32

Let $L$ be a context-free language and $M$ a regular language. Then the language $L \cap M$ is

A. always regular
B. never regular
C. always a deterministic context-free language
D. always a context-free language

gateit-2006 theory-of-computation closure-property easy

Answer key

## 6.2 Context Free Grammar (2)

✍ **Practice Tests:** Test 1 (11Q) Weekly Quiz 10 (10Q) Weekly Quiz 4 (10Q)

### 6.2.1 Context Free Grammar: GATE CSE 2025 | Set 1 | Question: 9

Consider the following context-free grammar $G$, where $S, A$, and $B$ are the variables (non-terminals), $a$ and $b$ are the terminal symbols, $S$ is the start variable, and the rules of $G$ are described as:

$$S \rightarrow aaB \mid Abb$$
$$A \rightarrow a \mid aA$$
$$B \rightarrow b \mid bB$$

Which ONE of the languages $L(G)$ is accepted by $G$ ?

A. $L(G) = \{a^2b^n \mid n \geq 1\} \cup \{a^nb^2 \mid n \geq 1\}$
B. $L(G) = \{a^nb^{2n} \mid n \geq 1\} \cup \{a^{2n}b^n \mid n \geq 1\}$
C. $L(G) = \{a^nb^n \mid n \geq 1\}$
D. $L(G) = \{a^{2n}b^{2n} \mid n \geq 1\}$

gatecse2025-set1 theory-of-computation context-free-grammar easy one-mark

Answer key

### 6.2.2 Context Free Grammar: GATE CSE 2026 | Set 1 | Question: 15

Consider the following grammar where $S$ is the start symbol, and $a$ and $b$ are terminal symbols.

$$S \rightarrow aSbS \mid bS \mid \epsilon$$

Which of the following statements is/are true?

A. The grammar is ambiguous
B. The string $abb$ has two distinct derivations in this grammar
C. The string $abab$ has only one rightmost derivation
D. The language generated by the grammar is undecidable

gatecse-2026-set1 theory-of-computation context-free-grammar multiple-selects one-mark

Answer key

## 6.3 Context Free Language (35)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (14Q)

### 6.3.1 Context Free Language: GATE CSE 1987 | Question: 2k

State whether the following statements are TRUE or FALSE:

The intersection of two CFL's is also a CFL.

gate1987 theory-of-computation context-free-language true-false

Answer key

## 6.3.2 Context Free Language: GATE CSE 1992 | Question: 02,xix

Context-free languages are:

A. closed under union
B. closed under complementation
C. closed under intersection
D. closed under Kleene closure

gate1992 context-free-language theory-of-computation normal multiple-selects

Answer key

## 6.3.3 Context Free Language: GATE CSE 1995 | Question: 2.20

Which of the following definitions below generate the same language as $L$, where $L = \{x^n y^n \text{ such that } n \geq 1\}$?

I. $E \to xEy \mid xy$
II. $xy \mid (x^+ xyy^+)$
III. $x^+ y^+$

A. I only
B. I and II
C. II and III
D. II only

gate1995 theory-of-computation easy context-free-language

Answer key

## 6.3.4 Context Free Language: GATE CSE 1996 | Question: 2.8

If $L_1$ and $L_2$ are context free languages and $R$ a regular set, one of the languages below is not necessarily a context free language. Which one?

A. $L_1 . L_2$
B. $L_1 \cap L_2$
C. $L_1 \cap R$
D. $L_1 \cup L_2$

gate1996 theory-of-computation context-free-language easy

Answer key

## 6.3.5 Context Free Language: GATE CSE 1996 | Question: 2.9

Define a context free languages $L \in \{0,1\}^*$, $\text{init}(L) = \{u \mid uv \in L$ for some $v$ in $\{0,1\}^*\}$ ( in other words, $\text{init}(L)$ is the set of prefixes of $L$)

Let $L = \{w \mid w \text{ is nonempty and has an equal number of 0's and 1's}\}$

Then $\text{init}(L)$ is:

A. the set of all binary strings with unequal number of $0$'s and $1$'s
B. the set of all binary strings including null string
C. the set of all binary strings with exactly one more $0$ than the number of $1$'s or one more $1$ than the number of $0$'s
D. None of the above

gate1996 theory-of-computation context-free-language normal

Answer key

## 6.3.6 Context Free Language: GATE CSE 1999 | Question: 1.5

Context-free languages are closed under:

A. Union, intersection
B. Union, Kleene closure
C. Intersection, complement
D. Complement, Kleene closure

gate1999 theory-of-computation context-free-language easy

Answer key

## 6.3.7 Context Free Language: GATE CSE 1999 | Question: 7

Show that the language

$$L = \{xcx \mid x \in \{0,1\}^* \text{ and } c \text{ is a terminal symbol}\}$$

is not context free. $c$ is not $0$ or $1$.

gate1999 theory-of-computation context-free-language normal proof

Answer key

## 6.3.8 Context Free Language: GATE CSE 2000 | Question: 7

A. Construct as minimal finite state machine that accepts the language, over $\{0,1\}$, of all strings that contain neither the substring $00$ nor the substring $11$.
B. Consider the grammar
   - $S \to aSAb$
   - $S \to \epsilon$
   - $A \to bA$
   - $A \to \epsilon$

   where $S$, $A$ are non-terminal symbols with $S$ being the start symbol; $a, b$ are terminal symbols and $\epsilon$ is the empty string. This grammar generates strings of the form $a^i b^j$ for some $i, j \geq 0$, where $i$ and $j$ satisfy some condition. What is the condition on the values of $i$ and $j$?

gatecse-2000 theory-of-computation descriptive regular-language context-free-language

Answer key

## 6.3.9 Context Free Language: GATE CSE 2001 | Question: 1.5

Which of the following statements is true?

A. If a language is context free it can always be accepted by a deterministic push-down automaton
B. The union of two context free languages is context free
C. The intersection of two context free languages is a context free
D. The complement of a context free language is a context free

gatecse-2001 theory-of-computation context-free-language easy

Answer key

## 6.3.10 Context Free Language: GATE CSE 2003 | Question: 51

Let $G = (\{S\}, \{a, b\}, R, S)$ be a context free grammar where the rule set R is $S \to aSb \mid SS \mid \epsilon$
Which of the following statements is true?

A. $G$ is not ambiguous
B. There exist $x, y \in L(G)$ such that $xy \notin L(G)$
C. There is a deterministic pushdown automaton that accepts $L(G)$
D. We can find a deterministic finite state automaton that accepts $L(G)$

gatecse-2003 theory-of-computation context-free-language normal

Answer key

## 6.3.11 Context Free Language: GATE CSE 2005 | Question: 57

Consider the languages:

- $L_1 = \left\{ww^R \mid w \in \{0,1\}^*\right\}$
- $L_2 = \left\{w\#w^R \mid w \in \{0,1\}^*\right\}$, where $\#$ is a special symbol
- $L_3 = \{ww \mid w \in \{0,1\}^*\}$

Which one of the following is TRUE?

A. $L_1$ is a deterministic CFL
B. $L_2$ is a deterministic CFL
C. $L_3$ is a CFL, but not a deterministic CFL
D. $L_3$ is a deterministic CFL

gatecse-2005 theory-of-computation context-free-language easy

Answer key☟

### 6.3.12 Context Free Language: GATE CSE 2006 | Question: 19

Let

$L_1 = \{0^{n+m}1^n0^m \mid n, m \geq 0\}$,

$L_2 = \{0^{n+m}1^{n+m}0^m \mid n, m \geq 0\}$ and

$L_3 = \{0^{n+m}1^{n+m}0^{n+m} \mid n, m \geq 0\}$.

Which of these languages are NOT context free?

A. $L_1$ only
B. $L_3$ only
C. $L_1$ and $L_2$
D. $L_2$ and $L_3$

gatecse-2006 theory-of-computation context-free-language normal

Answer key☟

### 6.3.13 Context Free Language: GATE CSE 2009 | Question: 12, ISRO2016-37

$$S \to aSa \mid bSb \mid a \mid b$$

The language generated by the above grammar over the alphabet $\{a, b\}$ is the set of:

A. all palindromes
B. all odd length palindromes
C. strings that begin and end with the same symbol
D. all even length palindromes

gatecse-2009 theory-of-computation context-free-language easy isro2016

Answer key☟

### 6.3.14 Context Free Language: GATE CSE 2015 | Set 3 | Question: 32

Which of the following languages are context-free?

$L_1 : \{a^m b^n a^n b^m \mid m, n \geq 1\}$

$L_2 : \{a^m b^n a^m b^n \mid m, n \geq 1\}$

$L_3 : \{a^m b^n \mid m = 2n + 1\}$

A. $L_1$ and $L_2$ only
B. $L_1$ and $L_3$ only
C. $L_2$ and $L_3$ only
D. $L_3$ only

gatecse-2015-set3 theory-of-computation context-free-language normal

Answer key☟

### 6.3.15 Context Free Language: GATE CSE 2016 | Set 1 | Question: 16

Which of the following languages is generated by the given grammar?

$$S \to aS \mid bS \mid \varepsilon$$

A. $\{a^n b^m \mid n, m \geq 0\}$
B. $\{w \in \{a, b\}^* \mid w \text{ has equal number of a's and b's}\}$
C. $\{a^n \mid n \geq 0\} \cup \{b^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$

D. $\{a, b\}^*$

gatecse-2016-set1 theory-of-computation context-free-language normal

Answer key

### 6.3.16 Context Free Language: GATE CSE 2016 | Set 1 | Question: 42

Consider the following context-free grammars;

$G_1 : S \to aS \mid B, B \to b \mid bB$

$G_2 : S \to aA \mid bB, A \to aA \mid B \mid \varepsilon, B \to bB \mid \varepsilon$

Which one of the following pairs of languages is generated by $G_1$ and $G_2$ ,respectively?

A. $\{a^m b^n \mid m > 0 \text{ or } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ and } n > 0\}$
B. $\{a^m b^n \mid m > 0 \text{ and } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ or } n \geq 0\}$
C. $\{a^m b^n \mid m \geq 0 \text{ or } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ and } n > 0\}$
D. $\{a^m b^n \mid m \geq 0 \text{ and } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ or } n > 0\}$

gatecse-2016-set1 theory-of-computation context-free-language normal

Answer key

### 6.3.17 Context Free Language: GATE CSE 2016 | Set 2 | Question: 43

Consider the following languages:

- $L_1 = \{a^n b^m c^{n+m} : m, n \geq 1\}$
- $L_2 = \{a^n b^n c^{2n} : n \geq 1\}$

Which one of the following is TRUE?

A. Both $L_1$ and $L_2$ are context-free.
B. $L_1$ is context-free while $L_2$ is not context-free.
C. $L_2$ is context-free while $L_1$ is not context-free.
D. Neither $L_1$ nor $L_2$ is context-free.

gatecse-2016-set2 theory-of-computation context-free-language normal

Answer key

### 6.3.18 Context Free Language: GATE CSE 2017 | Set 1 | Question: 10

Consider the following context-free grammar over the alphabet $\Sigma = \{a, b, c\}$ with $S$ as the start symbol:

$$S \to abScT \mid abcT$$

$$T \to bT \mid b$$

Which one of the following represents the language generated by the above grammar?

A. $\{(ab)^n (cb)^n \mid n \geq 1\}$
B. $\{(ab)^n cb^{m_1} cb^{m_2} \ldots cb^{m_n} \mid n, m_1, m_2, \ldots, m_n \geq 1\}$
C. $\{(ab)^n (cb^m)^n \mid m, n \geq 1\}$
D. $\{(ab)^n (cb^n)^m \mid m, n \geq 1\}$

gatecse-2017-set1 theory-of-computation context-free-language normal

Answer key

### 6.3.19 Context Free Language: GATE CSE 2017 | Set 1 | Question: 34

If $G$ is a grammar with productions

$S \rightarrow SaS \mid aSb \mid bSa \mid SS \mid \epsilon$

where $S$ is the start variable, then which one of the following strings is not generated by $G$?

A. $abab$ B. $aaab$ C. $abbaa$ D. $babba$

gatecse-2017-set1 theory-of-computation context-free-language normal

Answer key☟

## 6.3.20 Context Free Language: GATE CSE 2017 | Set 1 | Question: 38

Consider the following languages over the alphabet $\Sigma = \{a, b, c\}$. Let $L_1 = \{a^n b^n c^m \mid m, n \geq 0\}$ and $L_2 = \{a^m b^n c^n \mid m, n \geq 0\}$.

Which of the following are context-free languages?

I. $L_1 \cup L_2$
II. $L_1 \cap L_2$

A. I only B. II only C. I and II D. Neither I nor II

gatecse-2017-set1 theory-of-computation context-free-language normal

Answer key☟

## 6.3.21 Context Free Language: GATE CSE 2017 | Set 2 | Question: 16

Identify the language generated by the following grammar, where $S$ is the start variable.

- $S \rightarrow XY$
- $X \rightarrow aX \mid a$
- $Y \rightarrow aYb \mid \epsilon$

A. $\{a^m b^n \mid m \geq n, n > 0\}$
B. $\{a^m b^n \mid m \geq n, n \geq 0\}$
C. $\{a^m b^n \mid m > n, n \geq 0\}$
D. $\{a^m b^n \mid m > n, n > 0\}$

gatecse-2017-set2 theory-of-computation context-free-language

Answer key☟

## 6.3.22 Context Free Language: GATE CSE 2019 | Question: 31

Which one of the following languages over $\Sigma = \{a, b\}$ is NOT context-free?

A. $\{ww^R \mid w \in \{a, b\}^*\}$
B. $\{wa^n b^n w^R \mid w \in \{a, b\}^*, n \geq 0\}$
C. $\{wa^n w^R b^n \mid w \in \{a, b\}^*, n \geq 0\}$
D. $\{a^n b^i \mid i \in \{n, 3n, 5n\}, n \geq 0\}$

gatecse-2019 theory-of-computation context-free-language two-marks

Answer key☟

## 6.3.23 Context Free Language: GATE CSE 2021 | Set 1 | Question: 1

Suppose that $L_1$ is a regular language and $L_2$ is a context-free language. Which one of the following languages is $\text{NOT}$ necessarily context-free?

A. $L_1 \cap L_2$ B. $L_1 \cdot L_2$ C. $L_1 - L_2$ D. $L_1 \cup L_2$

gatecse-2021-set1 context-free-language theory-of-computation one-mark

Answer key☟

## 6.3.24 Context Free Language: GATE CSE 2021 | Set 2 | Question: 41

For a string $w$, we define $w^R$ to be the reverse of $w$. For example, if $w = 01101$ then $w^R = 10110$.

Which of the following languages is/are context-free?

A. $\{wxw^Rx^R \mid w, x \in \{0,1\}^*\}$
B. $\{ww^Rxx^R \mid w, x \in \{0,1\}^*\}$
C. $\{wxw^R \mid w, x \in \{0,1\}^*\}$
D. $\{wxx^Rw^R \mid w, x \in \{0,1\}^*\}$

gatecse-2021-set2 multiple-selects theory-of-computation context-free-language two-marks

Answer key

### 6.3.25 Context Free Language: GATE CSE 2022 | Question: 38

Consider the following languages:

- $L_1 = \{ww | w \in \{a,b\}^*\}$
- $L_2 = \{a^nb^nc^m | m, n \geq 0\}$
- $L_3 = \{a^mb^nc^n | m, n \geq 0\}$

Which of the following statements is/are $\text{FALSE}$?

A. $L_1$ is not context-free but $L_2$ and $L_3$ are deterministic context-free.
B. Neither $L_1$ nor $L_2$ is context-free.
C. $L_2, L_3$ and $L_2 \cap L_3$ all are context-free.
D. Neither $L_1$ nor its complement is context-free.

gatecse-2022 theory-of-computation context-free-language multiple-selects two-marks

Answer key

### 6.3.26 Context Free Language: GATE CSE 2023 | Question: 29

Consider the context-free grammar $G$ below

$$S \rightarrow aSb \mid X$$
$$X \rightarrow aX \mid Xb \mid a \mid b,$$

where $S$ and $X$ are non-terminals, and $a$ and $b$ are terminal symbols. The starting non-terminal is $S$.
Which one of the following statements is $\text{CORRECT}$?

A. The language generated by $G$ is $(a+b)^*$
B. The language generated by $G$ is $a^*(a+b)b^*$
C. The language generated by $G$ is $a^*b^*(a+b)$
D. The language generated by $G$ is not a regular language

gatecse-2023 theory-of-computation context-free-language two-marks

Answer key

### 6.3.27 Context Free Language: GATE CSE 2025 | Set 1 | Question: 35

Consider the following two languages over the alphabet $\{a, b, c\}$, where $m$ and $n$ are natural numbers.

$$L_1 = \{a^mb^mc^{m+n} \mid m, n \geq 1\}$$
$$L_2 = \{a^mb^nc^{m+n} \mid m, n \geq 1\}$$

Which ONE of the following statements is CORRECT?

A. Both $L_1$ and $L_2$ are context-free languages.
B. $L_1$ is a context-free language but $L_2$ is not a context-free language.
C. $L_1$ is not a context-free language but $L_2$ is a context-free language.
D. Neither $L_1$ nor $L_2$ are context-free languages.

gatecse2025-set1 theory-of-computation context-free-language two-marks

Answer key

### 6.3.28 Context Free Language: GATE CSE 2026 | Set 1 | Question: 42

Consider the following context-free grammar $G$.

$$S \to abaABAbba$$
$$A \to aaBBAb \mid bBabaa$$
$$B \to aBb \mid ab$$

In the above grammar, $S$ is the start symbol, $a$ and $b$ are terminal symbols, and $A$ and $B$ are non-terminal symbols.

Let $L(G)$ be the language generated by the grammar $G$. For a string $s \in L(G)$, let $n_1(s)$ be the number of $a$ 's in $s$ and $n_2(s)$ be the number of $b$ 's in $s$.

Which of the following statements is/are true?

A. There is a string $s \in L(G)$ such that $n_1(s) < n_2(s)$
B. For every string $s \in L(G), n_1(s) \geq n_2(s)$
C. There is a string $s \in L(G)$ such that $n_1(s) > 2n_2(s)$
D. For every string $s \in L(G), n_1(s) \leq 2n_2(s)$

gatecse-2026-set1 theory-of-computation context-free-language two-marks multiple-selects

Answer key

### 6.3.29 Context Free Language: GATE CSE 2026 | Set 2 | Question: 38

Let $\Sigma = \{a, b, c, d\}$ and let $L = \left\{a^i b^j c^k d^\ell \mid i, j, k, \ell \geq 0\right\}$.
Which of the following constraints ensure(s) that the language $L$ is context-free?

A. $i + k = j + \ell$ B. $i = k$ and $j = \ell$ C. $i = \ell$ and $j = k$ D. $i + j = k + \ell$

gatecse-2026-set2 theory-of-computation context-free-language multiple-selects two-marks

Answer key

### 6.3.30 Context Free Language: GATE IT 2006 | Question: 34

In the context-free grammar below, $S$ is the start symbol, $a$ and $b$ are terminals, and $\epsilon$ denotes the empty string.

- $S \to aSAb \mid \epsilon$
- $A \to bA \mid \epsilon$

The grammar generates the language

A. $((a + b)^* b)$
B. $\{a^m b^n \mid m \leq n\}$
C. $\{a^m b^n \mid m = n)$
D. $a^* b^*$

gateit-2006 theory-of-computation context-free-language normal

Answer key

### 6.3.31 Context Free Language: GATE IT 2006 | Question: 4

In the context-free grammar below, $S$ is the start symbol, $a$ and $b$ are terminals, and $\epsilon$ denotes the empty string.

$S \to aSa \mid bSb \mid a \mid b \mid \epsilon$

Which of the following strings is NOT generated by the grammar?

A. $aaaa$ B. $baba$ C. $abba$ D. $babaaabab$

gateit-2006 theory-of-computation context-free-language easy

Answer key

### 6.3.32 Context Free Language: GATE IT 2007 | Question: 46

The two grammars given below generate a language over the alphabet $\{x, y, z\}$

- $G1 : S \rightarrow x \mid z \mid x\,S \mid z\,S \mid y\,B$
  $\quad\;\; B \rightarrow y \mid z \mid y\,B \mid z\,B$
- $G2 : S \rightarrow y \mid z \mid y\,S \mid z\,S \mid x\,B$
  $\quad\;\; B \rightarrow y \mid y\,S$

Which one of the following choices describes the properties satisfied by the strings in these languages?

A. $G1$ : No $y$ appears before any $x$
   $G2$ : Every $x$ is followed by at least one $y$
B. $G1$ : No $y$ appears before any $x$
   $G2$ : No $x$ appears before any $y$
C. $G1$ : No $y$ appears after any $x$
   $G2$ : Every $x$ is followed by at least one $y$
D. $G1$ : No $y$ appears after any $x$
   $G2$ : Every $y$ is followed by at least one $x$

gateit-2007 theory-of-computation normal context-free-language

Answer key

### 6.3.33 Context Free Language: GATE IT 2007 | Question: 48

Consider the grammar given below:

$S \rightarrow x\,B \mid y\,A$
$A \rightarrow x \mid x\,S \mid y\,A\,A$
$B \rightarrow y \mid y\,S \mid x\,B\,B$

Consider the following strings.

i. $xxyyx$
ii. $xxyyxy$
iii. $xyxy$
iv. $yxxy$
v. $yxx$
vi. $xyx$

Which of the above strings are generated by the grammar ?

A. i, ii and iii B. ii, v and vi
C. ii, iii and iv D. i, iii and iv

gateit-2007 theory-of-computation context-free-language normal

Answer key

### 6.3.34 Context Free Language: GATE IT 2007 | Question: 49

Consider the following grammars. Names representing terminals have been specified in capital letters.

```
\begin{array}{|llll|}\hline G1 :  &  \text{stmnt} & \rightarrow & \text{WHILE (expr) stmnt} \\%\hline  \text{}  &  \text{stmnt} & \rightarrow & \t
```

Which one of the following statements is true?

A. $G_1$ is context-free but not regular and $G_2$ is regular
B. $G_2$ is context-free but not regular and $G_1$ is regular

C. Both $G_1$ and $G_2$ are regular
D. Both $G_1$ and $G_2$ are context-free but neither of them is regular

gateit-2007 theory-of-computation context-free-language normal

Answer key

### 6.3.35 Context Free Language: GATE IT 2008 | Question: 34

Consider a CFG with the following productions.

$S \rightarrow AA \mid B$
$A \rightarrow 0A \mid A0 \mid 1$
$B \rightarrow 0B00 \mid 1$

$S$ is the start symbol, $A$ and $B$ are non-terminals and 0 and 1 are the terminals. The language generated by this grammar is:

A. $\{0^n 10^{2n} \mid n \geq 1\}$
B. $\{0^i 10^j 10^k \mid i, j, k \geq 0\} \cup \{0^n 10^{2n} \mid n \geq 0\}$
C. $\{0^i 10^j \mid i, j \geq 0\} \cup \{0^n 10^{2n} \mid n \geq 0\}$
D. The set of all strings over $\{0, 1\}$ containing at least two $0$'s

gateit-2008 theory-of-computation context-free-language normal

Answer key

## 6.4 Countable Uncountable Set (3)

✍ **Practice Test:** Test 1 (9Q)

### 6.4.1 Countable Uncountable Set: GATE CSE 1997 | Question: 3.4

Given $\Sigma = \{a, b\}$, which one of the following sets is not countable?

A. Set of all strings over $\Sigma$
B. Set of all languages over $\Sigma$
C. Set of all regular languages over $\Sigma$
D. Set of all languages over $\Sigma$ accepted by Turing machines

gate1997 theory-of-computation normal countable-uncountable-set

Answer key

### 6.4.2 Countable Uncountable Set: GATE CSE 2014 | Set 3 | Question: 16

Let $\Sigma$ be a finite non-empty alphabet and let $2^{\Sigma^*}$ be the power set of $\Sigma^*$. Which one of the following is **TRUE**?

A. Both $2^{\Sigma^*}$ and $\Sigma^*$ are countable
B. $2^{\Sigma^*}$ is countable and $\Sigma^*$ is uncountable
C. $2^{\Sigma^*}$ is uncountable and $\Sigma^*$ is countable
D. Both $2^{\Sigma^*}$ and $\Sigma^*$ are uncountable

gatecse-2014-set3 theory-of-computation normal countable-uncountable-set

Answer key

### 6.4.3 Countable Uncountable Set: GATE CSE 2019 | Question: 34

Consider the following sets:

S1: Set of all recursively enumerable languages over the alphabet $\{0, 1\}$

S2: Set of all syntactically valid C programs

S3: Set of all languages over the alphabet $\{0, 1\}$

S4: Set of all non-regular languages over the alphabet $\{0, 1\}$

Which of the above sets are uncountable?

A. S1 and S2  B. S3 and S4  C. S2 and S3  D. S1 and S4

gatecse-2019 theory-of-computation countable-uncountable-set two-marks

Answer key

## 6.5 Decidability (30)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (8Q) Weekly Quiz 6 (10Q)

### 6.5.1 Decidability: GATE CSE 1987 | Question: 2l

State whether the following statement are TRUE or FALSE.

$A$ is recursive if both $A$ and its complement are accepted by Turing machines.

gate1987 theory-of-computation turing-machine decidability true-false

Answer key

### 6.5.2 Decidability: GATE CSE 1987 | Question: 2m

State whether the following statements are TRUE or FALSE:

The problem as to whether a Turing machine $M$ accepts input $w$ is undecidable.

gate1987 theory-of-computation turing-machine decidability true-false

Answer key

### 6.5.3 Decidability: GATE CSE 1988 | Question: 2viii

State the halting problem of the Turing machine.

gate1988 theory-of-computation descriptive decidability turing-machine

Answer key

### 6.5.4 Decidability: GATE CSE 1989 | Question: 3-iii

Which of the following problems are undecidable?

A. Membership problem in context-free languages.
B. Whether a given context-free language is regular.
C. Whether a finite state automation halts on all inputs.
D. Membership problem for type $0$ languages.

gate1989 normal theory-of-computation decidability multiple-selects

Answer key

### 6.5.5 Decidability: GATE CSE 1990 | Question: 3-vii

It is undecidable whether:

A. An arbitrary Turing machine halts after $100$ steps.
B. A Turing machine prints a specific letter.
C. A Turing machine computes the products of two numbers
D. None of the above.

gate1990 normal theory-of-computation decidability multiple-selects

Answer key

### 6.5.6 Decidability: GATE CSE 1995 | Question: 11

Let $L$ be a language over $\Sigma$ i.e., $L \subseteq \Sigma^*$. Suppose $L$ satisfies the two conditions given below.

i. $L$ is in NP and
ii. For every $n$, there is exactly one string of length $n$ that belongs to $L$.

Let $L^c$ be the complement of $L$ over $\Sigma^*$. Show that $L^c$ is also in NP.

gate1995 theory-of-computation normal decidability proof descriptive

Answer key

### 6.5.7 Decidability: GATE CSE 1996 | Question: 1.9

Which of the following statements is false?

A. The Halting Problem of Turing machines is undecidable
B. Determining whether a context-free grammar is ambiguous is undecidable
C. Given two arbitrary context-free grammars $G_1$ and $G_2$ it is undecidable whether $L(G_1) = L(G_2)$
D. Given two regular grammars $G_1$ and $G_2$ it is undecidable whether $L(G_1) = L(G_2)$

gate1996 theory-of-computation decidability easy

Answer key

### 6.5.8 Decidability: GATE CSE 1997 | Question: 6.5

Which one of the following is not decidable?

A. Given a Turing machine $M$, a string $s$ and an integer $k$, $M$ accepts $s$ within $k$ steps
B. Equivalence of two given Turing machines
C. Language accepted by a given finite state machine is not empty
D. Language generated by a context free grammar is non-empty

gate1997 theory-of-computation decidability easy

Answer key

### 6.5.9 Decidability: GATE CSE 1999 | Question: 10

Suppose we have a function HALTS which when applied to any arbitrary function $f$ and its arguments will say TRUE if function $f$ terminates for those arguments and FALSE otherwise. Example: Given the following function definition.

FACTORIAL (N) = IF (N=0) THEN 1 ELSE N*FACTORIAL (N-1)

Then HALTS (FACTORIAL, 4) = TRUE and HALTS (FACTORIAL, -5) = FALSE

Let us define the function FUNNY (f) = IF HALTS (f) THEN not (f) ELSE TRUE

a. Show that FUNNY terminates for all functions $f$.
b. use (a) to prove (by contradiction) that it is not possible to have a function like HALTS which for arbitrary functions and inputs says whether it will terminate on that input or not.

gate1999 theory-of-computation descriptive decidability

Answer key

### 6.5.10 Decidability: GATE CSE 2000 | Question: 2.9

Consider the following decision problems:

$(P1)$ : Does a given finite state machine accept a given string?

$(P2)$ : Does a given context free grammar generate an infinite number of strings?

Which of the following statements is true?

A. Both $(P1)$ and $(P2)$ are decidable
B. Neither $(P1)$ nor $(P2)$ is decidable
C. Only $(P1)$ is decidable
D. Only $(P2)$ is decidable

gatecse-2000 theory-of-computation decidability normal

Answer key☟

### 6.5.11 Decidability: GATE CSE 2001 | Question: 2.7

Consider the following problem $X$.

Given a Turing machine $M$ over the input alphabet $\Sigma$, any state $q$ of $M$ and a word $w \in \Sigma^*$, does the computation of $M$ on $w$ visit the state of $q$?

Which of the following statements about $X$ is correct?

A. $X$ is decidable
B. $X$ is undecidable but partially decidable
C. $X$ is undecidable and not even partially decidable
D. $X$ is not a decision problem

gatecse-2001 theory-of-computation decidability normal

Answer key☟

### 6.5.12 Decidability: GATE CSE 2001 | Question: 7

Let a decision problem $X$ be defined as follows:

$X$: Given a Turing machine $M$ over $\Sigma$ and any word $w \in \Sigma$, does $M$ loop forever on $w$?

You may assume that the halting problem of Turing machine is undecidable but partially decidable.

A. Show that $X$ is undecidable
B. Show that $X$ is not even partially decidable

gatecse-2001 theory-of-computation decidability turing-machine easy descriptive

Answer key☟

### 6.5.13 Decidability: GATE CSE 2002 | Question: 14

The aim of the following question is to prove that the language $\{M \mid M \text{ is the code of the Turing Machine which, irrespective of the input, halts and outputs a } 1\}$, is undecidable. This is to be done by reducing from the language $\{M', x \mid M' \text{ halts on } x\}$, which is known to be undecidable. In parts (a) and (b) describe the $2$ main steps in the construction of $M$. In part (c) describe the key property which relates the behaviour of $M$ on its input w to the behaviour of $M'$ on $x$.

A. On input $w$, what is the first step that $M$ must make?
B. On input $w$, based on the outcome of the first step, what is the second step $M$ must make?
C. What key property relates the behaviour of $M$ on $w$ to the behaviour of $M'$ on $x$?

gatecse-2002 theory-of-computation decidability normal turing-machine descriptive difficult

Answer key☟

### 6.5.14 Decidability: GATE CSE 2003 | Question: 52

Consider two languages $L_1$ and $L_2$ each on the alphabet $\Sigma$. Let $f : \Sigma^* \to \Sigma^*$ be a polynomial time computable bijection such that $(\forall x)[x \in L_1 \text{ iff } f(x) \in L_2]$. Further, let $f^{-1}$ be also polynomial time computable.

Which of the following **CANNOT** be true?

A. $L_1 \in P$ and $L_2$ is finite
B. $L_1 \in NP$ and $L_2 \in P$
C. $L_1$ is undecidable and $L_2$ is decidable
D. $L_1$ is recursively enumerable and $L_2$ is recursive

gatecse-2003 theory-of-computation difficult decidability

Answer key☟

### 6.5.15 Decidability: GATE CSE 2003 | Question: 53

A single tape Turing Machine $M$ has two states $q0$ and $q1$, of which $q0$ is the starting state. The tape alphabet of $M$ is $\{0, 1, B\}$ and its input alphabet is $\{0, 1\}$. The symbol $B$ is the blank symbol used to indicate end of an input string. The transition function of $M$ is described in the following table.

| | $0$ | $1$ | $B$ |
|---|---|---|---|
| $q0$ | $q1, 1, R$ | $q1, 1, R$ | $Halt$ |
| $q1$ | $q1, 1, R$ | $q0, 1, L$ | $q0, B, L$ |

The table is interpreted as illustrated below.

The entry $(q1, 1, R)$ in row $q0$ and column $1$ signifies that if $M$ is in state $q0$ and reads $1$ on the current page square, then it writes $1$ on the same tape square, moves its tape head one position to the right and transitions to state $q1$.

Which of the following statements is true about $M$?

A. $M$ does not halt on any string in $(0+1)^+$
B. $M$ does not halt on any string in $(00+1)^*$
C. $M$ halts on all strings ending in a $0$
D. $M$ halts on all strings ending in a $1$

gatecse-2003 theory-of-computation decidability normal

Answer key☟

### 6.5.16 Decidability: GATE CSE 2007 | Question: 6

Which of the following problems is undecidable?

A. Membership problem for CFGs
B. Ambiguity problem for CFGs
C. Finiteness problem for FSAs
D. Equivalence problem for FSAs

gatecse-2007 theory-of-computation decidability normal

Answer key☟

### 6.5.17 Decidability: GATE CSE 2008 | Question: 10

Which of the following are decidable?

I. Whether the intersection of two regular languages is infinite
II. Whether a given context-free language is regular
III. Whether two push-down automata accept the same language
IV. Whether a given grammar is context-free

A. I and II
B. I and IV
C. II and III
D. II and IV

gatecse-2008 theory-of-computation decidability easy

Answer key☟

### 6.5.18 Decidability: GATE CSE 2012 | Question: 24

Which of the following problems are decidable?

1. Does a given program ever produce an output?
2. If $L$ is a context-free language, then, is $\bar{L}$ also context-free?
3. If $L$ is a regular language, then, is $\bar{L}$ also regular?
4. If $L$ is a recursive language, then, is $\bar{L}$ also recursive?

A. $1,2,3,4$
B. $1,2$
C. $2,3,4$
D. $3,4$

gatecse-2012 theory-of-computation decidability normal

Answer key

### 6.5.19 Decidability: GATE CSE 2013 | Question: 41

Which of the following is/are undecidable?

1. $G$ is a CFG. Is $L(G) = \phi$?
2. $G$ is a CFG. Is $L(G) = \Sigma^*$?
3. $M$ is a Turing machine. Is $L(M)$ regular?
4. $A$ is a DFA and $N$ is an NFA. Is $L(A) = L(N)$?

A. $3$ only
B. $3$ and $4$ only
C. $1, 2$ and $3$ only
D. $2$ and $3$ only

gatecse-2013 theory-of-computation decidability normal

Answer key

### 6.5.20 Decidability: GATE CSE 2014 | Set 3 | Question: 35

Which one of the following problems is undecidable?

A. Deciding if a given context-free grammar is ambiguous.
B. Deciding if a given string is generated by a given context-free grammar.
C. Deciding if the language generated by a given context-free grammar is empty.
D. Deciding if the language generated by a given context-free grammar is finite.

gatecse-2014-set3 theory-of-computation context-free-language decidability normal

Answer key

### 6.5.21 Decidability: GATE CSE 2015 | Set 2 | Question: 21

Consider the following statements.

I. The complement of every Turing decidable language is Turing decidable
II. There exists some language which is in NP but is not Turing decidable
III. If L is a language in NP, L is Turing decidable

Which of the above statements is/are true?

A. Only II
B. Only III
C. Only I and II
D. Only I and III

gatecse-2015-set2 theory-of-computation decidability easy

Answer key

### 6.5.22 Decidability: GATE CSE 2015 | Set 3 | Question: 53

Language $L_1$ is polynomial time reducible to language $L_2$. Language $L_3$ is polynomial time reducible to language $L_2$, which in turn polynomial time reducible to language $L_4$. Which of the following is/are true?

I. $\text{if } L_4 \in P, \text{ then } L_2 \in P$
II. $\text{if } L_1 \in P \text{ or } L_3 \in P, \text{ then } L_2 \in P$
III. $L_1 \in P, \text{ if and only if } L_3 \in P$
IV. $\text{if } L_4 \in P, \text{ then } L_3 \in P$

A. II only
B. III only
C. I and IV only
D. I only

gatecse-2015-set3 theory-of-computation decidability normal

Answer key

### 6.5.23 Decidability: GATE CSE 2016 | Set 1 | Question: 17

Which of the following decision problems are undecidable?

I. Given NFAs $N_1$ and $N_2$ , is $L(N_1) \cap L(N_2) = \Phi$
II. Given a CFG $G = (N, \Sigma, P, S)$ and a string $x \in \Sigma^*$, does $x \in L(G)$} ?
III. Given CFGs $G_1$ and $G_2$, is $L(G_1) = L(G_2)$?
IV. Given a TM $M$, is $L(M) = \Phi$ ?

A. I and IV only
B. II and III only
C. III and IV only
D. II and IV only

gatecse-2016-set1 theory-of-computation decidability easy

Answer key

### 6.5.24 Decidability: GATE CSE 2017 | Set 1 | Question: 39

Let $A$ and $B$ be finite alphabets and let $\#$ be a symbol outside both $A$ and $B$. Let $f$ be a total function from $A^*$ to $B^*$. We say $f$ is *computable* if there exists a Turing machine $M$ which given an input $x \in A^*$, always halts with $f(x)$ on its tape. Let $L_f$ denote the language $\left\{x\#f(x) \mid x \in A^*\right\}$. Which of the following statements is true:

A. $f$ is computable if and only if $L_f$ is recursive.
B. $f$ is computable if and only if $L_f$ is recursively enumerable.
C. If $f$ is computable then $L_f$ is recursive, but not conversely.
D. If $f$ is computable then $L_f$ is recursively enumerable, but not conversely.

gatecse-2017-set1 theory-of-computation decidability difficult difficult

Answer key

### 6.5.25 Decidability: GATE CSE 2017 | Set 2 | Question: 41

Let $L(R)$ be the language represented by regular expression $R$. Let $L(G)$ be the language generated by a context free grammar $G$. Let $L(M)$ be the language accepted by a Turing machine $M$. Which of the following decision problems are undecidable?

I. Given a regular expression $R$ and a string $w$, is $w \in L(R)$?
II. Given a context-free grammar $G$, is $L(G) = \emptyset$
III. Given a context-free grammar $G$, is $L(G) = \Sigma^*$ for some alphabet $\Sigma$?
IV. Given a Turing machine $M$ and a string $w$, is $w \in L(M)$?

A. I and IV only
B. II and III only
C. II, III and IV only
D. III and IV only

gatecse-2017-set2 theory-of-computation decidability

Answer key

### 6.5.26 Decidability: GATE CSE 2018 | Question: 36

Consider the following problems. $L(G)$ denotes the language generated by a grammar $G$. L(M) denotes the language accepted by a machine $M$.

I. For an unrestricted grammar $G$ and a string $w$, whether $w \in L(G)$
II. Given a Turing machine $M$, whether $L(M)$ is regular
III. Given two grammar $G_1$ and $G_2$, whether $L(G_1) = L(G_2)$
IV. Given an NFA $N$, whether there is a deterministic PDA $P$ such that $N$ and $P$ accept the same language

Which one of the following statement is correct?

A. Only I and II are undecidable
B. Only II is undecidable
C. Only II and IV are undecidable
D. Only I, II and III are undecidable

gatecse-2018 theory-of-computation decidability easy two-marks

Answer key

### 6.5.27 Decidability: GATE CSE 2020 | Question: 26

Which of the following languages are undecidable? Note that $\langle M \rangle$ indicates encoding of the Turing machine M.

- $L_1 = \{\langle M \rangle \mid L(M) = \varnothing\}$
- $L_2 = \{\langle M, w, q \rangle \mid M \text{ on input } w \text{ reaches state } q \text{ in exactly } 100 \text{ steps}\}$
- $L_3 = \{\langle M \rangle \mid L(M) \text{ is not recursive}\}$
- $L_4 = \{\langle M \rangle \mid L(M) \text{contains at least } 21 \text{ members}\}$

A. $L_1$, $L_3$, and $L_4$ only
B. $L_1$ and $L_3$ only
C. $L_2$ and $L_3$ only
D. $L_2$, $L_3$, and $L_4$ only

gatecse-2020 theory-of-computation decidability two-marks

Answer key

### 6.5.28 Decidability: GATE CSE 2021 | Set 2 | Question: 36

Consider the following two statements about regular languages:

- $S_1$: Every infinite regular language contains an undecidable language as a subset.
- $S_2$: Every finite language is regular.

Which one of the following choices is correct?

A. Only $S_1$ is true
B. Only $S_2$ is true
C. Both $S_1$ and $S_2$ are true
D. Neither $S_1$ nor $S_2$ is true

gatecse-2021-set2 theory-of-computation regular-language decidability two-marks

Answer key

### 6.5.29 Decidability: GATE CSE 2022 | Question: 36

Which of the following is/are undecidable?

A. Given two Turing machines $M_1$ and $M_2$, decide if $L(M_1) = L(M_2)$.
B. Given a Turing machine $M$, decide if $L(M)$ is regular.
C. Given a Turing machine $M$, decide if $M$ accepts all strings.
D. Given a Turing machine $M$, decide if $M$ takes more than $1073$ steps on every string.

gatecse-2022 theory-of-computation turing-machine decidability multiple-selects two-marks

Answer key

### 6.5.30 Decidability: GATE CSE 2025 | Set 2 | Question: 15

Let $G_1, G_2$ be Context Free Grammars (CFGs) and $R$ be a regular expression. For a grammar $G$, let $L(G)$ denote the language generated by $G$.

Which ONE among the following questions is decidable?

A. Is $L(G_1) = L(G_2)$?
B. Is $L(G_1) \cap L(G_2) = \emptyset$ ?
C. Is $L(G_1) = L(R)$?
D. Is $L(G_1) = \emptyset$ ?

gatecse2025-set2 theory-of-computation decidability easy one-mark

Answer key

## 6.6 Dpda (1)

### 6.6.1 Dpda: GATE CSE 2025 | Set 2 | Question: 14

Which ONE of the following languages is accepted by a deteministic pushdown automaton?

A. Any regular language
B. Any context-free language
C. Any language accepted by a non-deterministic pushdown automaton

D. Any decidable language

gatecse2025-set2 theory-of-computation dpda one-mark easy

Answer key

## 6.7 Finite Automata (43)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (8Q) Weekly Quiz 1 (15Q)

### 6.7.1 Finite Automata: GATE CSE 1988 | Question: 15

Consider the DFA $M$ and NFA $M_2$ as defined below. Let the language accepted by machine $M$ be $L$. What language machine $M_2$ accepts, if

i. $F2 = A$?
ii. $F2 = B$?
iii. $F2 = C$?
iv. $F2 = D$?

- $M = (Q, \Sigma, \delta, q_0, F)$
- $M_2 = (Q2, \Sigma, \delta_2, q_{00}, F2)$

Where,

$Q2 = (Q \times Q \times Q) \cup \{q_{00}\}$
$\delta_2(q_{00}, \epsilon) = \{\langle q_0, q, q \rangle \mid q \in Q\}$
$\delta_2(\langle p, q, r \rangle, \sigma) = \langle \delta(p, \sigma), \delta(q, \sigma), r \rangle$
for all $p, q, r \in Q$ and $\sigma \in \Sigma$
$A = \{\langle p, q, r \rangle \mid p \in F; q, r \in Q\}$
$B = \{\langle p, q, r \rangle \mid q \in F; p, r \in Q\}$
$C = \{\langle p, q, r \rangle \mid p, q, r \in Q; \exists s \in \Sigma^*, \delta(p, s) \in F\}$
$D = \{\langle p, q, r \rangle \mid p, q \in Q; r \in F\}$

gate1988 descriptive theory-of-computation finite-automata difficult

Answer key

### 6.7.2 Finite Automata: GATE CSE 1991 | Question: 17,b

Let $L$ be the language of all binary strings in which the third symbol from the right is a $1$. Give a non-deterministic finite automaton that recognizes $L$. How many states does the minimized equivalent deterministic finite automaton have? Justify your answer briefly?

gate1991 theory-of-computation finite-automata normal descriptive

Answer key

### 6.7.3 Finite Automata: GATE CSE 1993 | Question: 27

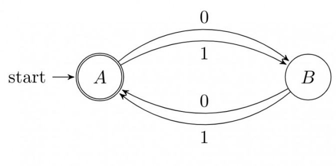
Draw the state transition of a deterministic finite state automaton which accepts all strings from the alphabet $\{a, b\}$, such that no string has $3$ consecutive occurrences of the letter $b$.

gate1993 theory-of-computation finite-automata easy descriptive

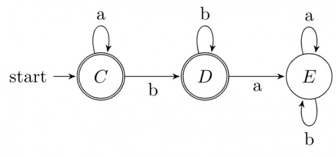
Answer key

### 6.7.4 Finite Automata: GATE CSE 1996 | Question: 12

Given below are the transition diagrams for two finite state machines $M_1$ and $M_2$ recognizing languages $L_1$ and $L_2$ respectively.




$\boxed{M_1}$




$\boxed{M_2}$

A. Display the transition diagram for a machine that recognizes $L_1 . L_2$, obtained from transition diagrams for $M_1$ and $M_2$ by adding only $\varepsilon$ transitions and no new states.

B. Modify the transition diagram obtained in part (a) obtain a transition diagram for a machine that recognizes $(L_1 . L_2)^*$ by adding only $\varepsilon$ transitions and no new states.

(Final states are enclosed in double circles).

gate1996 theory-of-computation finite-automata normal descriptive

Answer key

### 6.7.5 Finite Automata: GATE CSE 1997 | Question: 21

Given that $L$ is a language accepted by a finite state machine, show that $L^P$ and $L^R$ are also accepted by some finite state machines, where

$$L^P = \{s \mid ss' \in L \text{ some string } s'\}$$

$$L^R = \{s \mid s \text{ obtained by reversing some string in } L\}$$

gate1997 theory-of-computation finite-automata proof

Answer key

### 6.7.6 Finite Automata: GATE CSE 1998 | Question: 1.10

Which of the following set can be recognized by a Deterministic Finite state Automaton?

A. The numbers $1, 2, 4, 8, \dots 2^n, \dots$ written in binary
B. The numbers $1, 2, 4, 8, \dots 2^n, \dots$ written in unary
C. The set of binary string in which the number of zeros is the same as the number of ones.
D. The set $\{1, 101, 11011, 1110111, \dots\}$

gate1998 theory-of-computation finite-automata normal

Answer key

### 6.7.7 Finite Automata: GATE CSE 2001 | Question: 5

Construct DFA's for the following languages:

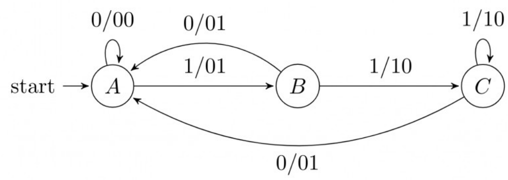
A. $L = \{w \mid w \in \{a, b\}^*, \text{ w has baab as a substring }\}$
B. $L = \{w \mid w \in \{a, b\}^*, \text{ w has an odd number of a's and an odd number of b's }\}$

gatecse-2001 theory-of-computation easy descriptive finite-automata normal

Answer key

### 6.7.8 Finite Automata: GATE CSE 2002 | Question: 2.5

The finite state machine described by the following state diagram with $A$ as starting state, where an arc label is $x/y$, and $x$ stands for $1$-bit input and $y$ stands for $2$-bit output




A. outputs the sum of the present and the previous bits of the input
B. outputs $01$ whenever the input sequence contains $11$
C. outputs $00$ whenever the input sequence contains $10$
D. none of the above

gatecse-2002 theory-of-computation normal finite-automata

Answer key

## 6.7.9 Finite Automata: GATE CSE 2002 | Question: 21

We require a four state automaton to recognize the regular expression $(a \mid b)^*abb$

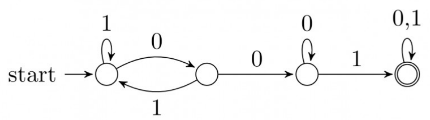
A. Give an NFA for this purpose
B. Give a DFA for this purpose

gatecse-2002 theory-of-computation finite-automata normal descriptive

Answer key

## 6.7.10 Finite Automata: GATE CSE 2003 | Question: 50

Consider the following deterministic finite state automaton $M$.




Let $S$ denote the set of seven bit binary strings in which the first, the fourth, and the last bits are$1$. The number of strings in $S$ that are accepted by $M$ is

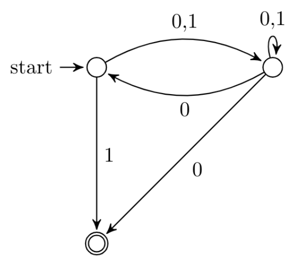
A. $1$
B. $5$
C. $7$
D. $8$

gatecse-2003 theory-of-computation finite-automata normal

Answer key

## 6.7.11 Finite Automata: GATE CSE 2003 | Question: 55

Consider the NFA $M$ shown below.




Let the language accepted by $M$ be $L$. Let $L_1$ be the language accepted by the NFA $M_1$ obtained by changing the accepting state of $M$ to a non-accepting state and by changing the non-accepting states of$M$ to accepting states. Which of the following statements is true?

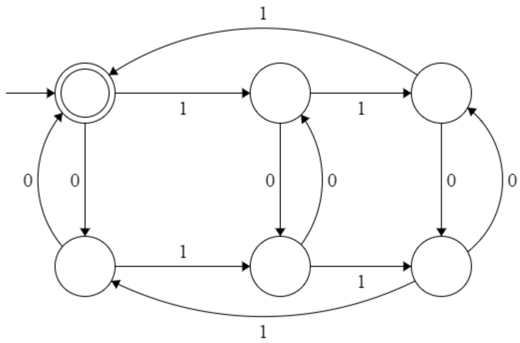
A. $L_1 = \{0,1\}^* - L$
B. $L_1 = \{0,1\}^*$
C. $L_1 \subseteq L$
D. $L_1 = L$

gatecse-2003 theory-of-computation finite-automata normal

Answer key

## 6.7.12 Finite Automata: GATE CSE 2004 | Question: 86

The following finite state machine accepts all those binary strings in which the number of $1$'s and $0$'s are respectively:



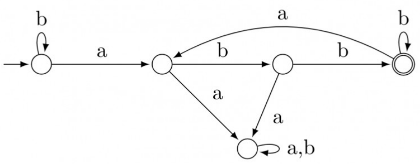
A. divisible by $3$ and $2$
B. odd and even
C. even and odd
D. divisible by $2$ and $3$

gatecse-2004 theory-of-computation finite-automata easy

Answer key

## 6.7.13 Finite Automata: GATE CSE 2005 | Question: 53

Consider the machine $M$:




The language recognized by $M$ is:

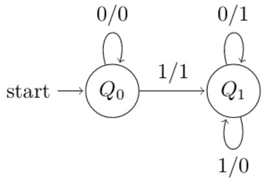
A. $\{w \in \{a, b\}^* \mid \text{every a in } w \text{ is followed by exactly two } b\text{'s}\}$
B. $\{w \in \{a, b\}^* \mid \text{every a in } w \text{ is followed by at least two } b\text{'s}\}$
C. $\{w \in \{a, b\}^* \mid w \text{ contains the substring '}abb\text{'}\}$
D. $\{w \in \{a, b\}^* \mid w \text{ does not contain '}aa\text{' as a substring}\}$

gatecse-2005 theory-of-computation finite-automata normal

Answer key

## 6.7.14 Finite Automata: GATE CSE 2005 | Question: 63

The following diagram represents a finite state machine which takes as input a binary number from the least significant bit.




Which of the following is TRUE?

A. It computes $1$'s complement of the input number

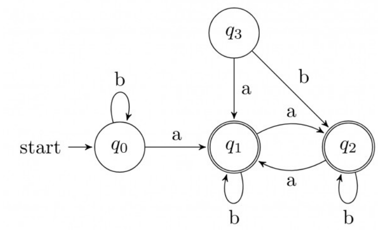
B. It computes $2$'s complement of the input number
C. It increments the input number
D. it decrements the input number

gatecse-2005 theory-of-computation finite-automata easy

Answer key☟

## 6.7.15 Finite Automata: GATE CSE 2007 | Question: 74

Consider the following Finite State Automaton:




The language accepted by this automaton is given by the regular expression

A. $b^*ab^*ab^*ab^*$ B. $(a+b)^*$ C. $b^*a(a+b)^*$ D. $b^*ab^*ab^*$

gatecse-2007 theory-of-computation finite-automata normal

Answer key☟

## 6.7.16 Finite Automata: GATE CSE 2008 | Question: 49

Given below are two finite state automata ( $\rightarrow$ indicates the start state and $F$ indicates a final state)

$Y$

| | a | b |
|---|---|---|
| $\rightarrow 1$ | 1 | 2 |
| $2(F)$ | 2 | 1 |

$Z$

| | a | b |
|---|---|---|
| $\rightarrow 1$ | 2 | 2 |
| $2(F)$ | 1 | 1 |

Which of the following represents the product automaton $Z \times Y$?

A.

| | a | b |
|---|---|---|
| $\rightarrow P$ | S | R |
| Q | R | S |
| R(F) | Q | P |
| S | Q | P |

B.

| | a | b |
|---|---|---|
| $\rightarrow P$ | S | Q |
| Q | R | S |
| R(F) | Q | P |
| S | P | Q |

C.

| | a | b |
|---|---|---|
| $\rightarrow P$ | Q | S |
| Q | R | S |
| R(F) | Q | P |
| S | Q | P |

D.

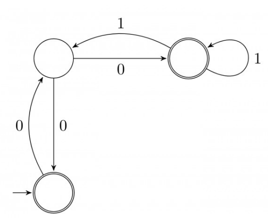
| | a | b |
|---|---|---|
| $\rightarrow P$ | S | Q |
| Q | S | R |
| R(F) | Q | P |
| S | Q | P |

gatecse-2008 normal theory-of-computation finite-automata

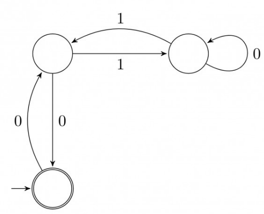
Answer key☟

## 6.7.17 Finite Automata: GATE CSE 2008 | Question: 52

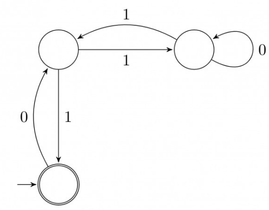
Match the following NFAs with the regular expressions they correspond to:



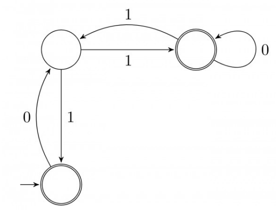

P




Q



R



S

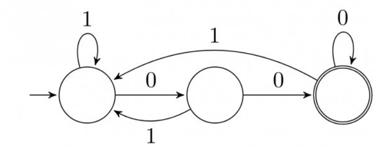
1. $\epsilon + 0(01^*1 + 00)^*01^*$
2. $\epsilon + 0(10^*1 + 00)^*0$
3. $\epsilon + 0(10^*1 + 10)^*1$
4. $\epsilon + 0(10^*1 + 10)^*10^*$

A. $P-2, Q-1, R-3, S-4$
B. $P-1, Q-3, R-2, S-4$
C. $P-1, Q-2, R-3, S-4$
D. $P-3, Q-2, R-1, S-4$

gatecse-2008 theory-of-computation finite-automata normal

Answer key☟

## 6.7.18 Finite Automata: GATE CSE 2009 | Question: 41



The above DFA accepts the set of all strings over $\{0, 1\}$ that

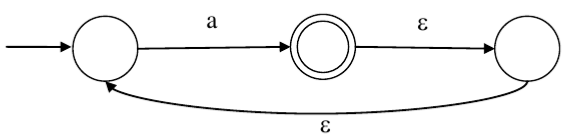
A. begin either with $0$ or $1$.
B. end with $0$.
C. end with $00$.
D. contain the substring $00$.

gatecse-2009 theory-of-computation finite-automata easy

Answer key☟

## 6.7.19 Finite Automata: GATE CSE 2012 | Question: 12

What is the complement of the language accepted by the NFA shown below?
Assume $\Sigma = \{a\}$ and $\epsilon$ is the empty string.



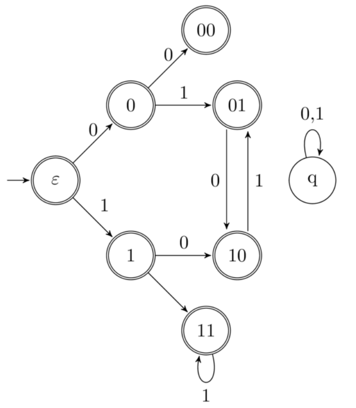
A. $\phi$
B. $\{\epsilon\}$
C. $a^*$
D. $\{a, \epsilon\}$

gatecse-2012 finite-automata easy theory-of-computation

Answer key☟

## 6.7.20 Finite Automata: GATE CSE 2012 | Question: 46

Consider the set of strings on $\{0, 1\}$ in which, *every substring of* $3$ *symbols* has at most *two* zeros. For example, $001110$ and $011001$ are in the language, but $100010$ is not. All strings of length less than $3$ are also in the language. A partially completed DFA that accepts this language is shown below.




The missing arcs in the DFA are:

A.

| | 00 | 01 | 10 | 11 | q |
|---|---|---|---|---|---|
| **00** | 1 | 0 | | | |
| **01** | | | | 1 | |
| **10** | 0 | | | | |
| **11** | | | 0 | | |

B.

| | 00 | 01 | 10 | 11 | q |
|---|---|---|---|---|---|
| **00** | | 0 | | | 1 |
| **01** | | 1 | | | |
| **10** | | | | 0 | |
| **11** | | 0 | | | |

C.

| | 00 | 01 | 10 | 11 | q |
|---|---|---|---|---|---|
| **00** | | 1 | | | 0 |
| **01** | | 1 | | | |
| **10** | | | 0 | | |
| **11** | | 0 | | | |

D.

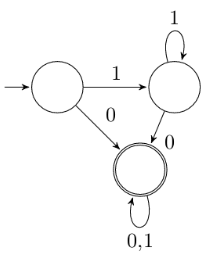
| | 00 | 01 | 10 | 11 | q |
|---|---|---|---|---|---|
| **00** | | 1 | | | 0 |
| **01** | | | | 1 | |
| **10** | 0 | | | | |
| **11** | | | 0 | | |

gatecse-2012 theory-of-computation finite-automata normal

Answer key

## 6.7.21 Finite Automata: GATE CSE 2013 | Question: 33

Consider the DFA $A$ given below.




Which of the following are **FALSE**?

1. Complement of $L(A)$ is context-free.
2. $L(A) = L((11^*0+0)(0+1)^*0^*1^*)$
3. For the language accepted by $A, A$ is the minimal DFA.
4. $A$ accepts all strings over $\{0,1\}$ of length at least $2$.

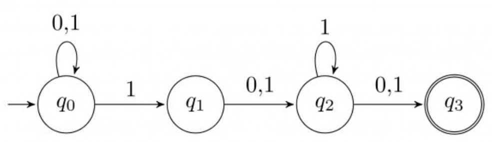
A. 1 and 3 only B. 2 and 4 only C. 2 and 3 only D. 3 and 4 only

gatecse-2013 theory-of-computation finite-automata normal

Answer key

## 6.7.22 Finite Automata: GATE CSE 2014 | Set 1 | Question: 16

Consider the finite automaton in the following figure:




What is the set of reachable states for the input string $0011$?

A. $\{q_0, q_1, q_2\}$ B. $\{q_0, q_1\}$ C. $\{q_0, q_1, q_2, q_3\}$ D. $\{q_3\}$

gatecse-2014-set1 theory-of-computation finite-automata easy

Answer key☟

## 6.7.23 Finite Automata: GATE CSE 2016 | Set 2 | Question: 42

Consider the following two statements:

I. If all states of an **NFA** are accepting states then the language accepted by the **NFA** is $\Sigma^*$.
II. There exists a regular language $A$ such that for all languages $B$, $A \cap B$ is regular.

Which one of the following is **CORRECT**?

A. Only **I** is true
B. Only **II** is true
C. Both **I** and **II** are true
D. Both **I** and **II** are false

gatecse-2016-set2 theory-of-computation finite-automata normal

Answer key☟

## 6.7.24 Finite Automata: GATE CSE 2017 | Set 2 | Question: 39

Let $\delta$ denote the transition function and $\hat{\delta}$ denote the extended transition function of the $\epsilon$-NFA whose transition table is given below:

| $\delta$ | $\epsilon$ | $a$ | $b$ |
|---|---|---|---|
| $\$ \rightarrow \$q_0$ | $\{q_2\}$ | $\{q_1\}$ | $\{q_0\}$ |
| $q_1$ | $\{q_2\}$ | $\{q_2\}$ | $\{q_3\}$ |
| $q_2$ | $\{q_0\}$ | $\emptyset$ | $\emptyset$ |
| $q_3$ | $\emptyset$ | $\emptyset$ | $\{q_2\}$ |

Then $\hat{\delta}(q_2, aba)$ is

A. $\emptyset$
B. $\{q_0, q_1, q_3\}$
C. $\{q_0, q_1, q_2\}$
D. $\{q_0, q_2, q_3\}$

gatecse-2017-set2 theory-of-computation finite-automata

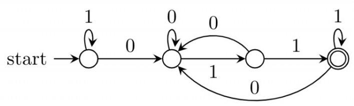
Answer key☟

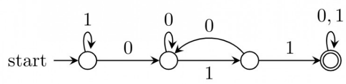
## 6.7.25 Finite Automata: GATE CSE 2021 | Set 1 | Question: 38

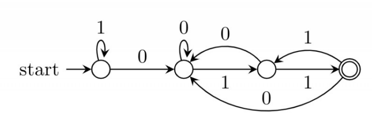
Consider the following language:

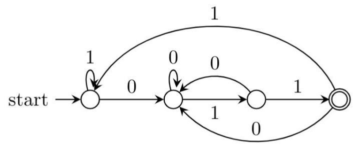
$$L = \{w \in \{0,1\}^* \mid w \text{ ends with the substring } 011\}$$

Which one of the following deterministic finite automata accepts $L$?

A. 


B. 

C. 

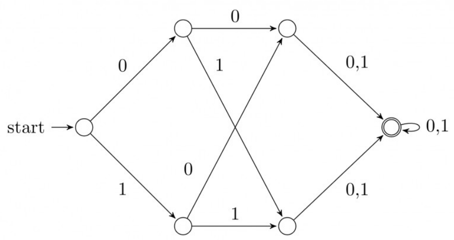

D. 


gatecse-2021-set1 theory-of-computation finite-automata two-marks

Answer key

### 6.7.26 Finite Automata: GATE CSE 2021 | Set 2 | Question: 17

Consider the following deterministic finite automaton $(\mathrm{DFA})$




The number of strings of length $8$ accepted by the above automaton is ___________

gatecse-2021-set2 numerical-answers theory-of-computation finite-automata one-mark

Answer key

### 6.7.27 Finite Automata: GATE CSE 2021 | Set 2 | Question: 28

Suppose we want to design a synchronous circuit that processes a string of $0$'s and $1$'s. Given a string, it produces another string by replacing the first $1$ in any subsequence of consecutive $1$'s by a $0$. Consider the following example.

$$\begin{array}{ll} \text{Input sequence:} & 00100011000011100 \\ \text{Output sequence:} & 00000001000001100 \end{array}$$

A *Mealy Machine* is a state machine where both the next state and the output are functions of the present state and the current input.

The above mentioned circuit can be designed as a two-state Mealy machine. The states in the Mealy machine can be represented using Boolean values $0$ and $1$. We denote the current state, the next state, the next incoming bit, and the output bit of the Mealy machine by the variables $s, t, b$ and $y$ respectively.

Assume the initial state of the Mealy machine is $0$.

What are the Boolean expressions corresponding to $t$ and $y$ in terms of $s$ and $b$?

A. $t = s + b$
$y = sb$

B. $t = b$
$y = sb$

C. $t = b$
$y = s\bar{b}$

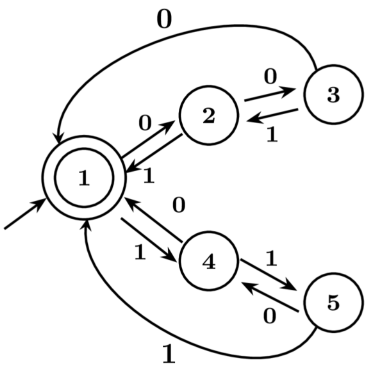
D. $t = s + b$
$y = s\bar{b}$

gatecse-2021-set2 theory-of-computation finite-automata two-marks

Answer key

### 6.7.28 Finite Automata: GATE CSE 2024 | Set 1 | Question: 40

Consider the $5$ -state $\mathrm{DFA}$. $M$ accepting the language $L(M) \subset (0+1)^*$ shown below. For any string $w \in (0+1)^*$ let $n_0(w)$ be the number of $0's$ in $w$ and $n_1(w)$ be the number of 1 's in $w$.




Which of the following statements is/are FALSE?

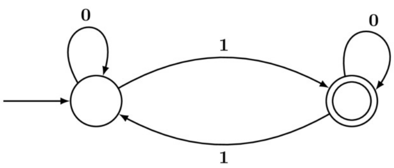
A. States $2$ and $4$ are distinguishable in $M$
B. States $3$ and $4$ are distinguishable in $M$
C. States $2$ and $5$ are distinguishable in $M$
D. Any string $w$ with $n_0(w) = n_1(w)$ is in $L(M)$

gatecse-2024-set1 multiple-selects theory-of-computation finite-automata two-marks

Answer key☟

## 6.7.29 Finite Automata: GATE CSE 2024 | Set 2 | Question: 12

Which one of the following regular expressions is equivalent to the language accepted by the $\mathrm{DFA}$ given below?



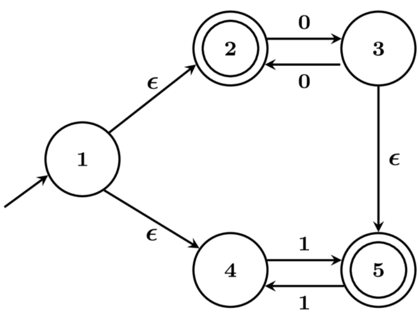

A. $0^*1(0+10^*1)^*$
B. $0^*(10^*11)^*0^*$
C. $0^*1(010^*1)^*0^*$
D. $0(1+0^*10^*1)^*0^*$

gatecse-2024-set2 theory-of-computation finite-automata one-mark

Answer key☟

## 6.7.30 Finite Automata: GATE CSE 2024 | Set 2 | Question: 31

Let $\mathrm{M}$ be the $5$-state $\mathrm{NFA}$ with $\epsilon$-transitions shown in the diagram below.




Which one of the following regular expressions represents the language accepted by $\mathrm{M}$?

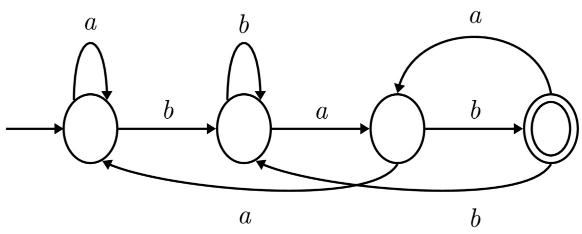
A. $(00)^* + 1(11)^*$
B. $0^* + (1 + 0(00)^*)\,(11)^*$
C. $(00)^* + (1 + (00)^*)\,(11)^*$
D. $0^+ + 1(11)^* + 0(11)^*$

gatecse-2024-set2 theory-of-computation finite-automata two-marks

Answer key

## 6.7.31 Finite Automata: GATE CSE 2025 | Set 1 | Question: 40

Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.



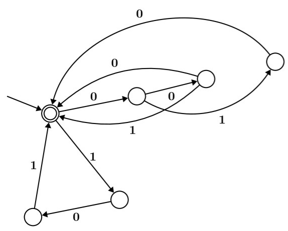
A. The set of all strings containing an even number of $b$ 's.
B. The set of all strings containing the pattern $bab$.
C. The set of all strings ending with the pattern $bab$.
D. The set of all strings not containing the pattern $aba$.

gatecse2025-set1 theory-of-computation finite-automata multiple-selects easy two-marks

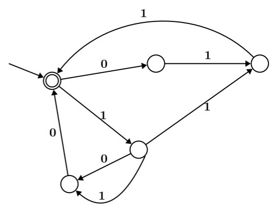
Answer key

## 6.7.32 Finite Automata: GATE CSE 2026 | Set 2 | Question: 37

Consider the following two finite automata $D_1$ and $D_2$.



$D_1$



$D_2$

Which of the following statements is/are true?

A. $L(D_1) = L(D_2)$
B. $L(D_1)$ is a proper subset of $L(D_2)$
C. $L(D_1) \cap L(D_2) = \{\epsilon\}$
D. $(L(D_1) \cup L(D_2))^*$ consists of all strings in $\{0, 1\}^*$ whose length is divisible by $3$

gatecse-2026-set2 theory-of-computation finite-automata multiple-selects two-marks

Answer key

## 6.7.33 Finite Automata: GATE IT 2004 | Question: 41

Let $M = (K, \Sigma, \sigma, s, F)$ be a finite state automaton, where

$K = \{A, B\}, \Sigma = \{a, b\}, s = A, F = \{B\},$
$\sigma(A, a) = A, \sigma(A, b) = B, \sigma(B, a) = B$ and $\sigma(B, b) = A$

A grammar to generate the language accepted by $M$ can be specified as $G = (V, \Sigma, R, S)$, where $V = K \cup \Sigma$, and $S = A$.

Which one of the following set of rules will make $L(G) = L(M)$ ?

A. $\{A \rightarrow aB, A \rightarrow bA, B \rightarrow bA, B \rightarrow aA, B \rightarrow \epsilon)$

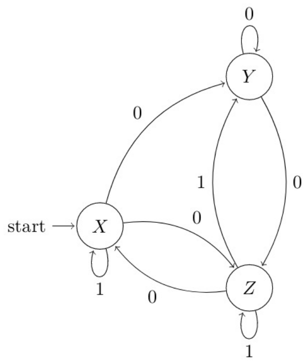
B. $\{A \rightarrow aA, A \rightarrow bB, B \rightarrow aB, B \rightarrow bA, B \rightarrow \epsilon)$
C. $\{A \rightarrow bB, A \rightarrow aB, B \rightarrow aA, B \rightarrow bA, B \rightarrow \epsilon)$
D. $\{A \rightarrow aA, A \rightarrow bA, B \rightarrow aB, B \rightarrow bA, A \rightarrow \epsilon)$

gateit-2004 theory-of-computation finite-automata normal

Answer key

## 6.7.34 Finite Automata: GATE IT 2005 | Question: 37

Consider the non-deterministic finite automaton (NFA) shown in the figure.




State $X$ is the starting state of the automaton. Let the language accepted by the NFA with $Y$ as the only accepting state be $L1$. Similarly, let the language accepted by the NFA with $Z$ as the only accepting state be $L2$. Which of the following statements about $L1$ and $L2$ is TRUE?

A. $L1 = L2$
B. $L1 \subset L2$
C. $L2 \subset L1$
D. None of the above

gateit-2005 theory-of-computation finite-automata normal

Answer key

## 6.7.35 Finite Automata: GATE IT 2005 | Question: 39

Consider the regular grammar:

- $S \rightarrow Xa \mid Ya$
- $X \rightarrow Za$
- $Z \rightarrow Sa \mid \epsilon$
- $Y \rightarrow Wa$
- $W \rightarrow Sa$

where $S$ is the starting symbol, the set of terminals is $\{a\}$ and the set of non-terminals is $\{S, W, X, Y, Z\}$.
We wish to construct a deterministic finite automaton (DFA) to recognize the same language. What is the minimum number of states required for the DFA?

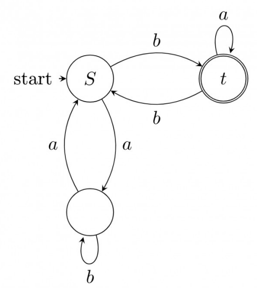
A. $2$
B. $3$
C. $4$
D. $5$

gateit-2005 theory-of-computation finite-automata normal

Answer key

## 6.7.36 Finite Automata: GATE IT 2006 | Question: 3

In the automaton below, $s$ is the start state and $t$ is the only final state.




Consider the strings $u = abbaba, v = bab, \text{and} w = aabb$. Which of the following statements is true?

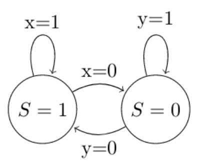
A. The automaton accepts $u$ and $v$ but not $w$
B. The automaton accepts each of $u, v,$ and $w$
C. The automaton rejects each of $u, v,$ and $w$
D. The automaton accepts $u$ but rejects $v$ and $w$

gateit-2006 theory-of-computation finite-automata easy

Answer key☟

## 6.7.37 Finite Automata: GATE IT 2006 | Question: 37

For a state machine with the following state diagram the expression for the next state $S^+$ in terms of the current state $S$ and the input variables $x$ and $y$ is



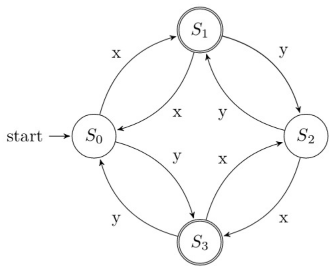

A. $S^+ = S'.y' + S.x$
B. $S^+ = S.x.y' + S'.y.x'$
C. $S^+ = x.y'$
D. $S^+ = S'.y + S.x'$

gateit-2006 theory-of-computation finite-automata normal

Answer key☟

## 6.7.38 Finite Automata: GATE IT 2007 | Question: 47

Consider the following DFA in which $S_0$ is the start state and $S_1$, $S_3$ are the final states.




What language does this DFA recognize?

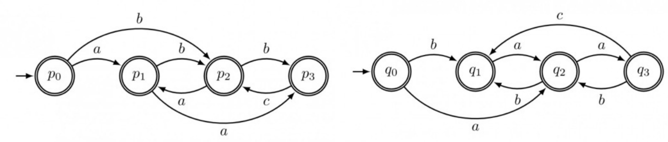
A. All strings of $x$ and $y$
B. All strings of $x$ and $y$ which have either even number of $x$ and even number of $y$ or odd number of $x$ and odd number of $y$
C. All strings of $x$ and $y$ which have equal number of $x$ and $y$
D. All strings of $x$ and $y$ with either even number of $x$ and odd number of $y$ or odd number of $x$ and even number of $y$

gateit-2007 theory-of-computation finite-automata normal

Answer key☟

## 6.7.39 Finite Automata: GATE IT 2007 | Question: 50

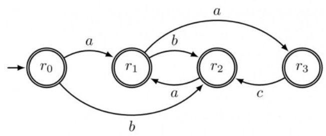
Consider the following finite automata $P$ and $Q$ over the alphabet $\{a, b, c\}$. The start states are indicated by a double arrow and final states are indicated by a double circle. Let the languages recognized by them be denoted by $L(P)$ and $L(Q)$ respectively.



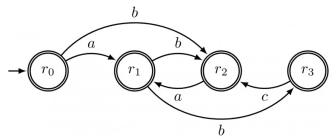

(P) (Q)

The automation which recognizes the language $L(P) \cap L(Q)$ is :

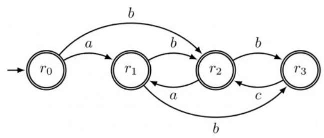
A.



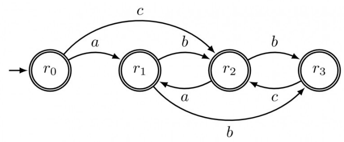
B.



C.



D.



gateit-2007 theory-of-computation finite-automata normal

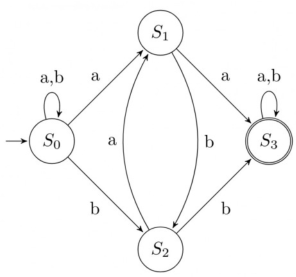
Answer key

## 6.7.40 Finite Automata: GATE IT 2007 | Question: 71

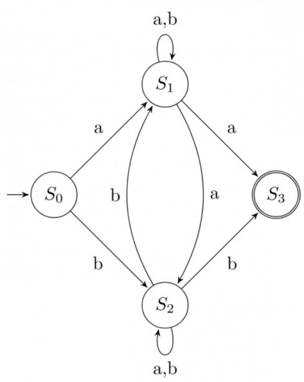
Consider the regular expression $R = (a+b)^*(aa+bb)(a+b)^*$

Which of the following non-deterministic finite automata recognizes the language defined by the regular expression $R$? Edges labeled $\lambda$ denote transitions on the empty string.

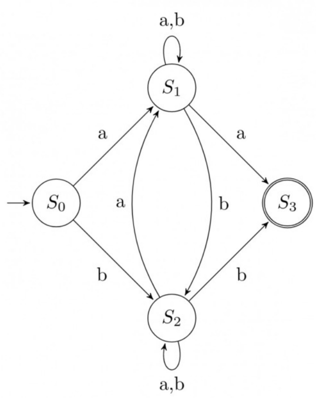
A.



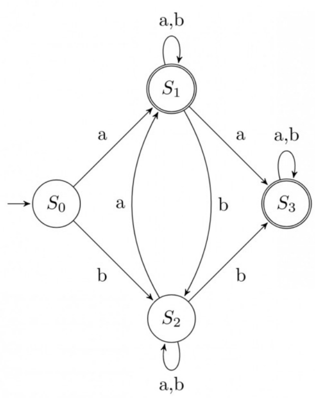
B.



C.



D.



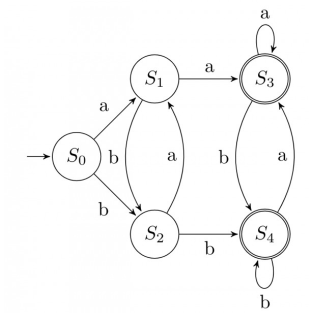
gateit-2007 theory-of-computation finite-automata normal

Answer key☟

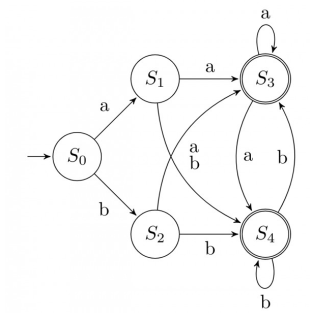
## 6.7.41 Finite Automata: GATE IT 2007 | Question: 72

Consider the regular expression $R = (a+b)^*(aa+bb)(a+b)^*$

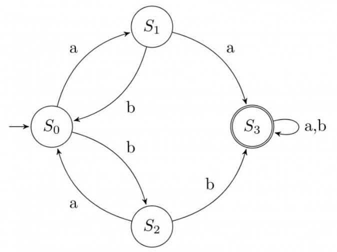
Which deterministic finite automaton accepts the language represented by the regular expression $R$?



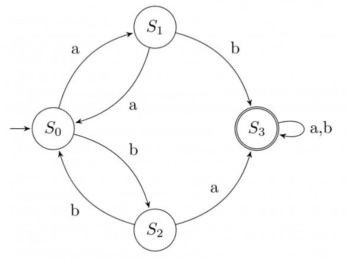

A.




B.




C.



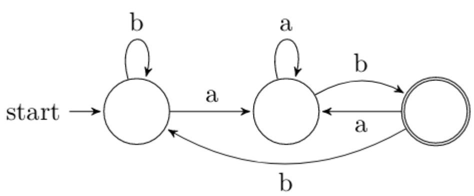

D.

gateit-2007 theory-of-computation finite-automata normal

Answer key☟

## 6.7.42 Finite Automata: GATE IT 2008 | Question: 32

If the final states and non-final states in the DFA below are interchanged, then which of the following languages over the alphabet $\{a, b\}$ will be accepted by the new DFA?



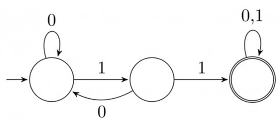

A. Set of all strings that do not end with $ab$
B. Set of all strings that begin with either an $a$ or $a\ b$
C. Set of all strings that do not contain the substring $ab$,
D. The set described by the regular expression $b^*aa^*(ba)^*b^*$

gateit-2008 theory-of-computation finite-automata normal

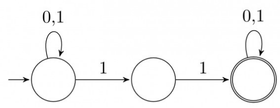
Answer key☟

## 6.7.43 Finite Automata: GATE IT 2008 | Question: 36

Consider the following two finite automata. $M_1$ accepts $L_1$ and $M_2$ accepts $L_2$.




$M_1$




$M_2$

Which one of the following is TRUE?

A. $L_1 = L_2$
B. $L_1 \subset L_2$
C. $L_1 \cap L_2^C = \varnothing$
D. $L_1 \cup L_2 \neq L_1$

gateit-2008 theory-of-computation finite-automata normal

Answer key

## 6.8 Finite State Machines (1)

### 6.8.1 Finite State Machines: GATE CSE 2025 | Set 1 | Question: 49

Consider a finite state machine (FSM) with one input $X$ and one output $f$, represented by the given state transition table. The minimum number of states required to realize this FSM is ________. (Answer in integer).

| Present state | Next state | | Output f | |
|---|---|---|---|---|
| | X=0 | X=1 | X=0 | X=1 |
| A | F | B | 0 | 0 |
| B | D | C | 0 | 0 |
| C | F | E | 0 | 0 |
| D | G | A | 1 | 0 |
| E | D | C | 0 | 0 |
| F | F | B | 1 | 1 |
| G | G | H | 0 | 1 |
| H | G | A | 1 | 0 |

gatecse2025-set1 theory-of-computation finite-state-machines numerical-answers two-marks

Answer key

## 6.9 Identify Class Language (31)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (13Q) Weekly Quiz 8 (10Q)

### 6.9.1 Identify Class Language: GATE CSE 1987 | Question: 1-xiii

FORTRAN is a:

A. Regular language.
B. Context-free language.
C. Context-senstive language.
D. None of the above.

gate1987 theory-of-computation identify-class-language

Answer key

### 6.9.2 Identify Class Language: GATE CSE 1988 | Question: 2ix

What is the type of the language $L$, where $L = \{a^n b^n \mid 0 < n < 327\text{-th prime number}\}$

gate1988 normal descriptive theory-of-computation identify-class-language

Answer key

### 6.9.3 Identify Class Language: GATE CSE 1991 | Question: 17,a

Show that the Turing machines, which have a read only input tape and constant size work tape, recognize precisely the class of regular languages.

gate1991 theory-of-computation descriptive identify-class-language proof

Answer key☟

### 6.9.4 Identify Class Language: GATE CSE 1994 | Question: 19

A. Given a set:

$$S = \{x \mid \text{ there is an x-block of 5's in the decimal expansion of } \pi\}$$

(Note: $x$-$block$ is a maximal block of $x$ successive 5's)
Which of the following statements is true with respect to $S$? No reason to be given for the answer.

i. $S$ is regular
ii. $S$ is recursively enumerable
iii. $S$ is not recursively enumerable
iv. $S$ is recursive

B. Given that a language $L_1$ is regular and that the language $L_1 \cup L_2$ is regular, is the language $L_2$ always regular? Prove your answer.

gate1994 theory-of-computation identify-class-language normal descriptive

Answer key☟

### 6.9.5 Identify Class Language: GATE CSE 1999 | Question: 2.4

If $L1$ is context free language and $L2$ is a regular language which of the following is/are false?

A. $L1 - L2$ is not context free
B. $L1 \cap L2$ is context free
C. $\sim L1$ is context free
D. $\sim L2$ is regular

gate1999 theory-of-computation identify-class-language normal multiple-selects

Answer key☟

### 6.9.6 Identify Class Language: GATE CSE 2000 | Question: 1.5

Let $L$ denote the languages generated by the grammar $S \rightarrow 0S0 \mid 00$.
Which of the following is TRUE?

A. $L = 0^+$
B. $L$ is regular but not $0^+$
C. $L$ is context free but not regular
D. $L$ is not context free

gatecse-2000 theory-of-computation easy identify-class-language

Answer key☟

### 6.9.7 Identify Class Language: GATE CSE 2002 | Question: 1.7

The language accepted by a Pushdown Automaton in which the stack is limited to $10$ items is best described as

A. Context free
B. Regular
C. Deterministic Context free
D. Recursive

gatecse-2002 theory-of-computation easy identify-class-language

Answer key☟

### 6.9.8 Identify Class Language: GATE CSE 2004 | Question: 87

The language $\{a^m b^n c^{m+n} \mid m, n \geq 1\}$ is

A. regular
B. context-free but not regular
C. context-sensitive but not context free
D. type-0 but not context sensitive

gatecse-2004 theory-of-computation normal identify-class-language

Answer key☟

### 6.9.9 Identify Class Language: GATE CSE 2005 | Question: 55

Consider the languages:

$L_1 = \{a^n b^n c^m \mid n, m > 0\}$ and $L_2 = \{a^n b^m c^m \mid n, m > 0\}$

Which one of the following statements is FALSE?

A. $L_1 \cap L_2$ is a context-free language
B. $L_1 \cup L_2$ is a context-free language
C. $L_1 \text{ and } L_2$ are context-free languages
D. $L_1 \cap L_2$ is a context sensitive language

gatecse-2005 theory-of-computation identify-class-language normal

Answer key☟

### 6.9.10 Identify Class Language: GATE CSE 2006 | Question: 30

For $s \in (0+1)^*$ let $d(s)$ denote the decimal value of $s$(e.g. $d(101) = 5$). Let

$$L = \{s \in (0+1)^* \mid d(s) \bmod 5 = 2 \text{ and } d(s) \bmod 7 \neq 4\}$$

Which one of the following statements is true?

A. $L$ is recursively enumerable, but not recursive
B. $L$ is recursive, but not context-free
C. $L$ is context-free, but not regular
D. $L$ is regular

gatecse-2006 theory-of-computation normal identify-class-language

Answer key☟

### 6.9.11 Identify Class Language: GATE CSE 2006 | Question: 33

Let $L_1$ be a regular language, $L_2$ be a deterministic context-free language and $L_3$ a recursively enumerable, but not recursive, language. Which one of the following statements is false?

A. $L_1 \cap L_2$ is a deterministic CFL
B. $L_3 \cap L_1$ is recursive
C. $L_1 \cup L_2$ is context free
D. $L_1 \cap L_2 \cap L_3$ is recursively enumerable

gatecse-2006 theory-of-computation normal identify-class-language

Answer key☟

### 6.9.12 Identify Class Language: GATE CSE 2007 | Question: 30

The language $L = \left\{0^i 21^i \mid i \geq 0\right\}$ over the alphabet $\{0, 1, 2\}$ is:

A. not recursive
B. is recursive and is a deterministic CFL
C. is a regular language
D. is not a deterministic CFL but a CFL

gatecse-2007 theory-of-computation normal identify-class-language

Answer key☟

### 6.9.13 Identify Class Language: GATE CSE 2008 | Question: 51

Match the following:

| | | | |
|---|---|---|---|
| E. | Checking that identifiers are declared before their use | P. | $L = \{a^n b^m c^n d^m \mid n \geq 1, m \geq 1\}$ |
| F. | Number of formal parameters in the declaration of a function agrees with the number of actual parameters in a use of that function | Q. | $X \rightarrow XbX \mid XcX \mid dXf \mid g$ |
| G. | Arithmetic expressions with matched pairs of parentheses | R. | $L = \{wcw \mid w \in (a \mid b)^*\}$ |
| H. | Palindromes | S. | $X \rightarrow bXb \mid cXc \mid \epsilon$ |

A. E-P, F-R, G-Q, H-S
B. E-R, F-P, G-S, H-Q
C. E-R, F-P, G-Q, H-S
D. E-P, F-R, G-S, H-Q

gatecse-2008 normal theory-of-computation identify-class-language match-the-following

Answer key

### 6.9.14 Identify Class Language: GATE CSE 2008 | Question: 9

Which of the following is true for the language

$$\{a^p \mid p \text{ is a prime }\}?$$

A. It is not accepted by a Turing Machine
B. It is regular but not context-free
C. It is context-free but not regular
D. It is neither regular nor context-free, but accepted by a Turing machine

gatecse-2008 theory-of-computation easy identify-class-language

Answer key

### 6.9.15 Identify Class Language: GATE CSE 2009 | Question: 40

Let $L = L_1 \cap L_2$, where $L_1$ and $L_2$ are languages as defined below:

$L_1 = \{a^m b^m c a^n b^n \mid m, n \geq 0\}$

$L_2 = \left\{a^i b^j c^k \mid i, j, k \geq 0\right\}$

Then $L$ is

A. Not recursive
B. Regular
C. Context free but not regular
D. Recursively enumerable but not context free.

gatecse-2009 theory-of-computation easy identify-class-language

Answer key

### 6.9.16 Identify Class Language: GATE CSE 2010 | Question: 40

Consider the languages

$L1 = \{0^i 1^j \mid i \neq j\}$,

$L2 = \{0^i 1^j \mid i = j\}$,

$L3 = \{0^i 1^j \mid i = 2j + 1\}$,

$L4 = \{0^i 1^j \mid i \neq 2j\}$

A. Only $L2$ is context free.
B. Only $L2$ and $L3$ are context free.
C. Only $L1$ and $L2$ are context free.
D. All are context free

gatecse-2010 theory-of-computation context-free-language identify-class-language normal

Answer key

### 6.9.17 Identify Class Language: GATE CSE 2011 | Question: 26

Consider the languages $L1,\ L2\,and\,L3$ as given below.

$L1 = \{0^p1^q \mid p,q \in N\}$,

$L2 = \{0^p1^q \mid p,q \in N\,and\,p = q\}\,and,$

$L3 = \{0^p1^q0^r \mid p,q,r \in N\,and\,p = q = r\}.$

Which of the following statements is **NOT TRUE**?

A. Push Down Automata (PDA) can be used to recognize $L1$ and $L2$
B. $L1$ is a regular language
C. All the three languages are context free
D. Turing machines can be used to recognize all the languages

gatecse-2011 theory-of-computation identify-class-language normal

Answer key☟

### 6.9.18 Identify Class Language: GATE CSE 2013 | Question: 32

Consider the following languages.

$L_1 = \{0^p1^q0^r \mid p,q,r \geq 0\}$

$L_2 = \{0^p1^q0^r \mid p,q,r \geq 0, p \neq r\}$

Which one of the following statements is **FALSE**?

A. $L_2$ is context-free.
B. $L_1 \cap L_2$ is context-free.
C. Complement of $L_2$ is recursive.
D. Complement of $L_1$ is context-free but not regular.

gatecse-2013 theory-of-computation identify-class-language normal

Answer key☟

### 6.9.19 Identify Class Language: GATE CSE 2014 | Set 3 | Question: 36

Consider the following languages over the alphabet $\sum = \{0,1,c\}$

$L_1 = \{0^n1^n \mid n \geq 0\}$

$L_2 = \{wcw^r \mid w \in \{0,1\}^*\}$

$L_3 = \{ww^r \mid w \in \{0,1\}^*\}$

Here, $w^r$ is the reverse of the string $w$. Which of these languages are deterministic Context-free languages?

A. None of the languages
B. Only $L_1$
C. Only $L_1$ and $L_2$
D. All the three languages

gatecse-2014-set3 theory-of-computation identify-class-language context-free-language normal

Answer key☟

### 6.9.20 Identify Class Language: GATE CSE 2017 | Set 1 | Question: 37

Consider the context-free grammars over the alphabet $\{a,b,c\}$ given below. $S$ and $T$ are non-terminals.

$G_1 : S \rightarrow aSb \mid T, T \rightarrow cT \mid \epsilon$

$G_2 : S \rightarrow bSa \mid T, T \rightarrow cT \mid \epsilon$

The language $L\,(G_1) \cap L(G_2)$ is

A. Finite
B. Not finite but regular
C. Context-Free but not regular
D. Recursive but not context-free

gatecse-2017-set1 theory-of-computation context-free-language identify-class-language normal

Answer key☟

### 6.9.21 Identify Class Language: GATE CSE 2017 | Set 2 | Question: 40

Consider the following languages.

- $L_1 = \{a^p \mid p \text{ is a prime number}\}$
- $L_2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$
- $L_3 = \{a^n b^n c^{2n} \mid n \geq 0\}$
- $L_4 = \{a^n b^n \mid n \geq 1\}$

Which of the following are CORRECT?

I. $L_1$ is context free but not regular
II. $L_2$ is not context free
III. $L_3$ is not context free but recursive
IV. $L_4$ is deterministic context free

A. I, II and IV only
B. II and III only
C. I and IV only
D. III and IV only

gatecse-2017-set2 theory-of-computation identify-class-language

Answer key

### 6.9.22 Identify Class Language: GATE CSE 2018 | Question: 35

Consider the following languages:

I. $\{a^m b^n c^p d^q \mid m+p=n+q, \text{ where } m,n,p,q \geq 0\}$
II. $\{a^m b^n c^p d^q \mid m=n \text{ and } p=q, \text{ where } m,n,p,q \geq 0\}$
III. $\{a^m b^n c^p d^q \mid m=n=p \text{ and } p \neq q, \text{ where } m,n,p,q \geq 0\}$
IV. $\{a^m b^n c^p d^q \mid mn=p+q, \text{ where } m,n,p,q \geq 0\}$

Which of the above languages are context-free?

A. I and IV only
B. I and II only
C. II and III only
D. II and IV only

gatecse-2018 theory-of-computation identify-class-language context-free-language normal two-marks

Answer key

### 6.9.23 Identify Class Language: GATE CSE 2020 | Question: 10

Consider the language $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and the following statements.

I. $L$ is deterministic context-free.
II. $L$ is context-free but not deterministic context-free.
III. $L$ is not $LL(k)$ for any $k$.

Which of the above statements is/are TRUE?

A. I only
B. II only
C. I and III only
D. III only

gatecse-2020 theory-of-computation identify-class-language one-mark

Answer key

### 6.9.24 Identify Class Language: GATE CSE 2020 | Question: 32

Consider the following languages.

$$L_1 = \{wxyx \mid w, x, y \in (0+1)^+\}$$
$$L_2 = \{xy \mid x, y \in (a+b)^*, \mid x \mid = \mid y \mid, x \neq y\}$$

Which one of the following is TRUE?

A. $L_1$ is regular and $L_2$ is context- free.
B. $L_1$ context- free but not regular and $L_2$ is context-free.

C. Neither $L_1$ nor $L_2$ is context- free.
D. $L_1$ context- free but $L_2$ is not context-free.

gatecse-2020 theory-of-computation identify-class-language two-marks

Answer key

### 6.9.25 Identify Class Language: GATE CSE 2021 | Set 2 | Question: 12

Let $L_1$ be a regular language and $L_2$ be a context-free language. Which of the following languages is/are context-free?

A. $L_1 \cap \overline{L_2}$
B. $\overline{\overline{L_1} \cup \overline{L_2}}$
C. $L_1 \cup (L_2 \cup \overline{L_2})$
D. $(L_1 \cap L_2) \cup (\overline{L_1} \cap L_2)$

gatecse-2021-set2 multiple-selects theory-of-computation identify-class-language one-mark

Answer key

### 6.9.26 Identify Class Language: GATE CSE 2022 | Question: 13

Which of the following statements is/are $\text{TRUE}$?

A. Every subset of a recursively enumerable language is recursive.
B. If a language $L$ and its complement $\bar{L}$ are both recursively enumerable, then $L$ must be recursive.
C. Complement of a context-free language must be recursive.
D. If $L_1$ and $L_2$ are regular, then $L_1 \cap L_2$ must be deterministic context-free.

gatecse-2022 theory-of-computation identify-class-language recursive-and-recursively-enumerable-languages multiple-selects one-mark

Answer key

### 6.9.27 Identify Class Language: GATE CSE 2022 | Question: 37

Consider the following languages:

$L_1 = \{a^n w a^n | w \in \{a, b\}^*\}$

$L_2 = \{w x w^R | w, x \in \{a, b\}^*, |w|, |x| > 0\}$

Note that $w^R$ is the reversal of the string $w$. Which of the following is/are $\text{TRUE}$?

A. $L_1$ and $L_2$ are regular.
B. $L_1$ and $L_2$ are context-free.
C. $L_1$ is regular and $L_2$ is context-free.
D. $L_1$ and $L_2$ are context-free but not regular.

gatecse-2022 theory-of-computation identify-class-language context-free-language multiple-selects two-marks

Answer key

### 6.9.28 Identify Class Language: GATE CSE 2023 | Question: 14

Which of the following statements is/are $\text{CORRECT}$?

A. The intersection of two regular languages is regular.
B. The intersection of two context-free languages is context-free.
C. The intersection of two recursive languages is recursive.
D. The intersection of two recursively enumerable languages is recursively enumerable.

gatecse-2023 theory-of-computation identify-class-language multiple-selects one-mark

Answer key

### 6.9.29 Identify Class Language: GATE IT 2005 | Question: 4

Let $L$ be a regular language and $M$ be a context-free language, both over the alphabet $\Sigma$. Let $L^c$ and $M^c$ denote the complements of $L$ and $M$ respectively. Which of the following statements about the language $L^c \cup M^c$ is TRUE?

A. It is necessarily regular but not necessarily context-free.
B. It is necessarily context-free.
C. It is necessarily non-regular.
D. None of the above

gateit-2005 theory-of-computation normal identify-class-language

Answer key

### 6.9.30 Identify Class Language: GATE IT 2005 | Question: 6

The language $\{0^n 1^n 2^n \mid 1 \leq n \leq 10^6\}$ is

A. regular
B. context-free but not regular
C. context-free but its complement is not context-free
D. not context-free

gateit-2005 theory-of-computation easy identify-class-language

Answer key

### 6.9.31 Identify Class Language: GATE IT 2008 | Question: 33

Consider the following languages.

- $L_1 = \{a^i b^j c^k \mid i = j, k \geq 1\}$
- $L_2 = \{a^i b^j \mid j = 2i, i \geq 0\}$

Which of the following is true?

A. $L_1$ is not a CFL but $L_2$ is
B. $L_1 \cap L_2 = \varnothing$ and $L_1$ is non-regular
C. $L_1 \cup L_2$ is not a CFL but $L_2$ is
D. There is a $4$-state PDA that accepts $L_1$, but there is no DPDA that accepts $L_2$.

gateit-2008 theory-of-computation normal identify-class-language

Answer key

## 6.10 Medium (1)

### 6.10.1 Medium: GATE CSE 2006 | Question: 29

If $s$ is a string over $(0+1)^*$ then let $n_0(s)$ denote the number of $0$'s in $s$ and $n_1(s)$ the number of $1$'s in $s$. Which one of the following languages is not regular?

A. $L = \{s \in (0+1)^* \mid n_0(s) \text{ is a 3-digit prime } \}$
B. $L = \{s \in (0+1)^* \mid \text{ for every prefix s' of s, } \mid n_0(s') - n_1(s') \mid \leq 2\}$
C. $L = \{s \in (0+1)^* \mid n_0(s) - n_1(s) \mid \leq 4\}$
D. $L = \{s \in (0+1)^* \mid n_0(s) \mod 7 = n_1(s) \mod 5 = 0\}$

gatecse-2006 theory-of-computation medium regular-language

Answer key

## 6.11 Minimal State Automata (25)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q)

### 6.11.1 Minimal State Automata: GATE CSE 1987 | Question: 2j

State whether the following statements are TRUE or FALSE:

A minimal DFA that is equivalent to an NDFA with $n$ nodes has always $2^n$ states.

gate1987 theory-of-computation finite-automata minimal-state-automata

Answer key

## 6.11.2 Minimal State Automata: GATE CSE 1997 | Question: 20

Construct a finite state machine with minimum number of states, accepting all strings over $(a, b)$ such that the number of $a$'s is divisible by two and the number of $b$'s is divisible by three.

gate1997 theory-of-computation finite-automata normal minimal-state-automata descriptive

Answer key

## 6.11.3 Minimal State Automata: GATE CSE 1997 | Question: 70

Following is a state table for time finite state machine.

| **Present State** | **Next State Output** | |
|---|---|---|
| | **Input- 0** | **Input-1** |
| A | B.1 | H.1 |
| B | F.1 | D.1 |
| C | D.0 | E.1 |
| D | C.0 | F.1 |
| E | D.1 | C.1 |
| F | C.1 | C.1 |
| G | C.1 | D.1 |
| H | C.0 | A.1 |

A. Find the equivalence partition on the states of the machine.

B. Give the state table for the minimal machine. (Use appropriate names for the equivalent states. For example if states $X$ and $Y$ are equivalent then use $XY$ as the name for the equivalent state in the minimal machine).

gate1997 theory-of-computation minimal-state-automata descriptive

Answer key

## 6.11.4 Minimal State Automata: GATE CSE 1998 | Question: 2.5

Let $L$ be the set of all binary strings whose last two symbols are the same. The number of states in the minimal state deterministic finite state automaton accepting $L$ is

A. $2$ B. $5$ C. $8$ D. $3$

gate1998 theory-of-computation finite-automata normal minimal-state-automata

Answer key

## 6.11.5 Minimal State Automata: GATE CSE 1998 | Question: 4

Design a deterministic finite state automaton (using minimum number of states) that recognizes the following language:

$L = \{w \in \{0, 1\}^* \mid w$ interpreted as binary number (ignoring the leading zeros) is divisible by five$\}$.

gate1998 theory-of-computation finite-automata normal minimal-state-automata descriptive

Answer key

## 6.11.6 Minimal State Automata: GATE CSE 1999 | Question: 1.4

Consider the regular expression $(0+1)(0+1)\dots N$ times. The minimum state finite automaton that recognizes the language represented by this regular expression contains

A. $n$ states B. $n+1$ states C. $n+2$ states D. None of the above

gate1999 theory-of-computation finite-automata easy minimal-state-automata

Answer key

## 6.11.7 Minimal State Automata: GATE CSE 2001 | Question: 1.6

Given an arbitrary non-deterministic finite automaton (NFA) with $N$ states, the maximum number of states in an equivalent minimized DFA at least

A. $N^2$ B. $2^N$ C. $2N$ D. $N!$

gatecse-2001 finite-automata theory-of-computation easy minimal-state-automata

Answer key

## 6.11.8 Minimal State Automata: GATE CSE 2001 | Question: 2.5

Consider a DFA over $\Sigma = \{a, b\}$ accepting all strings which have number of a's divisible by $6$ and number of $b$'s divisible by $8$. What is the minimum number of states that the DFA will have?

A. $8$ B. $14$ C. $15$ D. $48$

gatecse-2001 theory-of-computation finite-automata minimal-state-automata

Answer key

## 6.11.9 Minimal State Automata: GATE CSE 2002 | Question: 2.13

The smallest finite automaton which accepts the language $\{x|$ length of $x$ is divisible by $3\}$ has

A. $2$ states B. $3$ states C. $4$ states D. $5$ states

gatecse-2002 theory-of-computation normal finite-automata minimal-state-automata

Answer key

## 6.11.10 Minimal State Automata: GATE CSE 2006 | Question: 34

Consider the regular language $L = (111 + 11111)^*$. The minimum number of states in any DFA accepting this languages is:

A. $3$ B. $5$ C. $8$ D. $9$

gatecse-2006 theory-of-computation finite-automata normal minimal-state-automata

Answer key

## 6.11.11 Minimal State Automata: GATE CSE 2007 | Question: 29

A minimum state deterministic finite automaton accepting the language

$L = \{w \mid w \in \{0, 1\}^*,$ number of $0$s and $1$s in $w$ are divisible by $3$ and $5$, respectively $\}$ has

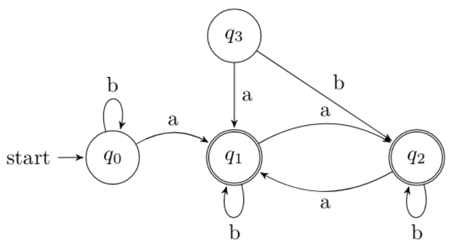
A. $15$ states B. $11$ states C. $10$ states D. $9$ states

gatecse-2007 theory-of-computation finite-automata normal minimal-state-automata

Answer key

## 6.11.12 Minimal State Automata: GATE CSE 2007 | Question: 75

Consider the following Finite State Automaton:




The minimum state automaton equivalent to the above FSA has the following number of states:

A. $1$ B. $2$ C. $3$ D. $4$

normal gatecse-2007 theory-of-computation finite-automata minimal-state-automata

Answer key

## 6.11.13 Minimal State Automata: GATE CSE 2010 | Question: 41

Let $w$ be any string of length $n$ in $\{0,1\}^*$. Let $L$ be the set of all substrings of $w$. What is the minimum number of states in non-deterministic finite automation that accepts $L$?

A. $n-1$ B. $n$ C. $n+1$ D. $2^{n-1}$

gatecse-2010 theory-of-computation finite-automata normal minimal-state-automata

Answer key

## 6.11.14 Minimal State Automata: GATE CSE 2011 | Question: 42

Definition of a language $L$ with alphabet $\{a\}$ is given as following.

$$L = \left\{a^{nk} \mid k > 0,\ \ and\ \ n \, \text{is a positive integer constant}\right\}$$

What is the minimum number of states needed in a DFA to recognize $L$?

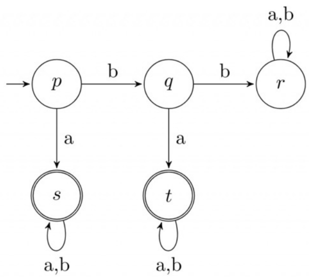
A. $k+1$ B. $n+1$ C. $2^{n+1}$ D. $2^{k+1}$

gatecse-2011 theory-of-computation finite-automata normal minimal-state-automata

Answer key

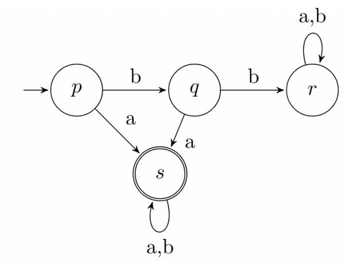
## 6.11.15 Minimal State Automata: GATE CSE 2011 | Question: 45

A deterministic finite automaton $(\mathrm{DFA})$ $D$ with alphabet $\Sigma = \{a,b\}$ is given below.

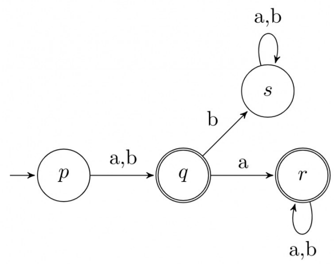



Which of the following finite state machines is a valid minimal $\mathrm{DFA}$ which accepts the same languages as $D$?

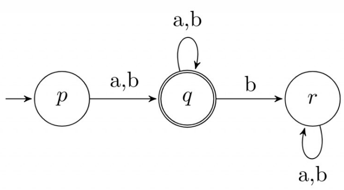
A.



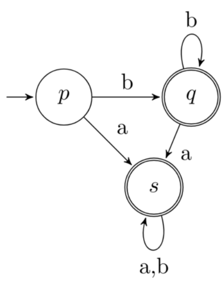

B.




C.



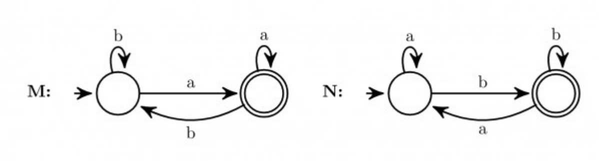

D.



gatecse-2011 theory-of-computation finite-automata easy minimal-state-automata

Answer key

## 6.11.16 Minimal State Automata: GATE CSE 2015 | Set 1 | Question: 52




Consider the DFAs $M$ and $N$ given above. The number of states in a minimal DFA that accept the language $L(M) \cap L(N)$ is____________.

gatecse-2015-set1 theory-of-computation finite-automata easy numerical-answers minimal-state-automata

Answer key

## 6.11.17 Minimal State Automata: GATE CSE 2015 | Set 2 | Question: 53

The number of states in the minimal deterministic finite automaton corresponding to the regular expression $(0+1)^*(10)$ is _____.

gatecse-2015-set2 theory-of-computation finite-automata normal numerical-answers minimal-state-automata

Answer key

## 6.11.18 Minimal State Automata: GATE CSE 2015 | Set 3 | Question: 18

Let $L$ be the language represented by the regular expression $\Sigma^*0011\Sigma^*$ where $\Sigma = \{0, 1\}$. What is the minimum number of states in a DFA that recognizes $\bar{L}$ (complement of $L$)?

A. $4$
B. $5$
C. $6$
D. $8$

gatecse-2015-set3 theory-of-computation finite-automata normal minimal-state-automata

Answer key

## 6.11.19 Minimal State Automata: GATE CSE 2016 | Set 2 | Question: 16

The number of states in the minimum sized DFA that accepts the language defined by the regular expression.

$$(0+1)^*(0+1)(0+1)^*$$

is ________.

gatecse-2016-set2 theory-of-computation finite-automata normal numerical-answers minimal-state-automata

Answer key

## 6.11.20 Minimal State Automata: GATE CSE 2017 | Set 1 | Question: 22

Consider the language $L$ given by the regular expression $(a+b)^*b(a+b)$ over the alphabet $\{a, b\}$. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting $L$ is ___________ .

gatecse-2017-set1 theory-of-computation finite-automata numerical-answers minimal-state-automata

Answer key

## 6.11.21 Minimal State Automata: GATE CSE 2017 | Set 2 | Question: 25

The minimum possible number of states of a deterministic finite automaton that accepts the regular language $L$ = $\{w_1 a w_2 \mid w_1, w_2 \in \{a, b\}^*, |w_1| = 2, |w_2| \geq 3\}$ is ______________ .

theory-of-computation gatecse-2017-set2 finite-automata numerical-answers minimal-state-automata

Answer key

### 6.11.22 Minimal State Automata: GATE CSE 2018 | Question: 6

Let $N$ be an NFA with $n$ states. Let $k$ be the number of states of a minimal DFA which is equivalent to $N$. Which one of the following is necessarily true?

A. $k \geq 2^n$
B. $k \geq n$
C. $k \leq n^2$
D. $k \leq 2^n$

gatecse-2018 theory-of-computation minimal-state-automata normal one-mark

Answer key☟

### 6.11.23 Minimal State Automata: GATE CSE 2019 | Question: 48

Let $\Sigma$ be the set of all bijections from $\{1, \dots, 5\}$ to $\{1, \dots, 5\}$, where $id$ denotes the identity function, i.e. $id(j) = j, \forall j$. Let $\circ$ denote composition on functions. For a string $x = x_1 x_2 \dots x_n \in \Sigma^n, n \geq 0$, let $\pi(x) = x_1 \circ x_2 \circ \dots \circ x_n$. Consider the language $L = \{x \in \Sigma^* \mid \pi(x) = id\}$. The minimum number of states in any DFA accepting $L$ is _______

gatecse-2019 numerical-answers theory-of-computation finite-automata minimal-state-automata difficult two-marks

Answer key☟

### 6.11.24 Minimal State Automata: GATE CSE 2023 | Question: 53

Consider the language $L$ over the alphabet $\{0, 1\}$, given below:

$$L = \{w \in \{0,1\}^* \mid w \text{ does not contain three or more consecutive } 1\text{ 's}\}.$$

The minimum number of states in a Deterministic Finite-State Automaton $(\mathrm{DFA})$ for $L$ is ___________.

gatecse-2023 theory-of-computation minimal-state-automata numerical-answers two-marks

Answer key☟

### 6.11.25 Minimal State Automata: GATE IT 2008 | Question: 6

Let $N$ be an NFA with $n$ states and let $M$ be the minimized DFA with m states recognizing the same language. Which of the following in NECESSARILY true?

A. $m \leq 2^n$
B. $n \leq m$
C. $M$ has one accept state
D. $m = 2^n$

gateit-2008 theory-of-computation finite-automata normal minimal-state-automata

Answer key☟

## 6.12 Non Determinism (6)

✍ **Practice Test:** Test 1 (6Q)

### 6.12.1 Non Determinism: GATE CSE 1992 | Question: 02,xx

In which of the cases stated below is the following statement true?
"For every non-deterministic machine $M_1$ there exists an equivalent deterministic machine $M_2$ recognizing the same language".

A. $M_1$ is non-deterministic finite automaton.
B. $M_1$ is non-deterministic PDA.
C. $M_1$ is a non-deterministic Turing machine.
D. For no machines $M_1$ and $M_2$, the above statement true.

gate1992 theory-of-computation easy non-determinism multiple-selects

Answer key☟

### 6.12.2 Non Determinism: GATE CSE 1994 | Question: 1.16

Which of the following conversions is not possible (algorithmically)?

A. Regular grammar to context free grammar
B. Non-deterministic FSA to deterministic FSA
C. Non-deterministic PDA to deterministic PDA
D. Non-deterministic Turing machine to deterministic Turing machine

gate1994 theory-of-computation easy non-determinism

Answer key

### 6.12.3 Non Determinism: GATE CSE 1998 | Question: 1.11

Regarding the power of recognition of languages, which of the following statements is false?

A. The non-deterministic finite-state automata are equivalent to deterministic finite-state automata.
B. Non-deterministic Push-down automata are equivalent to deterministic Push-down automata.
C. Non-deterministic Turing machines are equivalent to deterministic Turing machines.
D. Multi-tape Turing machines are available are equivalent to Single-tape Turing machines.

gate1998 theory-of-computation easy non-determinism

Answer key

### 6.12.4 Non Determinism: GATE CSE 2005 | Question: 54

Let $N_f$ and $N_p$ denote the classes of languages accepted by non-deterministic finite automata and non-deterministic push-down automata, respectively. Let $D_f$ and $D_p$ denote the classes of languages accepted by deterministic finite automata and deterministic push-down automata respectively. Which one of the following is TRUE?

A. $D_f \subset N_f \text{ and } D_p \subset N_p$
B. $D_f \subset N_f \text{ and } D_p = N_p$
C. $D_f = N_f \text{ and } D_p = N_p$
D. $D_f = N_f \text{ and } D_p \subset N_p$

gatecse-2005 theory-of-computation easy non-determinism

Answer key

### 6.12.5 Non Determinism: GATE CSE 2011 | Question: 8

Which of the following pairs have **DIFFERENT** expressive power?

A. Deterministic finite automata (DFA) and Non-deterministic finite automata (NFA)
B. Deterministic push down automata (DPDA) and Non-deterministic push down automata (NPDA)
C. Deterministic single tape Turing machine and Non-deterministic single tape Turing machine
D. Single tape Turing machine and multi-tape Turing machine

gatecse-2011 theory-of-computation easy non-determinism

Answer key

### 6.12.6 Non Determinism: GATE IT 2004 | Question: 9

Which one of the following statements is FALSE?

A. There exist context-free languages such that all the context-free grammars generating them are ambiguous
B. An unambiguous context-free grammar always has a unique parse tree for each string of the language generated by it
C. Both deterministic and non-deterministic pushdown automata always accept the same set of languages
D. A finite set of string from some alphabet is always a regular language

gateit-2004 theory-of-computation easy non-determinism

Answer key

## 6.13 Number of States (2)

### 6.13.1 Number of States: GATE CSE 2025 | Set 2 | Question: 50

Let $\Sigma = \{1, 2, 3, 4\}$. For $x \in \Sigma^*$, let $\text{prod}(x)$ be the product of symbols in $x$ modulo 7. We take $\text{prod}(\epsilon) = 1$, where $\epsilon$ is the null string.

For example, $\text{prod}(124) = (1 \times 2 \times 4) \bmod 7 = 1$.

Define $L = \{x \in \Sigma^* \mid \text{prod}(x) = 2\}$.

The number of states in a minimum state DFA for $L$ is __________ . (Answer in integer)

gatecse2025-set2 theory-of-computation finite-automata number-of-states numerical-answers two-marks

Answer key

### 6.13.2 Number of States: GATE CSE 2026 | Set 1 | Question: 16

Let $M$ be a nondeterministic finite automaton (NFA) with $6$ states over a finite alphabet.

Which of the following options CANNOT be the number of states in the minimal deterministic finite automaton (DFA) that is equivalent to $M$ ?

A. $32$
B. $65$
C. $1$
D. $128$

gatecse-2026-set1 theory-of-computation finite-automata number-of-states multiple-selects one-mark

Answer key

## 6.14 Pumping Lemma (2)

### 6.14.1 Pumping Lemma: GATE CSE 2019 | Question: 15

For $\Sigma = \{a, b\}$, let us consider the regular language $L = \{x \mid x = a^{2+3k} \text{ or } x = b^{10+12k}, k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for $L$ ?

A. $3$
B. $5$
C. $9$
D. $24$

gatecse-2019 theory-of-computation pumping-lemma one-mark

Answer key

### 6.14.2 Pumping Lemma: GATE IT 2005 | Question: 40

A language $L$ satisfies the Pumping Lemma for regular languages, and also the Pumping Lemma for context-free languages. Which of the following statements about $L$ is TRUE?

A. $L$ is necessarily a regular language.
B. $L$ is necessarily a context-free language, but not necessarily a regular language.
C. $L$ is necessarily a non-regular language.
D. None of the above

gateit-2005 theory-of-computation pumping-lemma easy

Answer key

## 6.15 Pushdown Automata (15)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (4Q)

### 6.15.1 Pushdown Automata: GATE CSE 1996 | Question: 13

Let $Q = (\{q_1, q_2\}, \{a, b\}, \{a, b, \perp\}, \delta, \perp, \phi)$ be a pushdown automaton accepting by empty stack for the language which is the set of all nonempty even palindromes over the set $\{a, b\}$. Below is an incomplete specification of the transitions $\delta$. Complete the specification. The top of the stack is assumed to be at the right end of the string representing stack contents.

1. $\delta(q_1, a, \perp) = \{(q_1, \perp a)\}$
2. $\delta(q_1, b, \perp) = \{(q_1, \perp b)\}$
3. $\delta(q_1, a, a) = \{(q_1, aa)\}$
4. $\delta(q_1, b, a) = \{(q_1, ab)\}$
5. $\delta(q_1, a, b) = \{(q_1, ba)\}$
6. $\delta(q_1, b, b) = \{(q_1, bb)\}$
7. $\delta(q_1, a, a) = \{(\dots, \dots)\}$
8. $\delta(q_1, b, b) = \{(\dots, \dots)\}$
9. $\delta(q_2, a, a) = \{(q_2, \epsilon)\}$
10. $\delta(q_2, b, b) = \{(q_2, \epsilon)\}$
11. $\delta(q_2, \epsilon, \perp) = \{(q_2, \epsilon)\}$

gate1996 theory-of-computation pushdown-automata normal descriptive

Answer key

### 6.15.2 Pushdown Automata: GATE CSE 1997 | Question: 6.6

Which of the following languages over $\{a, b, c\}$ is accepted by a deterministic pushdown automata?

A. $\{wcw^R \mid w \in \{a,b\}^*\}$
B. $\{ww^R \mid w \in \{a,b,c\}^*\}$
C. $\{a^n b^n c^n \mid n \geq 0\}$
D. $\{w \mid w \text{ is a palindrome over } \{a,b,c\}\}$

Note: $w^R$ is the string obtained by reversing $'w'$.

gate1997 theory-of-computation pushdown-automata easy

Answer key

### 6.15.3 Pushdown Automata: GATE CSE 1998 | Question: 13

Let $M = (\{q_0, q_1\}, \{0, 1\}, \{z_0, X\}, \delta, q_0, z_0, \phi)$ be a Pushdown automation where $\delta$ is given by

$\delta(q_0, 1, z_0) = \{(q_0, Xz_0)\}$

$\delta(q_0, \epsilon, z_0) = \{(q_0, \epsilon)\}$

$\delta(q_0, 1, X) = \{(q_0, XX)\}$

$\delta(q_1, 1, X) = \{(q_1, \epsilon)\}$

$\delta(q_0, 0, X) = \{(q_1, X)\}$

$\delta(q_0, 0, z_0) = \{(q_0, z_0)\}$

a. What is the language accepted by this PDA by empty stack?
b. Describe informally the working of the PDA

gate1998 theory-of-computation pushdown-automata descriptive

Answer key

### 6.15.4 Pushdown Automata: GATE CSE 1999 | Question: 1.6

Let $L_1$ be the set of all languages accepted by a PDA by final state and $L_2$ the set of all languages accepted by empty stack. Which of the following is true?

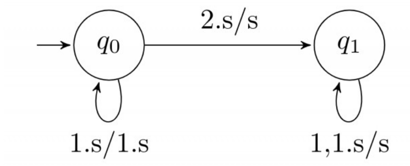
A. $L_1 = L_2$
B. $L_1 \supset L_2$
C. $L_1 \subset L_2$
D. None

normal theory-of-computation gate1999 pushdown-automata

Answer key

### 6.15.5 Pushdown Automata: GATE CSE 2000 | Question: 8

A push down automation (pda) is given in the following extended notation of finite state diagram:




The nodes denote the states while the edges denote the moves of the pda. The edge labels are of the form $d,\ s/s'$ where $d$ is the input symbol read and $s, s'$ are the stack contents before and after the move. For example the edge labeled $1, s/1.s$ denotes the move from state $q_0$ to $q_0$ in which the input symbol $1$ is read and pushed to the stack.

A. Introduce two edges with appropriate labels in the above diagram so that the resulting pda accepts the language $\{x2x^R \mid x \in \{0,1\}^*, x^R \text{ denotes reverse of x}\}$, by empty stack.
B. Describe a non-deterministic pda with three states in the above notation that accept the language $\{0^n1^m \mid n \leq m \leq 2n\}$ by empty stack

gatecse-2000 theory-of-computation descriptive pushdown-automata

Answer key

### 6.15.6 Pushdown Automata: GATE CSE 2001 | Question: 6

Give a deterministic PDA for the language $L = \{a^ncb^{2n} \mid n \geq 1\}$ over the alphabet $\Sigma = \{a, b, c\}$. Specify the acceptance state.

gatecse-2001 theory-of-computation normal pushdown-automata descriptive

Answer key

### 6.15.7 Pushdown Automata: GATE CSE 2009 | Question: 16, ISRO2017-12

Which one of the following is FALSE?

A. There is a unique minimal DFA for every regular language
B. Every NFA can be converted to an equivalent PDA.
C. Complement of every context-free language is recursive.
D. Every nondeterministic PDA can be converted to an equivalent deterministic PDA.

gatecse-2009 theory-of-computation easy isro2017 pushdown-automata

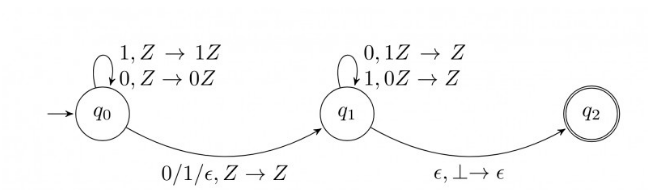
Answer key

### 6.15.8 Pushdown Automata: GATE CSE 2015 | Set 1 | Question: 51

Consider the NPDA

$$\langle Q = \{q_0, q_1, q_2\}, \Sigma = \{0, 1\}, \Gamma = \{0, 1, \perp\}, \delta, q_0, \perp, F = \{q_2\}\rangle$$

, where (as per usual convention) $Q$ is the set of states, $\Sigma$ is the input alphabet, $\Gamma$ is the stack alphabet, $\delta$ is the state transition function $q_0$ is the initial state, $\perp$ is the initial stack symbol, and $F$ is the set of accepting states. The state transition is as follows:




Which one of the following sequences must follow the string $101100$ so that the overall string is accepted by the automaton?

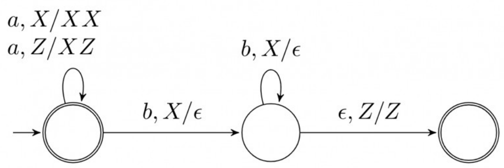
A. 10110 B. 10010 C. 01010 D. 01001

gatecse-2015-set1 theory-of-computation pushdown-automata normal

Answer key

## 6.15.9 Pushdown Automata: GATE CSE 2016 | Set 1 | Question: 43

Consider the transition diagram of a PDA given below with input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{X, Z\}$. $Z$ is the initial stack symbol. Let $L$ denote the language accepted by the PDA




Which one of the following is **TRUE**?

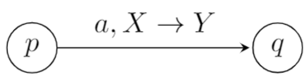
A. $L = \{a^n b^n \mid n \geq 0\}$ and is not accepted by any finite automata
B. $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and is not accepted by any deterministic PDA
C. $L$ is not accepted by any Turing machine that halts on every input
D. $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and is deterministic context-free

gatecse-2016-set1 theory-of-computation pushdown-automata normal

Answer key

## 6.15.10 Pushdown Automata: GATE CSE 2021 | Set 1 | Question: 51

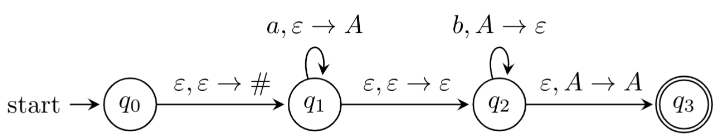
In a pushdown automaton $P = (Q, \Sigma, \Gamma, \delta, q_0, F)$, a transition of the form,




where $p, q \in Q$, $a \in \Sigma \cup \{\epsilon\}$, and $X, Y \in \Gamma \cup \{\epsilon\}$, represents

$$(q, Y) \in \delta(p, a, X).$$

Consider the following pushdown automaton over the input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{\#, A\}$.



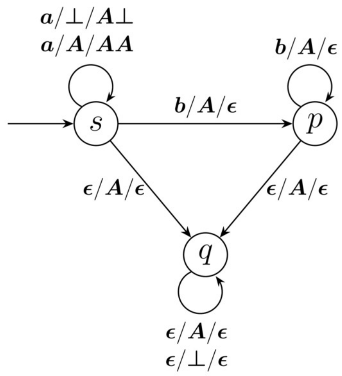

The number of strings of length $100$ accepted by the above pushdown automaton is ___________

gatecse-2021-set1 theory-of-computation pushdown-automata numerical-answers two-marks

Answer key

## 6.15.11 Pushdown Automata: GATE CSE 2023 | Question: 30

Consider the pushdown automaton $(\mathrm{PDA})$ $P$ below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{\perp, A\}$, and has three states $\{s, p, q\}$, with $s$ being the start state. A transition from state $u$ to state $v$, labelled $c/X/\gamma$, where $c$ is an input symbol or $\epsilon$, $X$ is a stack symbol, and $\gamma$ is a string of stack symbols, represents the fact that in state $u$, the $(\mathrm{PDA})$ can read $c$ from the input, with $X$ on the top of its stack, pop $X$ from the stack, push in the string $\gamma$ on the stack, and go to state $v$. In the initial configuration, the stack has only the symbol $\perp$ in it. The $(\mathrm{PDA})$ accepts by empty stack.




Which one of the following options correctly describes the language accepted by $P$?

A. $\{a^m b^n \mid 1 \leq m$ and $n < m\}$
B. $\{a^m b^n \mid 0 \leq n \leq m\}$
C. $\{a^m b^n \mid 0 \leq m$ and $0 \leq n\}$
D. $\{a^m \mid 0 \leq m\} \cup \{b^n \mid 0 \leq n\}$

gatecse-2023 theory-of-computation pushdown-automata two-marks

Answer key☟

## 6.15.12 Pushdown Automata: GATE IT 2004 | Question: 40

Let $M = (K, \Sigma, \Gamma, \Delta, s, F)$ be a pushdown automaton, where
$K = (s, f), F = \{f\}, \Sigma = \{a, b\}, \Gamma = \{a\}$ and
$\Delta = \{((s, a, \epsilon), (s, a)), ((s, b, \epsilon), (s, a)), ((s, a, a), (f, \epsilon)), ((f, a, a), (f, \epsilon)), ((f, b, a), (f, \epsilon))\}$.
Which one of the following strings is not a member of $L(M)$?

A. $aaa$ B. $aabab$ C. $baaba$ D. $bab$

gateit-2004 theory-of-computation pushdown-automata normal

Answer key☟

## 6.15.13 Pushdown Automata: GATE IT 2005 | Question: 38

Let $P$ be a non-deterministic push-down automaton (NPDA) with exactly one state, $q$, and exactly one symbol, $Z$, in its stack alphabet. State $q$ is both the starting as well as the accepting state of the PDA. The stack is initialized with one $Z$ before the start of the operation of the PDA. Let the input alphabet of the PDA be $\Sigma$. Let $L(P)$ be the language accepted by the PDA by reading a string and reaching its accepting state. Let $N(P)$ be the language accepted by the PDA by reading a string and emptying its stack.
Which of the following statements is TRUE?

A. $L(P)$ is necessarily $\Sigma^*$ but $N(P)$ is not necessarily $\Sigma^*$.
B. $N(P)$ is necessarily $\Sigma^*$ but $L(P)$ is not necessarily $\Sigma^*$.
C. Both $L(P)$ and $N(P)$ are necessarily $\Sigma^*$.
D. Neither $L(P)$ nor $N(P)$ are necessarily $\Sigma^*$

gateit-2005 theory-of-computation pushdown-automata normal

Answer key☟

## 6.15.14 Pushdown Automata: GATE IT 2006 | Question: 31

Which of the following languages is accepted by a non-deterministic pushdown automaton (PDA) but NOT by a deterministic PDA?

A. $\{a^n b^n c^n \mid n \geq 0\}$
B. $\{a^l b^m c^n \mid l \neq m \text{ or } m \neq n\}$
C. $\{a^n b^n \mid n \geq 0\}$
D. $\{a^m b^n \mid m, n \geq 0\}$

gateit-2006 theory-of-computation pushdown-automata normal

Answer key

### 6.15.15 Pushdown Automata: GATE IT 2006 | Question: 33

Consider the pushdown automaton (PDA) below which runs over the input alphabet $(a, b, c)$. It has the stack alphabet $\{Z_0, X\}$ where $Z_0$ is the bottom-of-stack marker. The set of states of the PDA is $(s, t, u, f)$ where $s$ is the start state and $f$ is the final state. The PDA accepts by final state. The transitions of the PDA given below are depicted in a standard manner. For example, the transition $(s, b, X) \rightarrow (t, XZ_0)$ means that if the PDA is in state $s$ and the symbol on the top of the stack is $X$, then it can read b from the input and move to state $t$ after popping the top of stack and pushing the symbols $Z_0$ and $X$ (in that order) on the stack.

$(s, a, Z_0) \rightarrow (s, XXZ_0)$
$(s, \epsilon, Z_0) \rightarrow (f, \epsilon)$
$(s, a, X) \rightarrow (s, XXX)$
$(s, b, X) \rightarrow (t, \epsilon)$
$(t, b, X) \rightarrow (t, \epsilon)$
$(t, c, X) \rightarrow (u, \epsilon)$
$(u, c, X) \rightarrow (u, \epsilon)$
$(u, \epsilon, Z_0) \rightarrow (f, \epsilon)$

The language accepted by the PDA is

A. $\{a^l b^m c^n \mid l = m = n\}$
B. $\{a^l b^m c^n \mid l = m\}$
C. $\{a^l b^m c^n \mid 2l = m + n\}$
D. $\{a^l b^m c^n \mid m = n\}$

gateit-2006 theory-of-computation pushdown-automata normal

Answer key

## 6.16 Recursive and Recursively Enumerable Languages (16)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (11Q)

### 6.16.1 Recursive and Recursively Enumerable Languages: GATE CSE 1990 | Question: 3-vi

Recursive languages are:

A. A proper superset of context free languages.
B. Always recognizable by pushdown automata.
C. Also called type $0$ languages.
D. Recognizable by Turing machines.

gate1990 normal theory-of-computation turing-machine recursive-and-recursively-enumerable-languages multiple-selects

Answer key

### 6.16.2 Recursive and Recursively Enumerable Languages: GATE CSE 2003 | Question: 13

Nobody knows yet if $P = NP$. Consider the language $L$ defined as follows.

$$L = \begin{cases} (0+1)^* & \text{if } P = NP \\ \phi & otherwise \end{cases}$$

Which of the following statements is true?

A. $L$ is recursive
B. $L$ is recursively enumerable but not recursive
C. $L$ is not recursively enumerable
D. Whether $L$ is recursively enumerable or not will be known after we find out if $P = NP$

gatecse-2003 theory-of-computation normal recursive-and-recursively-enumerable-languages

Answer key

### 6.16.3 Recursive and Recursively Enumerable Languages: GATE CSE 2003 | Question: 15

If the strings of a language $L$ can be effectively enumerated in lexicographic (i.e., alphabetic) order, which of the following statements is true?

A. $L$ is necessarily finite
B. $L$ is regular but not necessarily finite
C. $L$ is context free but not necessarily regular
D. $L$ is recursive but not necessarily context-free

theory-of-computation gatecse-2003 normal recursive-and-recursively-enumerable-languages

Answer key

### 6.16.4 Recursive and Recursively Enumerable Languages: GATE CSE 2003 | Question: 54

Define languages $L_0$ and $L_1$ as follows :

- $L_0 = \{\langle M, w, 0 \rangle \mid M \text{ halts on } w\}$
- $L_1 = \{\langle M, w, 1 \rangle \mid M \text{ does not halts on } w\}$

Here $\langle M, w, i \rangle$ is a triplet, whose first component $M$ is an encoding of a Turing Machine, second component $w$ is a string, and third component $i$ is a bit.

Let $L = L_0 \cup L_1$. Which of the following is true?

A. $L$ is recursively enumerable, but $L'$ is not
B. $L'$ is recursively enumerable, but $L$ is not
C. Both $L$ and $L'$ are recursive
D. Neither $L$ nor $L'$ is recursively enumerable

theory-of-computation recursive-and-recursively-enumerable-languages gatecse-2003 difficult

Answer key

### 6.16.5 Recursive and Recursively Enumerable Languages: GATE CSE 2004 | Question: 89

$L_1$ is a recursively enumerable language over $\Sigma$. An algorithm $A$ effectively enumerates its words as $\omega_1, \omega_2, \omega_3, \dots$. Define another language $L_2$ over $\Sigma \cup \{\#\}$ as $\{w_i \# w_j \mid w_i, w_j \in L_1, i < j\}$. Here # is new symbol. Consider the following assertions.

- $S_1$ : $L_1$ is recursive implies $L_2$ is recursive
- $S_2$ : $L_2$ is recursive implies $L_1$ is recursive

Which of the following statements is true?

A. Both $S_1$ and $S_2$ are true
B. $S_1$ is true but $S_2$ is not necessarily true
C. $S_2$ is true but $S_1$ is not necessarily true
D. Neither is necessarily true

gatecse-2004 theory-of-computation recursive-and-recursively-enumerable-languages difficult

Answer key

### 6.16.6 Recursive and Recursively Enumerable Languages: GATE CSE 2005 | Question: 56

Let $L_1$ be a recursive language, and let $L_2$ be a recursively enumerable but not a recursive language. Which one of the following is TRUE?

A. $L_1$' is recursive and $L_2$' is recursively enumerable
B. $L_1$' is recursive and $L_2$' is not recursively enumerable
C. $L_1$' and $L_2$' are recursively enumerable
D. $L_1$' is recursively enumerable and $L_2$' is recursive

gatecse-2005 theory-of-computation recursive-and-recursively-enumerable-languages easy

Answer key

## 6.16.7 Recursive and Recursively Enumerable Languages: GATE CSE 2008 | Question: 13, ISRO2016-36

If $L$ and $\overline{L}$ are recursively enumerable then $L$ is

A. regular
B. context-free
C. context-sensitive
D. recursive

gatecse-2008 theory-of-computation easy isro2016 recursive-and-recursively-enumerable-languages

Answer key

## 6.16.8 Recursive and Recursively Enumerable Languages: GATE CSE 2008 | Question: 48

Which of the following statements is false?

A. Every NFA can be converted to an equivalent DFA
B. Every non-deterministic Turing machine can be converted to an equivalent deterministic Turing machine
C. Every regular language is also a context-free language
D. Every subset of a recursively enumerable set is recursive

gatecse-2008 theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key

## 6.16.9 Recursive and Recursively Enumerable Languages: GATE CSE 2010 | Question: 17

Let $L_1$ be the recursive language. Let $L_2$ and $L_3$ be languages that are recursively enumerable but not recursive. Which of the following statements is not necessarily true?

A. $L_2 - L_1$ is recursively enumerable.
B. $L_1 - L_3$ is recursively enumerable.
C. $L_2 \cap L_3$ is recursively enumerable.
D. $L_2 \cup L_3$ is recursively enumerable.

gatecse-2010 theory-of-computation recursive-and-recursively-enumerable-languages decidability normal

Answer key

## 6.16.10 Recursive and Recursively Enumerable Languages: GATE CSE 2014 | Set 1 | Question: 35

Let $L$ be a language and $\bar{L}$ be its complement. Which one of the following is **NOT** a viable possibility?

A. Neither $L$ nor $\bar{L}$ is recursively enumerable $(r.e.)$.
B. One of $L$ and $\bar{L}$ is r.e. but not recursive; the other is not r.e.
C. Both $L$ and $\bar{L}$ are r.e. but not recursive.
D. Both $L$ and $\bar{L}$ are recursive.

gatecse-2014-set1 theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key

## 6.16.11 Recursive and Recursively Enumerable Languages: GATE CSE 2014 | Set 2 | Question: 16

Let $A \leq_m B$ denotes that language $A$ is mapping reducible (also known as many-to-one reducible) to language $B$. Which one of the following is FALSE?

A. If $A \leq_m B$ and $B$ is recursive then $A$ is recursive.
B. If $A \leq_m B$ and $A$ is undecidable then $B$ is undecidable.
C. If $A \leq_m B$ and $B$ is recursively enumerable then $A$ is recursively enumerable.
D. If $A \leq_m B$ and $B$ is not recursively enumerable then $A$ is not recursively enumerable.

gatecse-2014-set2 theory-of-computation recursive-and-recursively-enumerable-languages normal

Answer key

### 6.16.12 Recursive and Recursively Enumerable Languages: GATE CSE 2014 | Set 2 | Question: 35

Let $\langle M \rangle$ be the encoding of a Turing machine as a string over $\Sigma = \{0, 1\}$. Let

$$L = \{\langle M \rangle \mid M \text{ is a Turing machine} \\ \text{that accepts a string of length } 2014\}.$$

Then $L$ is:

A. decidable and recursively enumerable
B. undecidable but recursively enumerable
C. undecidable and not recursively enumerable
D. decidable but not recursively enumerable

gatecse-2014-set2 theory-of-computation recursive-and-recursively-enumerable-languages normal

Answer key

### 6.16.13 Recursive and Recursively Enumerable Languages: GATE CSE 2015 | Set 1 | Question: 3

For any two languages $L_1$ and $L_2$ such that $L_1$ is context-free and $L_2$ is recursively enumerable but not recursive, which of the following is/are necessarily true?

I. $\bar{L}_1$ ( Complement of $L_1$) is recursive
II. $\bar{L}_2$ ( Complement of $L_2$) is recursive
III. $\bar{L}_1$ is context-free
IV. $\bar{L}_1 \cup L_2$ is recursively enumerable

A. I only
B. III only
C. III and IV only
D. I and IV only

gatecse-2015-set1 theory-of-computation recursive-and-recursively-enumerable-languages normal

Answer key

### 6.16.14 Recursive and Recursively Enumerable Languages: GATE CSE 2016 | Set 2 | Question: 44

Consider the following languages.

- $L_1 = \{\langle M \rangle \mid M \text{ takes at least 2016 steps on some input}\}$,
- $L_2 = \{\langle M \rangle \mid M \text{ takes at least 2016 steps on all inputs}\}$ and
- $L_3 = \{\langle M \rangle \mid M \text{ accepts } \epsilon\}$,

where for each Turing machine $M$, $\langle M \rangle$ denotes a specific encoding of $M$. Which one of the following is TRUE?

A. $L_1$ is recursive and $L_2, L_3$ are not recursive
B. $L_2$ is recursive and $L_1, L_3$ are not recursive
C. $L_1, L_2$ are recursive and $L_3$ is not recursive
D. $L_1, L_2, L_3$ are recursive

gatecse-2016-set2 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

### 6.16.15 Recursive and Recursively Enumerable Languages: GATE CSE 2021 | Set 1 | Question: 12

Let $\langle M \rangle$ denote an encoding of an automaton $M$. Suppose that $\Sigma = \{0, 1\}$. Which of the following languages is/are $\text{NOT}$ recursive?

A. $L = \{\langle M \rangle \mid M \text{ is a DFA such that } L(M) = \emptyset\}$
B. $L = \{\langle M \rangle \mid M \text{ is a DFA such that } L(M) = \Sigma^*\}$
C. $L = \{\langle M \rangle \mid M \text{ is a PDA such that } L(M) = \emptyset\}$
D. $L = \{\langle M \rangle \mid M \text{ is a PDA such that } L(M) = \Sigma^*\}$

gatecse-2021-set1 multiple-selects theory-of-computation recursive-and-recursively-enumerable-languages one-mark

Answer key

### 6.16.16 Recursive and Recursively Enumerable Languages: GATE CSE 2021 | Set 1 | Question: 39

For a Turing machine $M$, $\langle M \rangle$ denotes an encoding of $M$. Consider the following two languages.

$$L_1 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on all inputs}\}$$
$$L_2 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on some input}\}$$

Which one of the following options is correct?

A. Both $L_1$ and $L_2$ are decidable
B. $L_1$ is decidable and $L_2$ is undecidable
C. $L_1$ is undecidable and $L_2$ is decidable
D. Both $L_1$ and $L_2$ are undecidable

gatecse-2021-set1 theory-of-computation recursive-and-recursively-enumerable-languages decidability easy two-marks

Answer key

## 6.17 Reduction (2)

### 6.17.1 Reduction: GATE CSE 2005 | Question: 45

Consider three decision problems $P_1$, $P_2$ and $P_3$. It is known that $P_1$ is decidable and $P_2$ is undecidable. Which one of the following is TRUE?

A. $P_3$ is decidable if $P_1$ is reducible to $P_3$
B. $P_3$ is undecidable if $P_3$ is reducible to $P_2$
C. $P_3$ is undecidable if $P_2$ is reducible to $P_3$
D. $P_3$ is decidable if $P_3$ is reducible to $P_2$'s complement

gatecse-2005 theory-of-computation decidability normal reduction

Answer key

### 6.17.2 Reduction: GATE CSE 2016 | Set 1 | Question: 44

Let $X$ be a recursive language and $Y$ be a recursively enumerable but not recursive language. Let $W$ and $Z$ be two languages such that $\overline{Y}$ reduces to $W$, and $Z$ reduces to $\overline{X}$ (reduction means the standard many-one reduction). Which one of the following statements is TRUE?

A. $W$ can be recursively enumerable and $Z$ is recursive.
B. $W$ can be recursive and $Z$ is recursively enumerable.
C. $W$ is not recursively enumerable and $Z$ is recursive.
D. $W$ is not recursively enumerable and $Z$ is not recursive.

gatecse-2016-set1 theory-of-computation easy recursive-and-recursively-enumerable-languages reduction

Answer key

## 6.18 Regular Expression (29)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (7Q)

### 6.18.1 Regular Expression: GATE CSE 1987 | Question: 10d

Give a regular expression over the alphabet $\{0, 1\}$ to denote the set of proper non-null substrings of the string $0110$.

gate1987 theory-of-computation regular-expression descriptive

Answer key

## 6.18.2 Regular Expression: GATE CSE 1991 | Question: 03,xiii

Let $r = 1(1+0)^*, s = 11^*0 \text{ and } t = 1^*0$ be three regular expressions. Which one of the following is true?

A. $L(s) \subseteq L(r)$ and $L(s) \subseteq L(t)$
B. $L(r) \subseteq L(s)$ and $L(s) \subseteq L(t)$
C. $L(s) \subseteq L(t)$ and $L(s) \subseteq L(r)$
D. $L(t) \subseteq L(s)$ and $L(s) \subseteq L(r)$
E. None of the above

gate1991 theory-of-computation regular-expression normal multiple-selects

Answer key

## 6.18.3 Regular Expression: GATE CSE 1992 | Question: 02,xvii

Which of the following regular expression identities is/are TRUE?

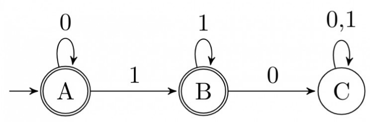
A. $r^{(*)} = r^*$
B. $(r^*s^*) = (r+s)^*$
C. $(r+s)^* = r^* + s^*$
D. $r^*s^* = r^* + s^*$

gate1992 theory-of-computation regular-expression easy

Answer key

## 6.18.4 Regular Expression: GATE CSE 1994 | Question: 2.10

The regular expression for the language recognized by the finite state automaton of figure is ________




gate1994 theory-of-computation finite-automata regular-expression easy fill-in-the-blanks

Answer key

## 6.18.5 Regular Expression: GATE CSE 1995 | Question: 1.9 , ISRO2017-13

In some programming language, an identifier is permitted to be a letter followed by any number of letters or digits. If $L$ and $D$ denote the sets of letters and digits respectively, which of the following expressions defines an identifier?

A. $(L+D)^+$
B. $(L.D)^*$
C. $L(L+D)^*$
D. $L(L.D)^*$

gate1995 theory-of-computation regular-expression easy isro2017

Answer key

## 6.18.6 Regular Expression: GATE CSE 1996 | Question: 1.8

Which two of the following four regular expressions are equivalent? ($\varepsilon$ is the empty string).

i. $(00)^*(\varepsilon + 0)$
ii. $(00)^*$
iii. $0^*$
iv. $0(00)^*$

A. (i) and (ii)
B. (ii) and (iii)
C. (i) and (iii)
D. (iii) and (iv)

gate1996 theory-of-computation regular-expression easy

Answer key

## 6.18.7 Regular Expression: GATE CSE 1997 | Question: 6.4

Which one of the following regular expressions over $\{0,1\}$ denotes the set of all strings not containing $100$ as substring?

A. $0^*(1+0)^*$
B. $0^*1010^*$
C. $0^*1^*01^*$
D. $0^*(10+1)^*$

gate1997 theory-of-computation regular-expression normal

Answer key

### 6.18.8 Regular Expression: GATE CSE 1998 | Question: 1.12

The string $1101$ does not belong to the set represented by

A. $110^*(0+1)$
B. $1(0+1)^*101$
C. $(10)^*(01)^*(00+11)^*$
D. $(00+(11)^*0)^*$

gate1998 theory-of-computation regular-expression easy multiple-selects

Answer key

### 6.18.9 Regular Expression: GATE CSE 1998 | Question: 1.9

If the regular set $A$ is represented by $A=(01+1)^*$ and the regular set $B$ is represented by $B=((01)^*1^*)^*$, which of the following is true?

A. $A \subset B$
B. $B \subset A$
C. $A$ and $B$ are incomparable
D. $A=B$

gate1998 theory-of-computation regular-expression normal

Answer key

### 6.18.10 Regular Expression: GATE CSE 1998 | Question: 3b

Give a regular expression for the set of binary strings where every $0$ is immediately followed by exactly $k$ $1$'s and preceded by at least $k$ $1$'s ($k$ is a fixed integer)

gate1998 theory-of-computation regular-expression easy descriptive

Answer key

### 6.18.11 Regular Expression: GATE CSE 2000 | Question: 1.4

Let $S$ and $T$ be languages over $\Sigma=\{a,b\}$ represented by the regular expressions $(a+b^*)^*$ and $(a+b)^*$, respectively. Which of the following is true?

A. $S \subset T$
B. $T \subset S$
C. $S=T$
D. $S \cap T=\phi$

gatecse-2000 theory-of-computation regular-expression easy

Answer key

### 6.18.12 Regular Expression: GATE CSE 2003 | Question: 14

The regular expression $0^*(10^*)^*$ denotes the same set as

A. $(1^*0)^*1^*$
B. $0+(0+10)^*$
C. $(0+1)^*10(0+1)^*$
D. None of the above

gatecse-2003 theory-of-computation regular-expression easy

Answer key

### 6.18.13 Regular Expression: GATE CSE 2009 | Question: 15

Which one of the following languages over the alphabet $\{0,1\}$ is described by the regular expression: $(0+1)^*0(0+1)^*0(0+1)^*$?

A. The set of all strings containing the substring $00$
B. The set of all strings containing at most two $0$'s
C. The set of all strings containing at least two $0$'s
D. The set of all strings that begin and end with either $0$ or $1$

gatecse-2009 theory-of-computation regular-expression easy

Answer key

## 6.18.14 Regular Expression: GATE CSE 2010 | Question: 39

Let $L = \{w \in (0+1)^* \mid w \text{ has even number of } 1s\}$. i.e., $L$ is the set of all the bit strings with even numbers of $1$s. Which one of the regular expressions below represents $L$?

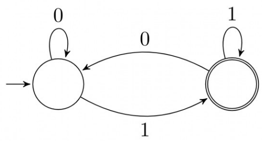
A. $(0^*10^*1)^*$
B. $0^*(10^*10^*)^*$
C. $0^*(10^*1)^*0^*$
D. $0^*1(10^*1)^*10^*$

gatecse-2010 theory-of-computation regular-expression normal

Answer key☟

## 6.18.15 Regular Expression: GATE CSE 2014 | Set 1 | Question: 36

Which of the regular expressions given below represent the following DFA?




I. $0^*1(1+00^*1)^*$
II. $0^*1^*1+11^*0^*1$
III. $(0+1)^*1$

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

gatecse-2014-set1 theory-of-computation regular-expression finite-automata easy

Answer key☟

## 6.18.16 Regular Expression: GATE CSE 2014 | Set 3 | Question: 15

The length of the shortest string NOT in the language (over $\Sigma = \{a, b\}$) of the following regular expression is _______.

$$a^*b^*(ba)^*a^*$$

gatecse-2014-set3 theory-of-computation regular-expression numerical-answers easy

Answer key☟

## 6.18.17 Regular Expression: GATE CSE 2016 | Set 1 | Question: 18

Which one of the following regular expressions represents the language: *the set of all binary strings having two consecutive $0$'s and two consecutive $1$'s*?

A. $(0+1)^*0011(0+1)^* + (0+1)^*1100(0+1)^*$
B. $(0+1)^*(00(0+1)^*11 + 11(0+1)^*00)(0+1)^*$
C. $(0+1)^*00(0+1)^* + (0+1)^*11(0+1)^*$
D. $00(0+1)^*11 + 11(0+1)^*00$

gatecse-2016-set1 theory-of-computation regular-expression normal

Answer key☟

## 6.18.18 Regular Expression: GATE CSE 2020 | Question: 7

Which one of the following regular expressions represents the set of all binary strings with an odd number of $1'$s?

A. $((0+1)^*1(0+1)^*1)^*10^*$
B. $(0^*10^*10^*)^*0^*1$
C. $10^*(0^*10^*10^*)^*$
D. $(0^*10^*10^*)^*10^*$

gatecse-2020 regular-expression normal theory-of-computation one-mark

Answer key☟

## 6.18.19 Regular Expression: GATE CSE 2021 | Set 2 | Question: 47

Which of the following regular expressions represent(s) the set of all binary numbers that are divisible by three? Assume that the string $\epsilon$ is divisible by three.

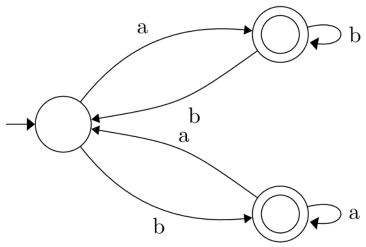
A. $(0+1(01^*0)^*1)^*$
B. $(0+11+10(1+00)^*01)^*$
C. $(0^*(1(01^*0)^*1)^*)^*$
D. $(0+11+11(1+00)^*00)^*$

gatecse-2021-set2 multiple-selects theory-of-computation regular-expression two-marks

Answer key

### 6.18.20 Regular Expression: GATE CSE 2022 | Question: 2

Which one of the following regular expressions correctly represents the language of the finite automaton given below?



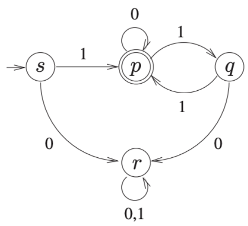

A. $ab^*bab^*+ba^*aba^*$
B. $(ab^*b)^*ab^*+(ba^*a)^*ba^*$
C. $(ab^*b+ba^*a)^*(a^*+b^*)$
D. $(ba^*a+ab^*b)^*(ab^*+ba^*)$

gatecse-2022 theory-of-computation finite-automata regular-expression one-mark

Answer key

### 6.18.21 Regular Expression: GATE CSE 2023 | Question: 4

Consider the Deterministic Finite-state Automaton $(\mathrm{DFA})$ $\mathcal{A}$ shown below. The $\mathrm{DFA}$ runs on the alphabet $\{0,1\}$, and has the set of states $\{s,p,q,r\}$, with $s$ being the start state and $p$ being the only final state.




Which one of the following regular expressions correctly describes the language accepted by $\mathcal{A}$?

A. $1(0^*11)^*$
B. $0(0+1)^*$
C. $1(0+11)^*$
D. $1(110^*)^*$

gatecse-2023 theory-of-computation regular-expression one-mark

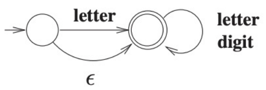
Answer key

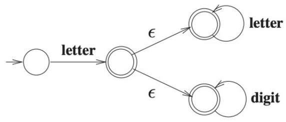
### 6.18.22 Regular Expression: GATE CSE 2023 | Question: 9

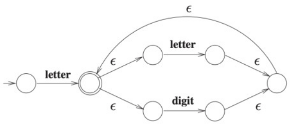
Consider the following definition of a lexical token **id** for an identifier in a programming language, using extended regular expressions:

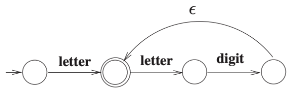
$$\begin{aligned}\textbf{letter} &\to [\mathrm{A-Za-z}]\\ \textbf{digit} &\to [0-9]\\ \textbf{id} &\to \textbf{letter}\ (\ \textbf{letter}\ \mid\ \textbf{digit}\ )^*\end{aligned}$$

Which one of the following Non-deterministic Finite-state Automata with $\epsilon$-transitions accepts the set of valid identifiers? (A double-circle denotes a final state)

A. 


B. 


C. 


D. 


gatecse-2023 theory-of-computation regular-expression one-mark

Answer key☟

### 6.18.23 Regular Expression: GATE CSE 2024 | Set 1 | Question: 51

Consider the following two regular expressions over the alphabet $\{0,1\}$ :

$r = 0^* + 1^*$

$s = 01^* + 10^*$

The total number of strings of length less than or equal to $5$, which are neither in $r$ nor in $s$, is ________.

gatecse-2024-set1 numerical-answers theory-of-computation regular-expression two-marks

Answer key☟

### 6.18.24 Regular Expression: GATE CSE 2024 | Set 2 | Question: 52

Let $L_1$ be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L_2 = \{w \in (a+b)^* || w | \leq 4\}$, where $|w|$ denotes the length of string $w$. The number of strings in $L_2$ which are also in $L_1$ is _________.

gatecse-2024-set2 numerical-answers theory-of-computation regular-expression two-marks

Answer key☟

### 6.18.25 Regular Expression: GATE IT 2004 | Question: 7

Which one of the following regular expressions is NOT equivalent to the regular expression $(a+b+c)^*$?

A. $(a^* + b^* + c^*)^*$
B. $(a^*b^*c^*)^*$
C. $((ab)^* + c^*)^*$
D. $(a^*b^* + c^*)^*$

gateit-2004 theory-of-computation regular-expression normal

Answer key☟

### 6.18.26 Regular Expression: GATE IT 2005 | Question: 5

Which of the following statements is TRUE about the regular expression $01^*0$?

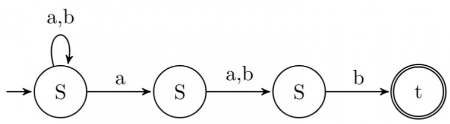
A. It represents a finite set of finite strings.
B. It represents an infinite set of finite strings.
C. It represents a finite set of infinite strings.
D. It represents an infinite set of infinite strings.

gateit-2005 theory-of-computation regular-expression easy

Answer key☟

### 6.18.27 Regular Expression: GATE IT 2006 | Question: 5

Which regular expression best describes the language accepted by the non-deterministic automaton below?




A. $(a+b)^*\ a(a+b)b$
B. $(abb)^*$
C. $(a+b)^*\ a(a+b)^*\ b(a+b)^*$
D. $(a+b)^*$

gateit-2006 theory-of-computation regular-expression normal

Answer key☟

### 6.18.28 Regular Expression: GATE IT 2007 | Question: 73

Consider the regular expression $R = (a+b)^*\ (aa+bb)\ (a+b)^*$

Which one of the regular expressions given below defines the same language as defined by the regular expression $R$ ?

A. $(a(ba)^* + b(ab)^*)(a+b)^+$
B. $(a(ba)^* + b(ab)^*)^*(a+b)^*$
C. $(a(ba)^*(a+bb) + b(ab)^*(b+aa))(a+b)^*$
D. $(a(ba)^*(a+bb) + b(ab)^*(b+aa))(a+b)^+$

gateit-2007 theory-of-computation regular-expression normal

Answer key☟

### 6.18.29 Regular Expression: GATE IT 2008 | Question: 5

Which of the following regular expressions describes the language over $\{0,1\}$ consisting of strings that contain exactly two $1$'s?

A. $(0+1)^*\ 11(0+1)^*$
B. $0^*\ 110^*$
C. $0^*10^*10^*$
D. $(0+1)^*1(0+1)^*1(0+1)^*$

gateit-2008 theory-of-computation regular-expression easy

Answer key☟

## 6.19 Regular Grammar (3)

### 6.19.1 Regular Grammar: GATE CSE 1990 | Question: 15a

Is the language generated by the grammar $G$ regular? If so, give a regular expression for it, else prove otherwise

- G:
  - $S \to aB$
  - $B \to bC$
  - $C \to xB$
  - $C \to c$

gate1990 descriptive theory-of-computation regular-language regular-grammar

Answer key☟

### 6.19.2 Regular Grammar: GATE CSE 2015 | Set 2 | Question: 35

Consider the alphabet $\Sigma = \{0,1\}$, the null/empty string $\lambda$ and the set of strings $X_0, X_1, \text{ and } X_2$ generated by the corresponding non-terminals of a regular grammar. $X_0, X_1, \text{ and } X_2$ are related as follows.

- $X_0 = 1X_1$
- $X_1 = 0X_1 + 1X_2$
- $X_2 = 0X_1 + \{\lambda\}$

Which one of the following choices precisely represents the strings in $X_0$?

A. $10(0^* + (10)^*)1$
B. $10(0^* + (10)^*)^*1$
C. $1(0 + 10)^*1$
D. $10(0 + 10)^*1 + 110(0 + 10)^*1$

gatecse-2015-set2 theory-of-computation regular-grammar normal

Answer key☟

### 6.19.3 Regular Grammar: GATE IT 2006 | Question: 29

Consider the regular grammar below

$S \to bS \mid aA \mid \epsilon$
$A \to aS \mid bA$

The Myhill-Nerode equivalence classes for the language generated by the grammar are

A. $\{w \in (a+b)^* \mid \#a(w) \text{ is even}) \text{ and} \{w \in (a+b)^* \mid \#a(w) \text{ is odd}\}$
B. $\{w \in (a+b)^* \mid \#a(w) \text{ is even}) \text{ and} \{w \in (a+b)^* \mid \#b(w) \text{ is odd}\}$
C. $\{w \in (a+b)^* \mid \#a(w) = \#b(w) \text{and } \{w \in (a+b)^* \mid \#a(w) \neq \#b(w)\}$
D. $\{\epsilon\}, \{wa \mid w \in (a+b)^* \text{and} \{wb \mid w \in (a+b)^*\}$

gateit-2006 theory-of-computation normal regular-grammar

Answer key☟

## 6.20 Regular Language (35)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (7Q)

### 6.20.1 Regular Language: GATE CSE 1987 | Question: 2h

State whether the following statements are TRUE or FALSE:

Regularity is preserved under the operation of string reversal.

gate1987 theory-of-computation regular-language true-false

Answer key☟

### 6.20.2 Regular Language: GATE CSE 1987 | Question: 2i

State whether the following statements are TRUE or FALSE:

All subsets of regular sets are regular.

gate1987 theory-of-computation regular-language true-false

Answer key☟

### 6.20.3 Regular Language: GATE CSE 1990 | Question: 3-viii

Let $R_1$ and $R_2$ be regular sets defined over the alphabet $\Sigma$ Then:

A. $R_1 \cap R_2$ is not regular.
B. $R_1 \cup R_2$ is regular.
C. $\Sigma^* - R_1$ is regular.
D. $R_1^*$ is not regular.

gate1990 normal theory-of-computation regular-language multiple-selects

Answer key☟

### 6.20.4 Regular Language: GATE CSE 1991 | Question: 03,xiv

Which of the following is the strongest correct statement about a finite language over some finite alphabet $\Sigma$?

A. It could be undecidable
B. It is Turing-machine recognizable
C. It is a context-sensitive language.
D. It is a regular language.
E. None of the above,

gate1991 theory-of-computation easy regular-language multiple-selects

Answer key

### 6.20.5 Regular Language: GATE CSE 1995 | Question: 2.24

Let $\Sigma = \{0, 1\}$, $L = \Sigma^*$ and $R = \{0^n 1^n \mid n > 0\}$ then the languages $L \cup R$ and $R$ are respectively

A. regular, regular
B. not regular, regular
C. regular, not regular
D. not regular, not regular

gate1995 theory-of-computation easy regular-language

Answer key

### 6.20.6 Regular Language: GATE CSE 1996 | Question: 1.10

Let $L \subseteq \Sigma^*$ where $\Sigma = \{a, b\}$. Which of the following is true?

a. $L = \{x \mid x \text{ has an equal number of } a\text{'s and } b\text{'s}\}$ is regular
b. $L = \{a^n b^n \mid n \geq 1\}$ is regular
c. $L = \{x \mid x \text{ has more number of } a\text{'s than } b\text{'s}\}$ is regular
d. $L = \{a^m b^n \mid m \geq 1, n \geq 1\}$ is regular

gate1996 theory-of-computation normal regular-language

Answer key

### 6.20.7 Regular Language: GATE CSE 1998 | Question: 2.6

Which of the following statements is false?

a. Every finite subset of a non-regular set is regular
b. Every subset of a regular set is regular
c. Every finite subset of a regular set is regular
d. The intersection of two regular sets is regular

gate1998 theory-of-computation easy regular-language

Answer key

### 6.20.8 Regular Language: GATE CSE 1999 | Question: 6

A. Given that $A$ is regular and $(A \cup B)$ is regular, does it follow that $B$ is necessarily regular? Justify your answer.
B. Given two finite automata $M1, M2$, outline an algorithm to decide if $L(M1) \subset L(M2)$. (note: strict subset)

gate1999 theory-of-computation normal regular-language descriptive

Answer key

### 6.20.9 Regular Language: GATE CSE 2000 | Question: 2.8

What can be said about a regular language $L$ over $\{a\}$ whose minimal finite state automaton has two states?

A. $L$ must be $\{a^n \mid n \text{ is odd}\}$
B. $L$ must be $\{a^n \mid n \text{ is even}\}$
C. $L$ must be $\{a^n \mid n \geq 0\}$
D. Either $L$ must be $\{a^n \mid n \text{ is odd}\}$, or $L$ must be $\{a^n \mid n \text{ is even}\}$

gatecse-2000 theory-of-computation easy regular-language

Answer key

### 6.20.10 Regular Language: GATE CSE 2001 | Question: 1.4

Consider the following two statements:

$S_1 : \{0^{2n} \mid n \geq 1\}$ is a regular language

$S_2 : \{0^m 1^n 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\}$ is a regular language

Which of the following statement is correct?

A. Only $S_1$ is correct
B. Only $S_2$ is correct
C. Both $S_1$ and $S_2$ are correct
D. None of $S_1$ and $S_2$ is correct

gatecse-2001 theory-of-computation easy regular-language

Answer key

### 6.20.11 Regular Language: GATE CSE 2001 | Question: 2.6

Consider the following languages:

- $L1 = \{ww \mid w \in \{a,b\}^*\}$
- $L2 = \{ww^R \mid w \in \{a,b\}^*, w^R \text{ is the reverse of w}\}$
- $L3 = \{0^{2i} \mid \text{ i is an integer}\}$
- $L4 = \{0^{i^2} \mid \text{ i is an integer}\}$

Which of the languages are regular?

A. Only $L1$ and $L2$
B. Only $L2, L3$ and $L4$
C. Only $L3$ and $L4$
D. Only $L3$

gatecse-2001 theory-of-computation normal regular-language

Answer key

### 6.20.12 Regular Language: GATE CSE 2007 | Question: 31

Which of the following languages is regular?

A. $\{ww^R \mid w \in \{0,1\}^+\}$
B. $\{ww^R x \mid x, w \in \{0,1\}^+\}$
C. $\{wxw^R \mid x, w \in \{0,1\}^+\}$
D. $\{xww^R \mid x, w \in \{0,1\}^+\}$

gatecse-2007 theory-of-computation normal regular-language

Answer key

### 6.20.13 Regular Language: GATE CSE 2007 | Question: 7

Which of the following is TRUE?

A. Every subset of a regular set is regular
B. Every finite subset of a non-regular set is regular
C. The union of two non-regular sets is not regular
D. Infinite union of finite sets is regular

gatecse-2007 theory-of-computation easy regular-language

Answer key

### 6.20.14 Regular Language: GATE CSE 2008 | Question: 53

Which of the following are regular sets?

I. $\{a^n b^{2m} \mid n \geq 0, m \geq 0\}$
II. $\{a^n b^m \mid n = 2m\}$
III. $\{a^n b^m \mid n \neq m\}$

IV. $\{xcy \mid x, y, \in \{a, b\}^*\}$

A. I and IV only
B. I and III only
C. I only
D. IV only

gatecse-2008 theory-of-computation normal regular-language

Answer key

### 6.20.15 Regular Language: GATE CSE 2011 | Question: 24

Let $P$ be a regular language and $Q$ be a context-free language such that $Q \subseteq P$. (For example, let $P$ be the language represented by the regular expression $p^*q^*$ and $Q$ be $\{p^nq^n \mid n \in N\}$). Then which of the following is **ALWAYS** regular?

A. $P \cap Q$
B. $P - Q$
C. $\Sigma^* - P$
D. $\Sigma^* - Q$

gatecse-2011 theory-of-computation easy regular-language

Answer key

### 6.20.16 Regular Language: GATE CSE 2012 | Question: 25

Given the language $L = \{ab, aa, baa\}$, which of the following strings are in $L^*$?

1. $abaabaaabaa$
2. $aaaabaaaa$
3. $baaaaabaaaab$
4. $baaaaabaa$

A. $1, 2 \text{ and } 3$
B. $2, 3 \text{ and } 4$
C. $1, 2 \text{ and } 4$
D. $1, 3 \text{ and } 4$

gatecse-2012 theory-of-computation easy regular-language

Answer key

### 6.20.17 Regular Language: GATE CSE 2013 | Question: 8

Consider the languages $L_1 = \phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1{L_2}^* \cup {L_1}^*$ ?

A. $\{\epsilon\}$
B. $\phi$
C. $a^*$
D. $\{\epsilon, a\}$

gatecse-2013 theory-of-computation normal regular-language

Answer key

### 6.20.18 Regular Language: GATE CSE 2014 | Set 1 | Question: 15

Which one of the following is **TRUE**?

A. The language $L = \{a^nb^n \mid n \geq 0\}$ is regular.
B. The language $L = \{a^n \mid n \text{ is prime }\}$ is regular.
C. The language $L$= $\{w \mid w \text{ has } 3k+1\ b's \text{ for some } k \in N \text{ with } \Sigma = \{a, b\}\}$ is regular.
D. The language $L = \{ww \mid w \in \Sigma^* \text{ with } \Sigma = \{0, 1\}\}$ is regular.

gatecse-2014-set1 theory-of-computation regular-language normal

Answer key

### 6.20.19 Regular Language: GATE CSE 2014 | Set 2 | Question: 15

If $L_1 = \{a^n \mid n \geq 0\}$ and $L_2 = \{b^n \mid n \geq 0\}$ , consider

I. $L_1 . L_2$ is a regular language
II. $L_1 . L_2 = \{a^nb^n \mid n \geq 0\}$

Which one of the following is CORRECT?

A. Only I
B. Only II
C. Both I and II
D. Neither I nor II

gatecse-2014-set2 theory-of-computation normal regular-language

Answer key☟

### 6.20.20 Regular Language: GATE CSE 2014 | Set 2 | Question: 36

L e t $L_1 = \{w \in \{0,1\}^* \mid w \text{ has at least as many occurrences of } (110)'\text{s as } (011)'\text{s}\}$. Let $L_2 = \{w \in \{0,1\}^* \mid w \text{ has at least as many occurrences of } (000)'\text{s as } (111)'\text{s}\}$. Which one of the following is TRUE?

A. $L_1$ is regular but not $L_2$
B. $L_2$ is regular but not $L_1$
C. Both $L_1$ and $L_2$ are regular
D. Neither $L_1$ nor $L_2$ are regular

gatecse-2014-set2 theory-of-computation normal regular-language

Answer key☟

### 6.20.21 Regular Language: GATE CSE 2015 | Set 2 | Question: 51

Which of the following is/are regular languages?

$L_1 : \left\{wxw^R \mid w, x \in \{a,b\}^* \text{ and } |w|, |x| > 0\right\}, w^R \text{ is the reverse of string } w$

$L_2 : \{a^n b^m \mid m \neq n \text{ and } m, n \geq 0\}$

$L_3 : \{a^p b^q c^r \mid p, q, r \geq 0\}$

A. $L_1$ and $L_3$ only
B. $L_2$ only
C. $L_2$ and $L_3$ only
D. $L_3$ only

gatecse-2015-set2 theory-of-computation normal regular-language

Answer key☟

### 6.20.22 Regular Language: GATE CSE 2016 | Set 2 | Question: 17

Language $L_1$ is defined by the grammar: $S_1 \to aS_1b \mid \varepsilon$

Language $L_2$ is defined by the grammar: $S_2 \to abS_2 \mid \varepsilon$

Consider the following statements:

- P: $L_1$ is regular
- Q: $L_2$ is regular

Which one of the following is **TRUE**?

A. Both $P$ and $Q$ are true.
B. $P$ is true and $Q$ is false.
C. $P$ is false and $Q$ is true.
D. Both $P$ and $Q$ are false.

gatecse-2016-set2 theory-of-computation easy regular-language

Answer key☟

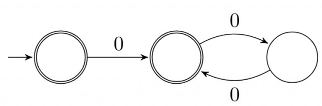
### 6.20.23 Regular Language: GATE CSE 2018 | Question: 52

Given a language $L$, define $L^i$ as follows:

$$L^0 = \{\varepsilon\}$$

$$L^i = L^{i-1} \bullet L \text{ for all } I > 0$$

The order of a language $L$ is defined as the smallest $k$ such that $L^k = L^{k+1}$. Consider the language $L_1$(over alphabet $0$) accepted by the following automaton.




The order of $L_1$ is ________.

gatecse-2018 theory-of-computation numerical-answers regular-language two-marks

Answer key

## 6.20.24 Regular Language: GATE CSE 2019 | Question: 7

If $L$ is a regular language over $\Sigma = \{a, b\}$, which one of the following languages is NOT regular?

A. $L. L^R = \{xy \mid x \in L, y^R \in L\}$
B. $\{ww^R \mid w \in L\}$
C. $\text{Prefix}\ (L) = \{x \in \Sigma^* \mid \exists y \in \Sigma^* \text{such that } xy \in L\}$
D. $\text{Suffix}\ (L) = \{y \in \Sigma^* \mid \exists x \in \Sigma^* \text{such that } xy \in L\}$

gatecse-2019 theory-of-computation regular-language one-mark

Answer key

## 6.20.25 Regular Language: GATE CSE 2020 | Question: 51

Consider the following language.

$L = \{x \in \{a, b\}^* \mid \text{number of } a\text{'s in } x \text{ divisible by } 2 \text{ but not divisible by } 3\}$

The minimum number of states in DFA that accepts $L$ is _________

gatecse-2020 numerical-answers theory-of-computation regular-language two-marks

Answer key

## 6.20.26 Regular Language: GATE CSE 2020 | Question: 8

Consider the following statements.

I. If $L_1 \cup L_2$ is regular, then both $L_1$ and $L_2$ must be regular.
II. The class of regular languages is closed under infinite union.

Which of the above statements is/are TRUE?

A. I only
B. II only
C. Both I and II
D. Neither I nor II

gatecse-2020 theory-of-computation regular-language one-mark

Answer key

## 6.20.27 Regular Language: GATE CSE 2021 | Set 2 | Question: 9

Let $L \subseteq \{0, 1\}^*$ be an arbitrary regular language accepted by a minimal $\text{DFA}$ with $k$ states. Which one of the following languages must necessarily be accepted by a minimal $\text{DFA}$ with $k$ states?

A. $L - \{01\}$
B. $L \cup \{01\}$
C. $\{0,1\}^* - L$
D. $L \cdot L$

gatecse-2021-set2 theory-of-computation finite-automata regular-language one-mark

Answer key

## 6.20.28 Regular Language: GATE CSE 2024 | Set 1 | Question: 13

Let $L_1, L_2$ be two regular languages and $L_3$ a language which is not regular.
Which of the following statements is/are always TRUE?

A. $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \phi$
B. $L_1 \cup L_3$ is not regular
C. $\overline{L_3}$ is not regular
D. $\overline{L_1} \cup \overline{L_2}$ is regular

gatecse-2024-set1 multiple-selects theory-of-computation regular-language one-mark

Answer key

### 6.20.29 Regular Language: GATE CSE 2025 | Set 1 | Question: 18

A regular language $L$ is accepted by a non-deterministic finite automaton (NFA) with $n$ states. Which of the following statement(s) is/are FALSE?

A. $L$ may have an accepting NFA with $< n$ states.
B. $L$ may have an accepting DFA with $< n$ states.
C. There exists a DFA with $\leq 2^n$ states that accepts $L$.
D. Every DFA that accepts $L$ has $> 2^n$ states.

gatecse2025-set1 theory-of-computation finite-automata regular-language multiple-selects one-mark

Answer key

### 6.20.30 Regular Language: GATE CSE 2025 | Set 1 | Question: 34

Consider the following two languages over the alphabet $\{a, b\}$:

$$L_1 = \{\alpha\beta\alpha \mid \alpha \in \{a,b\}^+ \text{AND } \beta \in \{a,b\}^+\}$$
$$L_2 = \{\alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{AND } \beta \in \{a,b\}^+\}$$

Which ONE of the following statements is CORRECT?

A. Both $L_1$ and $L_2$ are regular languages.
B. $L_1$ is a regular language but $L_2$ is not a regular language.
C. $L_1$ is not a regular language but $L_2$ is a regular language.
D. Neither $L_1$ nor $L_2$ is a regular language.

gatecse2025-set1 theory-of-computation regular-language two-marks

Answer key

### 6.20.31 Regular Language: GATE CSE 2025 | Set 2 | Question: 42

Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\#_\alpha(x)$ denote the number of occurrences of $\alpha$ in $x$.
Which one or more of the following option(s) define(s) regular language(s)?

A. $\{a^m b^n \mid m, n \geq 0\}$
B. $\{a,b\}^* \cap \{a^m b^n c^{m-n} \mid m \geq n \geq 0\}$
C. $\{w \mid w \in \{a,b\}^*, \#_a(w) \equiv 2 (\bmod 7)$, and $\#_b(w) \equiv 3 (\bmod 9)\}$
D. $\{w \mid w \in \{a,b\}^*, \#_a(w) \equiv 2 (\bmod 7)$, and $\#_a(w) = \#_b(w)\}$

gatecse2025-set2 theory-of-computation regular-language multiple-selects two-marks

Answer key

### 6.20.32 Regular Language: GATE IT 2006 | Question: 30

Which of the following statements about regular languages is NOT true ?

A. Every language has a regular superset
B. Every language has a regular subset
C. Every subset of a regular language is regular
D. Every subset of a finite language is regular

gateit-2006 theory-of-computation easy regular-language

Answer key

### 6.20.33 Regular Language: GATE IT 2006 | Question: 80

Let $L$ be a regular language. Consider the constructions on $L$ below:

I. repeat $(L) = \{ww \mid w \in L\}$
II. prefix $(L) = \{u \mid \exists v : uv \in L\}$
III. suffix $(L) = \{v \mid \exists u : uv \in L\}$
IV. half $(L) = \{u \mid \exists v : |v| = |u| \text{ and } uv \in L\}$

Which of the constructions could lead to a non-regular language?

A. Both I and IV
B. Only I
C. Only IV
D. Both II and III

gateit-2006 theory-of-computation normal regular-language

Answer key

### 6.20.34 Regular Language: GATE IT 2006 | Question: 81

Let $L$ be a regular language. Consider the constructions on $L$ below:

I. $\text{repeat}(L) = \{ww \mid w \in L\}$
II. $\text{prefix}(L) = \{u \mid \exists v : uv \in L\}$
III. $\text{suffix}(L) = \{v \mid \exists u : uv \in L\}$
IV. $\text{half}(L) = \{u \mid \exists v : |v| = |u| \text{ and } uv \in L\}$

Which of the constructions could lead to a non-regular language?

a. Both I and IV
b. Only I
c. Only IV
d. Both II and III

Which choice of $L$ is best suited to support your answer above?

A. $(a+b)^*$
B. $\{\epsilon, a, ab, bab\}$
C. $(ab)^*$
D. $\{a^n b^n \mid n \geq 0\}$

gateit-2006 theory-of-computation normal regular-language

Answer key

### 6.20.35 Regular Language: GATE IT 2008 | Question: 35

Which of the following languages is (are) non-regular?

- $L_1 = \{0^m 1^n \mid 0 \leq m \leq n \leq 10000\}$
- $L_2 = \{w \mid w \text{ reads the same forward and backward}\}$
- $L_3 = \{w \in \{0,1\}^* \mid w \text{ contains an even number of 0's and an even number of 1's}\}$

A. $L_2$ and $L_3$ only
B. $L_1$ and $L_2$ only
C. $L_3$ only
D. $L_2$ only

gateit-2008 theory-of-computation normal regular-language

Answer key

## 6.21 Turing Machine (1)

✍ **Practice Test:** Test 1 (13Q)

### 6.21.1 Turing Machine: GATE CSE 2026 | Set 2 | Question: 3

Which one of the following statements is equivalent to the following assertion?
Turing machine $M$ decides the language $L \subseteq \{0,1\}^*$

A. Turing machine $M$ halts on all input strings in $\{0,1\}^*$
B. Turing machine $M$ accepts all input strings in $L$
C. Turing machine $M$ rejects all input strings in $\{0,1\}^* - L$
D. Turing machine $M$ accepts all input strings in $L$ and rejects all input strings in $\{0,1\}^* - L$

gatecse-2026-set2 theory-of-computation turing-machine one-mark

Answer key☟

# Answer Keys

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 6.1.1 | A;C | 6.1.2 | A;B | 6.1.3 | A | 6.1.4 | C | 6.1.5 | D |
| 6.1.6 | B | 6.1.7 | B | 6.1.8 | B;C | 6.1.9 | C | 6.1.10 | D |
| 6.2.1 | A | 6.2.2 | A;B;C | 6.3.1 | False | 6.3.2 | A;D | 6.3.3 | A |
| 6.3.4 | B | 6.3.5 | B | 6.3.6 | B | 6.3.7 | N/A | 6.3.8 | N/A |
| 6.3.9 | B | 6.3.10 | C | 6.3.11 | B | 6.3.12 | D | 6.3.13 | B |
| 6.3.14 | B | 6.3.15 | D | 6.3.16 | D | 6.3.17 | B | 6.3.18 | B |
| 6.3.19 | D | 6.3.20 | A | 6.3.21 | C | 6.3.22 | C | 6.3.23 | C |
| 6.3.24 | B;C;D | 6.3.25 | B;C;D | 6.3.26 | B | 6.3.27 | C | 6.3.28 | B;D |
| 6.3.29 | A;C;D | 6.3.30 | B | 6.3.31 | B | 6.3.32 | A | 6.3.33 | C |
| 6.3.34 | D | 6.3.35 | B | 6.4.1 | B | 6.4.2 | C | 6.4.3 | B |
| 6.5.1 | True | 6.5.2 | True | 6.5.3 | N/A | 6.5.4 | B;D | 6.5.5 | B;C |
| 6.5.6 | N/A | 6.5.7 | D | 6.5.8 | B | 6.5.9 | N/A | 6.5.10 | A |
| 6.5.11 | B | 6.5.12 | N/A | 6.5.13 | N/A | 6.5.14 | C | 6.5.15 | A |
| 6.5.16 | B | 6.5.17 | B | 6.5.18 | D | 6.5.19 | D | 6.5.20 | A |
| 6.5.21 | D | 6.5.22 | C | 6.5.23 | C | 6.5.24 | A | 6.5.25 | D |
| 6.5.26 | D | 6.5.27 | A | 6.5.28 | C | 6.5.29 | A;B;C | 6.5.30 | D |
| 6.6.1 | A | 6.7.1 | N/A | 6.7.2 | N/A | 6.7.3 | N/A | 6.7.4 | N/A |
| 6.7.5 | N/A | 6.7.6 | A | 6.7.7 | N/A | 6.7.8 | A | 6.7.9 | N/A |
| 6.7.10 | C | 6.7.11 | B | 6.7.12 | A | 6.7.13 | X | 6.7.14 | B |
| 6.7.15 | C | 6.7.16 | A | 6.7.17 | C | 6.7.18 | C | 6.7.19 | B |
| 6.7.20 | D | 6.7.21 | D | 6.7.22 | A | 6.7.23 | B | 6.7.24 | C |
| 6.7.25 | D | 6.7.26 | 256 : 256 | 6.7.27 | B | 6.7.28 | B;C | 6.7.29 | A |
| 6.7.30 | B | 6.7.31 | C | 6.7.32 | C;D | 6.7.33 | B | 6.7.34 | A |
| 6.7.35 | B | 6.7.36 | D | 6.7.37 | A | 6.7.38 | D | 6.7.39 | A |
| 6.7.40 | A | 6.7.41 | A | 6.7.42 | A | 6.7.43 | X | 6.8.1 | 5:5 |
| 6.9.1 | C | 6.9.2 | N/A | 6.9.3 | N/A | 6.9.4 | N/A | 6.9.5 | A;C |
| 6.9.6 | B | 6.9.7 | B | 6.9.8 | B | 6.9.9 | A | 6.9.10 | D |
| 6.9.11 | B | 6.9.12 | B | 6.9.13 | C | 6.9.14 | D | 6.9.15 | C |
| 6.9.16 | D | 6.9.17 | C | 6.9.18 | D | 6.9.19 | C | 6.9.20 | B |
| 6.9.21 | D | 6.9.22 | B | 6.9.23 | C | 6.9.24 | A | 6.9.25 | B;C;D |
| 6.9.26 | B;C;D | 6.9.27 | A;B;C | 6.9.28 | A;C;D | 6.9.29 | D | 6.9.30 | A |
| 6.9.31 | B | 6.10.1 | C | 6.11.1 | 0 | 6.11.2 | N/A | 6.11.3 | N/A |
| 6.11.4 | B | 6.11.5 | N/A | 6.11.6 | B | 6.11.7 | B | 6.11.8 | D |
| 6.11.9 | B | 6.11.10 | D | 6.11.11 | A | 6.11.12 | B | 6.11.13 | C |
| 6.11.14 | B | 6.11.15 | A | 6.11.16 | 1 | 6.11.17 | 3 | 6.11.18 | B |
| 6.11.19 | 2 | 6.11.20 | 4 | 6.11.21 | 8 | 6.11.22 | D | 6.11.23 | 120 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 6.11.24 | 4 | 6.11.25 | A | 6.12.1 | A;C | 6.12.2 | C | 6.12.3 | B |
| 6.12.4 | D | 6.12.5 | B | 6.12.6 | C | 6.13.1 | 6:6 | 6.13.2 | B;D |
| 6.14.1 | D | 6.14.2 | D | 6.15.1 | N/A | 6.15.2 | A | 6.15.3 | N/A |
| 6.15.4 | A | 6.15.5 | N/A | 6.15.6 | N/A | 6.15.7 | D | 6.15.8 | B |
| 6.15.9 | D | 6.15.10 | 50 : 50 | 6.15.11 | A | 6.15.12 | B | 6.15.13 | D |
| 6.15.14 | B | 6.15.15 | C | 6.16.1 | A;D | 6.16.2 | A | 6.16.3 | D |
| 6.16.4 | D | 6.16.5 | A | 6.16.6 | B | 6.16.7 | D | 6.16.8 | D |
| 6.16.9 | B | 6.16.10 | C | 6.16.11 | D | 6.16.12 | B | 6.16.13 | D |
| 6.16.14 | C | 6.16.15 | D | 6.16.16 | A | 6.17.1 | C | 6.17.2 | C |
| 6.18.1 | N/A | 6.18.2 | A;C | 6.18.3 | A | 6.18.4 | N/A | 6.18.5 | C |
| 6.18.6 | C | 6.18.7 | D | 6.18.8 | C;D | 6.18.9 | D | 6.18.10 | N/A |
| 6.18.11 | C | 6.18.12 | A | 6.18.13 | C | 6.18.14 | B | 6.18.15 | B |
| 6.18.16 | 3 | 6.18.17 | B | 6.18.18 | X | 6.18.19 | A;B;C | 6.18.20 | D |
| 6.18.21 | C | 6.18.22 | C | 6.18.23 | 44 | 6.18.24 | 15 | 6.18.25 | C |
| 6.18.26 | B | 6.18.27 | A | 6.18.28 | C | 6.18.29 | C | 6.19.1 | N/A |
| 6.19.2 | C | 6.19.3 | A | 6.20.1 | True | 6.20.2 | False | 6.20.3 | B;C |
| 6.20.4 | D | 6.20.5 | C | 6.20.6 | D | 6.20.7 | B | 6.20.8 | N/A |
| 6.20.9 | X | 6.20.10 | A | 6.20.11 | D | 6.20.12 | C | 6.20.13 | B |
| 6.20.14 | A | 6.20.15 | C | 6.20.16 | C | 6.20.17 | A | 6.20.18 | C |
| 6.20.19 | A | 6.20.20 | A | 6.20.21 | A | 6.20.22 | C | 6.20.23 | 2 |
| 6.20.24 | B | 6.20.25 | 6 | 6.20.26 | D | 6.20.27 | C | 6.20.28 | C;D |
| 6.20.29 | D | 6.20.30 | C | 6.20.31 | A;C | 6.20.32 | C | 6.20.33 | B |
| 6.20.34 | A | 6.20.35 | D | 6.21.1 | D | | | | |